

Algebraic Geometry

Lecture Notes

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Chapter 1

Sheaves and Categories

The language of categories and functors is pervasive in modern algebraic geometry, nowhere more so than in the theory of sheaves. This chapter develops the categorical framework we shall need—monomorphisms, epimorphisms, Yoneda’s lemma, adjoint functors, kernels and cokernels in abelian categories—and applies it to two constructions that will reappear throughout: sheafification, and the pair of pushforward/inverse-image functors associated to a continuous map.

1.1 Categorical preliminaries

1.1.1 Categories, functors, natural transformations

Definition 1.1.1. A *category* (范畴) $\mathcal{C}$ consists of a class of *objects* $\text{Ob}(\mathcal{C})$ and, for every pair $A, B \in \text{Ob}(\mathcal{C})$, a set $\text{Hom}_{\mathcal{C}}(A, B)$ of *morphisms*, together with an associative composition law

$$\text{Hom}_{\mathcal{C}}(B, C) \times \text{Hom}_{\mathcal{C}}(A, B) \longrightarrow \text{Hom}_{\mathcal{C}}(A, C), \quad (\psi, \varphi) \longmapsto \psi \circ \varphi,$$

and for every A an identity $\text{id}_A \in \text{Hom}_{\mathcal{C}}(A, A)$ that is a two-sided unit for $\circ$.

Definition 1.1.2. A *functor* (函子) $F : \mathcal{C} \rightarrow \mathcal{D}$ consists of an assignment $A \mapsto F(A)$ on objects together with, for every pair A, B , a map

$$F : \text{Hom}_{\mathcal{C}}(A, B) \longrightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B))$$

preserving identities and composition: $F(\text{id}_A) = \text{id}_{F(A)}$ and $F(\psi \circ \varphi) = F(\psi) \circ F(\varphi)$. A functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$ (equivalently, one reversing the direction of arrows) is called *contravariant*.

Definition 1.1.3. Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be functors. A *natural transformation* (自然变换) $\eta : F \Rightarrow G$ is a family of morphisms

$$\eta_A : F(A) \longrightarrow G(A), \quad A \in \text{Ob}(\mathcal{C}),$$

such that for every morphism $\varphi : A \rightarrow B$ in $\mathcal{C}$ the square

$$\begin{array}{ccc} F(A) & \xrightarrow{F(\varphi)} & F(B) \\ \eta_A \downarrow & & \downarrow \eta_B \\ G(A) & \xrightarrow{G(\varphi)} & G(B) \end{array}$$

commutes. η is a *natural isomorphism* if each η_A is an isomorphism.

1.1.2 Monomorphisms and epimorphisms

Definition 1.1.4. A morphism $\varphi : A \rightarrow B$ in a category $\mathcal{C}$ is a *monomorphism* (单态射) if, for every object C and every pair $\varphi_1, \varphi_2 : C \rightarrow A$,

$$\varphi \circ \varphi_1 = \varphi \circ \varphi_2 \implies \varphi_1 = \varphi_2.$$

It is an *epimorphism* (满态射) if, dually, for every pair $\psi_1, \psi_2 : B \rightarrow C$,

$$\psi_1 \circ \varphi = \psi_2 \circ \varphi \implies \psi_1 = \psi_2.$$

Remark 1.1.5. 若 φ 有左逆则自动是 mono, 对偶地有右逆则是 epi; 但反过来未必成立。同时具有左右逆的态射就是 isomorphism。在 **Set** 和 **Ab** 中 mono/epi 恰好是通常意义的单/满射; 但在一般范畴中未必——例如 **Ring** 中 $\mathbb{Z} \hookrightarrow \mathbb{Q}$ 既是 mono 又是 epi, 却显然不是同构。

For presheaves and sheaves of abelian groups on a topological space, the categorical notions of mono and epi admit concrete characterizations.

Proposition 1.1.6. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism in $\text{PSh}(X)$ or $\text{Sh}(X)$.

- (1) φ is a monomorphism $\iff \ker \varphi = 0$.
- (2) In $\text{Sh}(X)$, φ is an epimorphism $\iff$ the induced map on every stalk $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is surjective.

We postpone the proof to §1.5, where it emerges naturally from the stalk adjunction; see [definitions 1.5.6](#) and [1.5.7](#).

1.1.3 Yoneda's lemma

Theorem 1.1.7 (Yoneda). *Let $\mathcal{C}$ be a category. For every $A \in \mathcal{C}$, the functor*

$$h_A := \text{Hom}_{\mathcal{C}}(A, -) : \mathcal{C} \longrightarrow \mathbf{Set}$$

determines A up to unique isomorphism. Concretely, if $A, B \in \mathcal{C}$ and there is a natural isomorphism

$$\eta : \text{Hom}_{\mathcal{C}}(A, -) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(B, -),$$

then there exists a unique isomorphism $A \cong B$ inducing η .

Proof. Set $u := \eta_B^{-1}(\text{id}_B) : A \rightarrow B$ and $v := \eta_A(\text{id}_A) : B \rightarrow A$. Naturality applied to $u : A \rightarrow B$ gives $\eta_A(v) \circ u = \eta_A(u \circ v)$ and $u = \eta_A(v \circ \text{id}_A) = \eta_A(v)$, so a short diagram chase shows $u \circ v = \text{id}_B$ and $v \circ u = \text{id}_A$. $\square$

Remark 1.1.8. Yoneda 的实际威力是“对象由其 Hom 仿函数唯一决定”。在实用上, 这个 corollary 就是核心: 要证两个对象同构, 只需证它们表示同一个函子。下面的 sheafification 的唯一性、ker/coker 的唯一性、fibre product 的唯一性, 都是这个套路。

1.1.4 Adjoint functors

Definition 1.1.9. Let $\mathcal{C}, \mathcal{D}$ be categories and $F : \mathcal{C} \rightarrow \mathcal{D}, G : \mathcal{D} \rightarrow \mathcal{C}$ functors. We say F is *left adjoint* to G (and G is *right adjoint* to F), written $F \dashv G$, if there is a bijection

$$\text{Hom}_{\mathcal{D}}(F(A), B) \cong \text{Hom}_{\mathcal{C}}(A, G(B))$$

natural in $A \in \mathcal{C}$ and $B \in \mathcal{D}$.

Remark 1.1.10. “Natural” 的意思是: 对任何 $A \rightarrow A'$ 和 $B \rightarrow B'$, 相应方块交换。等价地, 伴随可以用 $\text{unit } \eta : \text{id}_{\mathcal{C}} \Rightarrow GF$ 和 $\text{counit } \varepsilon : FG \Rightarrow \text{id}_{\mathcal{D}}$ 刻画。标准的例子: tensor 与 Hom 伴随、自由与遗忘伴随。

1.1.5 Additive categories

Definition 1.1.11. A category $\mathcal{C}$ is *additive* (加性范畴) if

- (i) for every pair A, B , $\text{Hom}_{\mathcal{C}}(A, B)$ carries the structure of an abelian group;
- (ii) composition is bilinear: $(\psi_1 + \psi_2) \circ \varphi = \psi_1 \circ \varphi + \psi_2 \circ \varphi$ and $\psi \circ (\varphi_1 + \varphi_2) = \psi \circ \varphi_1 + \psi \circ \varphi_2$.

We further require that $\mathcal{C}$ have a zero object and all finite direct sums, though both follow formally from (i)–(ii) plus mild assumptions.

1.2 Sheafification

1.2.1 The universal property

Definition 1.2.1. Let $F \in \text{PSh}(X)$. A *sheafification* (层化) of F is a pair (F^+, θ) where $F^+ \in \text{Sh}(X)$ and $\theta : F \rightarrow F^+$ is a morphism of presheaves, satisfying the following universal property: for every sheaf $\mathcal{G}$, the map

$$\text{Hom}_{\text{Sh}(X)}(F^+, \mathcal{G}) \xrightarrow{\sim} \text{Hom}_{\text{PSh}(X)}(F, \mathcal{G}), \quad \rho \longmapsto \rho \circ \theta,$$

is a bijection.

1.2.2 Uniqueness

Proposition 1.2.2. If sheafifications (F_1^+, θ_1) and (F_2^+, θ_2) exist, there is a unique isomorphism $F_1^+ \xrightarrow{\sim} F_2^+$ compatible with the θ_i .

Proof. Both F_i^+ represent the same functor $\mathcal{G} \mapsto \text{Hom}_{\text{PSh}(X)}(F, \mathcal{G})$ on $\text{Sh}(X)$. Apply Yoneda ([definition 1.1.7](#)). $\square$

1.2.3 Existence via compatible germs

Theorem 1.2.3. Every presheaf F on X admits a sheafification.

Proof. Define a presheaf F^+ by

$$F^+(U) := \left\{ f : U \rightarrow \prod_{x \in U} F_x \left| \begin{array}{l} f(x) \in F_x \text{ for all } x \in U, \text{ and for every } x \in U \\ \text{there exist } V \subseteq U \text{ open and } s \in F(V) \\ \text{with } f(y) = s_y \text{ for all } y \in V \end{array} \right. \right\},$$

with the obvious restriction maps. The sheaf axiom for F^+ is immediate from the pointwise definition (locality and gluing both reduce to pointwise checks). Define $\theta : F \rightarrow F^+$ on U by

$$\theta_U(s) : x \longmapsto s_x \in F_x.$$

It remains to verify three claims.

Claim 1: $\theta_x : F_x \rightarrow F_x^+$ is an isomorphism for every $x \in X$.

Surjectivity. Given $[(U, f)] \in F_x^+$ with $f \in F^+(U)$, by construction there exist $V \subseteq U$ and $s \in F(V)$ with $f|_V = \theta_V(s)$. Then

$$\theta_x([(V, s)]) = [(V, \theta_V(s))] = [(V, f|_V)] = [(U, f)] \text{ in } F_x^+.$$

Injectivity. Suppose $\theta_x([(U_1, f_1)]) = \theta_x([(U_2, f_2)])$, i.e. $[(U_1, \theta(f_1))] = [(U_2, \theta(f_2))]$ in F_x^+ . There exists $W \subseteq U_1 \cap U_2$ with $x \in W$ and $\theta(f_1)|_W = \theta(f_2)|_W$. By the definition of θ , this says $(f_1)_y = (f_2)_y$ in F_y for every $y \in W$. In particular $(f_1)_x = (f_2)_x$, so $[(U_1, f_1)] = [(U_2, f_2)]$ in F_x .

Claim 2: Universal property. Let $\mathcal{G}$ be a sheaf and $\alpha : F \rightarrow \mathcal{G}$ a morphism of presheaves. We construct $\alpha^+ : F^+ \rightarrow \mathcal{G}$ with $\alpha = \alpha^+ \circ \theta$.

Given $f \in F^+(U)$, cover U by opens $U_x \ni x$ on which $f|_{U_x} = \theta(s_x)$ for some $s_x \in F(U_x)$. On overlaps $U_x \cap U_y$, the sections s_x, s_y have equal germs at each point of $U_x \cap U_y$ (both equal f), so $\alpha(s_x)$ and $\alpha(s_y)$ have equal germs at each point of $U_x \cap U_y$; since $\mathcal{G}$ is a sheaf and this is a local condition, $\alpha(s_x)|_{U_x \cap U_y} = \alpha(s_y)|_{U_x \cap U_y}$.¹ The sheaf axiom for $\mathcal{G}$ glues the $\alpha(s_x)$ to a section $t \in \mathcal{G}(U)$; set $\alpha^+(f) := t$.

Claim 3: Uniqueness of α^+ . If $\beta_1, \beta_2 : F^+ \rightarrow \mathcal{G}$ both satisfy $\beta_i \circ \theta = \alpha$, then at every stalk

$$(\beta_1)_x \circ \theta_x = (\beta_2)_x \circ \theta_x,$$

and θ_x is an isomorphism by Claim 1, so $(\beta_1)_x = (\beta_2)_x$ for all x . A morphism of sheaves determined on every stalk must itself be determined ([definition 1.5.7](#)), so $\beta_1 = \beta_2$. $\square$

Remark 1.2.4. F^+ 的构造叫做“étalé-space”或“compatible germs”：先把所有 stalks 拼成一个“巨大的集合 $\coprod_x F_x$ ”，再把 $F^+(U)$ 定义为“局部来自某个 $s \in F(V)$ 的切面”。这本质上和流形上的 vector bundle 的截面是同一个构造。

1.2.4 Functoriality and the adjunction

Any morphism $\alpha : F_1 \rightarrow F_2$ in $\text{PSh}(X)$ composes with $\theta_2 : F_2 \rightarrow F_2^+$ to give $\theta_2 \circ \alpha : F_1 \rightarrow F_2^+$, a morphism from a presheaf to a sheaf. By the universal property, this factors through a unique morphism $\alpha^+ : F_1^+ \rightarrow F_2^+$:

$$\begin{array}{ccc} F_1 & \xrightarrow{\alpha} & F_2 \\ \theta_1 \downarrow & & \downarrow \theta_2 \\ F_1^+ & \xrightarrow{\alpha^+} & F_2^+ \end{array}$$

Functoriality (preservation of identities and composition) follows from the uniqueness clause of the universal property. Thus $(-)^+ : \text{PSh}(X) \rightarrow \text{Sh}(X)$ is a functor.

Proposition 1.2.5. *Let $\iota : \text{Sh}(X) \hookrightarrow \text{PSh}(X)$ be the inclusion (forgetful) functor. Then*

$$(-)^+ \dashv \iota,$$

¹A direct but subtle point: equal germs at every point plus $\mathcal{G}$ being a sheaf force equality of sections; see [definition 1.5.7](#).

i.e. for every $F \in \text{PSh}(X)$ and $\mathcal{G} \in \text{Sh}(X)$,

$$\text{Hom}_{\text{Sh}(X)}(F^+, \mathcal{G}) \cong \text{Hom}_{\text{PSh}(X)}(F, \iota \mathcal{G}),$$

naturally in F and $\mathcal{G}$.

Proof. This is exactly the universal property of [definition 1.2.1](#). □

1.3 Kernels, images, cokernels, exact sequences

Throughout this section, all presheaves and sheaves take values in **Ab**.

1.3.1 Recall: in abelian groups

For a homomorphism $f : A \rightarrow B$ in **Ab**:

$$\ker f = \{a \in A : f(a) = 0\}, \quad \text{Im } f = \{f(a) : a \in A\}, \quad \text{coker } f = B / \text{Im } f.$$

A sequence $A \xrightarrow{f} B \xrightarrow{g} C$ is *exact at B* (在 B 处正合) if $\ker g = \text{Im } f$.

1.3.2 In presheaves: everything sectionwise

For a morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ in $\text{PSh}(X)$, define presheaves sectionwise:

$$(\ker \varphi)(U) := \ker(\varphi_U), \quad (\text{Im } \varphi)(U) := \text{Im}(\varphi_U), \quad (\text{coker } \varphi)(U) := \text{coker}(\varphi_U).$$

A sequence $\mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H}$ in $\text{PSh}(X)$ is exact iff $\mathcal{F}(U) \rightarrow \mathcal{G}(U) \rightarrow \mathcal{H}(U)$ is exact in **Ab** for every open U .

1.3.3 In sheaves: kernel sectionwise, image and cokernel need sheafification

For a morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of sheaves:

- $(\ker \varphi)(U) := \ker(\varphi_U)$ is again a sheaf; this is the kernel in $\text{Sh}(X)$.
- The sectionwise image and cokernel are in general only presheaves; we define

$$\text{Im}_{\text{Sh}}(\varphi) := (\text{Im}_{\text{PSh}} \varphi)^+, \quad \text{coker}_{\text{Sh}}(\varphi) := (\text{coker}_{\text{PSh}} \varphi)^+.$$

A sequence of sheaves $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$ is *exact at $\mathcal{G}$* if $\ker g = \text{Im } f$ as subsheaves of $\mathcal{G}$. In particular

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H}$$

is exact iff $\mathcal{F} \rightarrow \mathcal{G}$ is injective and $\text{Im}(\mathcal{F} \rightarrow \mathcal{G}) = \ker(\mathcal{G} \rightarrow \mathcal{H})$.

1.3.4 Exactness on stalks

The following criterion makes stalks the tool of choice for testing exactness.

Theorem 1.3.1. *A sequence of sheaves $\mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H}$ on X is exact if and only if for every $x \in X$ the sequence of stalks*

$$\mathcal{F}_x \rightarrow \mathcal{G}_x \rightarrow \mathcal{H}_x$$

is exact in $\mathbf{Ab}$.

The proof uses the fact that taking stalks, being a filtered direct limit, is exact.

Theorem 1.3.2. *In $\mathbf{Ab}$ (more generally in $R\text{-}\mathbf{Mod}$), filtered direct limits are exact. Explicitly, if I is a filtered category and $A \rightarrow B \rightarrow C$ is a sequence of I -indexed systems in which each*

$$A(i) \rightarrow B(i) \rightarrow C(i)$$

is exact, then

$$\varinjlim_i A(i) \rightarrow \varinjlim_i B(i) \rightarrow \varinjlim_i C(i)$$

is exact.

Remark 1.3.3. 回想 I filtered 的含义: 任意两个对象被某个第三对象支配, 任意两条平行态射可以被“共尾等同”。这恰好是让“直接极限交换有限极限”成立的条件。因为 stalks $\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U)$ 是 filtered colimit, 所以“取 stalk”是精确函子——这就是为什么 [definition 1.3.1](#) 成立。

1.4 Abelian categories

Additive categories admit universal-property versions of $\ker$ and coker .

1.4.1 Kernels, cokernels, images in a category

Definition 1.4.1. Let $f : A \rightarrow B$ be a morphism in an additive category $\mathcal{C}$. A *kernel* of f is a morphism $g : K \rightarrow A$ with $f \circ g = 0$ satisfying the universal property: for every $h : D \rightarrow A$ with $f \circ h = 0$, there exists a unique $u : D \rightarrow K$ with $h = g \circ u$.

Dually, a *cokernel* of f is $g : B \rightarrow Q$ with $g \circ f = 0$ such that for every $h : B \rightarrow D$ with $h \circ f = 0$ there exists a unique $u : Q \rightarrow D$ with $h = u \circ g$.

Definition 1.4.2. When kernels and cokernels exist, the *image* of $f : A \rightarrow B$ is

$$\operatorname{Im} f := \ker(B \rightarrow \operatorname{coker} f).$$

Equivalently (in the setting of the next subsection), $\operatorname{Im} f = \operatorname{coker}(\ker f \rightarrow A)$.

1.4.2 Abelian categories

Definition 1.4.3. An additive category $\mathcal{C}$ is *abelian* (阿贝尔范畴) if

- (i) every morphism has a kernel and a cokernel;
- (ii) every monomorphism is the kernel of its cokernel;
- (iii) every epimorphism is the cokernel of its kernel.

Remark 1.4.4. 可以总结成一个口号:

additive + kernel, cokernel, finite sum exist + mono/epi normal = abelian.

条件 (ii)(iii) 是说 mono 和 epi “正常”——它们完全由其 coker 或 ker 决定。这保证了同构的第一定理 $\text{Im } f \cong A / \ker f$ 在抽象范畴中仍然成立。

Example 1.4.5.

- (1) **Ab** is abelian.
- (2) For a ring R , $R\text{-Mod}$ is abelian.
- (3) For a topological space X , both $\text{PSh}(X, \mathbf{Ab})$ and $\text{Sh}(X, \mathbf{Ab})$ are abelian.

1.4.3 Categorical = concrete in PSh and Sh

We now verify that the universal-property kernels/cokernels in $\text{PSh}(X)$ and $\text{Sh}(X)$ recover the sectionwise constructions (possibly followed by sheafification in the sheaf case).

Proposition 1.4.6. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism in $\text{PSh}(X)$ or $\text{Sh}(X)$.

- (1) In both $\text{PSh}(X)$ and $\text{Sh}(X)$, the categorical kernel is the sectionwise kernel $(\ker \varphi)(U) = \ker(\varphi_U)$.
- (2) In $\text{PSh}(X)$, the categorical cokernel is the sectionwise cokernel $(\text{coker } \varphi)(U) = \text{coker}(\varphi_U)$.
- (3) In $\text{Sh}(X)$, the categorical cokernel is the sheafification of the sectionwise cokernel:

$$\text{coker}_{\text{Sh}}(\varphi) = (\text{coker}_{\text{PSh}} \varphi)^+.$$

Proof. (1) *Kernel.* The sectionwise kernel $K(U) := \ker(\varphi_U)$ is a subpresheaf of $\mathcal{F}$, and $\varphi \circ (K \hookrightarrow \mathcal{F}) = 0$. If $f : \mathcal{D} \rightarrow \mathcal{F}$ satisfies $\varphi \circ f = 0$, then for every U , $\text{Im}(f_U) \subseteq \ker(\varphi_U) = K(U)$, so f_U factors uniquely through $K(U)$; compatibility with restriction

gives a unique factorization $\mathcal{D} \rightarrow K$. Moreover, if $\mathcal{F}, \mathcal{G}$ are sheaves, so is K , since $(\ker \varphi)(U) = \ker(\varphi_U)$ visibly respects local sections.

(2) *Cokernel in PSh.* Dual argument: $Q(U) := \operatorname{coker}(\varphi_U) = \mathcal{G}(U)/\operatorname{Im}(\varphi_U)$. Given $f : \mathcal{G} \rightarrow \mathcal{D}$ with $f \circ \varphi = 0$, each f_U factors through $Q(U)$; compatibility gives $Q \rightarrow \mathcal{D}$.

(3) *Cokernel in Sh.* The sectionwise cokernel is only a presheaf; sheafify. For a test sheaf $\mathcal{D}$,

$$\operatorname{Hom}_{\operatorname{Sh}}((\operatorname{coker}_{\operatorname{PSh}} \varphi)^+, \mathcal{D}) \cong \operatorname{Hom}_{\operatorname{PSh}}(\operatorname{coker}_{\operatorname{PSh}} \varphi, \mathcal{D}) \cong \{f \in \operatorname{Hom}_{\operatorname{PSh}}(\mathcal{G}, \mathcal{D}) : f \circ \varphi = 0\},$$

the first bijection by [definition 1.2.5](#), the second by (2). Since $\mathcal{D}$ is a sheaf, $\operatorname{Hom}_{\operatorname{PSh}}(\mathcal{G}, \mathcal{D}) = \operatorname{Hom}_{\operatorname{Sh}}(\mathcal{G}, \mathcal{D})$, so the composed bijection exhibits $(\operatorname{coker}_{\operatorname{PSh}} \varphi)^+$ as the categorical cokernel. $\square$

Example 1.4.7 (Forgetful $\operatorname{Sh}(X) \rightarrow \operatorname{PSh}(X)$ does not preserve cokernels). Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be an epimorphism in $\operatorname{Sh}(X)$, i.e. $\operatorname{coker}_{\operatorname{Sh}} \varphi = 0$. This means $(\operatorname{coker}_{\operatorname{PSh}} \varphi)^+ = 0$, or equivalently, every stalk of $\operatorname{coker}_{\operatorname{PSh}} \varphi$ vanishes. But $\operatorname{coker}_{\operatorname{PSh}} \varphi$ itself need not vanish: the map $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ can fail to be surjective, even though it is surjective on each stalk. A standard example is the exponential sequence on $\mathbb{C}^\times$:

$$0 \rightarrow 2\pi i\mathbb{Z} \rightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^\times \rightarrow 0,$$

where $\exp$ is surjective on stalks but not on global sections.

1.5 Pushforward and inverse image

Let $f : X \rightarrow Y$ be a continuous map of topological spaces. We associate to f two functors that will be fundamental throughout:

$$f_* : \operatorname{Sh}(X) \rightarrow \operatorname{Sh}(Y), \quad f^{-1} : \operatorname{Sh}(Y) \rightarrow \operatorname{Sh}(X).$$

1.5.1 Pushforward

Definition 1.5.1. For a sheaf $\mathcal{F}$ on X and $V \subseteq Y$ open, define

$$(f_*\mathcal{F})(V) := \mathcal{F}(f^{-1}(V)),$$

with restriction maps induced by $\mathcal{F}$'s.

Fact 1.5.2. $f_*\mathcal{F}$ is a sheaf on Y .

Proof. Given an open cover $V = \bigcup_i V_i \subseteq Y$, preimages give an open cover $f^{-1}(V) = \bigcup_i f^{-1}(V_i) \subseteq X$, with $f^{-1}(V_i) \cap f^{-1}(V_j) = f^{-1}(V_i \cap V_j)$. The sheaf axiom for $\mathcal{F}$ applied to this cover is exactly the sheaf axiom for $f_*\mathcal{F}$ applied to $\{V_i\}$. $\square$

1.5.2 Inverse image

Definition 1.5.3. For a sheaf $\mathcal{G}$ on Y , define the presheaf

$$\mathcal{P}(U) := \varinjlim_{V \supseteq f(U)} \mathcal{G}(V), \quad U \subseteq X \text{ open},$$

where the colimit runs over open $V \subseteq Y$ containing $f(U)$. Set

$$f^{-1}\mathcal{G} := \mathcal{P}^+,$$

the sheafification of $\mathcal{P}$.

Remark 1.5.4. 当 $f(U)$ 本身不一定是开集时，必须通过 $V \supseteq f(U)$ 的 colimit 来逼近——所以先写成 presheaf 再 sheafify。 f^{-1} 读作“ f 逆像”或“逆像层”；注意不要写成 f^* ——在概形理论里 f^* 通常指 $\mathcal{O}$ -模的拉回 $f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$ 。

1.5.3 Properties

Proposition 1.5.5. Let $f : X \rightarrow Y$ be continuous, $\mathcal{G}$ a sheaf on Y , and $x \in X$. Then

$$(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}.$$

Moreover, f_* and f^{-1} are functors, and

$$f^{-1} \dashv f_*.$$

Proof. Stalks. Sheafification preserves stalks (an isomorphism θ_x , by [definition 1.2.3](#)), so

$$(f^{-1}\mathcal{G})_x = \mathcal{P}_x = \varinjlim_{U \ni x} \mathcal{P}(U) = \varinjlim_{U \ni x} \varinjlim_{V \supseteq f(U)} \mathcal{G}(V) = \varinjlim_{V \ni f(x)} \mathcal{G}(V) = \mathcal{G}_{f(x)},$$

where the penultimate equality collapses the iterated colimit: the pairs (U, V) with $U \ni x$ and $V \supseteq f(U)$ are cofinal in the $V \ni f(x)$ system.

Adjunction. By the universal property of sheafification,

$$\mathrm{Hom}_{\mathrm{Sh}(X)}(f^{-1}\mathcal{G}, \mathcal{F}) \cong \mathrm{Hom}_{\mathrm{PSh}(X)}(\mathcal{P}, \mathcal{F}).$$

A morphism $\mathcal{P} \rightarrow \mathcal{F}$ amounts to, for every $U \subseteq X$, a map $\psi_U : \varinjlim_{V \supseteq f(U)} \mathcal{G}(V) \rightarrow \mathcal{F}(U)$ compatible with restrictions—equivalently, by the universal property of the colimit, a compatible family $\psi_{U,V} : \mathcal{G}(V) \rightarrow \mathcal{F}(U)$ for all pairs $U \subseteq f^{-1}(V)$. Such a family is determined by its values at $U = f^{-1}(V)$, i.e. by a compatible family $\phi_V : \mathcal{G}(V) \rightarrow \mathcal{F}(f^{-1}(V)) = (f_*\mathcal{F})(V)$. These are exactly morphisms $\mathcal{G} \rightarrow f_*\mathcal{F}$.

Functoriality follows from uniqueness clauses in the universal properties used throughout. $\square$

1.5.4 Applications: mono/epi in $\mathrm{Sh}(X)$

We can now complete the proof of [definition 1.1.6](#): mono iff sections are injective, epi iff stalks are surjective.

Proposition 1.5.6. *A morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ in $\mathrm{Sh}(X)$ is an epimorphism if and only if $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is surjective for every $x \in X$.*

Proof. φ is an epimorphism iff $\mathrm{coker}_{\mathrm{Sh}} \varphi = 0$, iff every stalk $(\mathrm{coker}_{\mathrm{Sh}} \varphi)_x = 0$ (by [definition 1.3.1](#)), iff $\mathrm{coker}(\varphi_x) = 0$ (since sheafification preserves stalks), iff φ_x is surjective. $\square$

Proposition 1.5.7. *A morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ in $\mathrm{Sh}(X)$ is a monomorphism if and only if $\varphi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective for every open $U \subseteq X$, if and only if $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is injective for every $x \in X$.*

Proof. Injectivity on sections is manifestly equivalent to $\ker \varphi = 0$ (sectionwise), which characterizes monomorphisms in $\mathrm{Sh}(X)$ by [definition 1.1.6\(1\)](#). For the stalk version: $\ker \varphi = 0$ iff every stalk $(\ker \varphi)_x = 0$, iff $\ker(\varphi_x) = 0$.

For the reverse implication "stalkwise injective $\Rightarrow$ sectionwise injective" directly: let $s \in \mathcal{F}(U)$ with $\varphi_U(s) = 0$. Then for every $x \in U$, $\varphi_x(s_x) = (\varphi_U(s))_x = 0$, so $s_x = 0$. By definition of zero germ, there is an open $U_x \ni x$ with $s|_{U_x} = 0$. The U_x cover U ; the sheaf axiom forces $s = 0$. $\square$

Remark 1.5.8. 上面的证明揭示了一个关键对称性: sheaf 范畴里 mono 和 epi 的“正常” (normal) 结构——mono 可以用 sectionwise 条件或 stalkwise 条件来检测, epi 只能用 stalkwise。前者因为 sheaf 公理保证“stalkwise 为零 $\Rightarrow$ 整体为零”; 后者失败则给出 [definition 1.4.7](#) 中的指数层序列等经典例子。

Chapter 2

scheme

2.1 affine scheme

In this section, a ring is a commutative ring with unity.

Let A be a ring.

Recall localization. Let $S \subseteq A$ be a multiplicative subset, i.e.

$$a, b \in S \implies ab \in S, \quad 1 \in S, \quad 0 \notin S.$$

Define

$$S^{-1}A := \left\{ \frac{a}{s} \mid a \in A, s \in S \right\}.$$

We declare

$$\frac{a}{s} = \frac{a_1}{s_1} \iff \exists t \in S \text{ such that } ts_1a = ts_1a_1.$$

For $f \in A$, define

$$A_f := S^{-1}A, \quad S = \{f^n \mid n \in \mathbb{Z}_{\geq 0}\}.$$

For a prime ideal $\mathfrak{p} \subseteq A$, define

$$A_{\mathfrak{p}} := S^{-1}A, \quad S = A \setminus \mathfrak{p}.$$

There is a one-to-one correspondence

$$\left\{ \begin{array}{l} \text{prime ideals } \mathfrak{p} \subseteq A \\ \text{such that } \mathfrak{p} \cap S = \emptyset \end{array} \right\} \longleftrightarrow \{\text{prime ideals of } S^{-1}A\}.$$

Let A be a ring.

Definition 2.1.1. Define

$$\operatorname{Spec} A := \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ is a prime ideal}\}.$$

For an ideal $I \subseteq A$, define

$$V(I) := \{\mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \supseteq I\}.$$

The closed subsets of $\operatorname{Spec} A$ are the subsets of the form $V(I)$, where $I \subseteq A$ is an ideal. The open subsets are the complements of closed subsets, namely

$$D(I) := \operatorname{Spec} A \setminus V(I).$$

Lemma 2.1.2. *$\operatorname{Spec} A$ is a topological space with respect to this collection of closed sets. This topology is called the Zariski topology.*

1. For ideals $I_1, I_2 \subseteq A$, one has

$$V(I_1, I_2) = V(I_1) \cup V(I_2), \quad D(I_1, I_2) = D(I_1) \cap D(I_2).$$

2. for a family of ideals $\{I_i\}_{i \in \Lambda}$,

$$V\left(\sum_{i \in \Lambda} I_i\right) = \bigcap_{i \in \Lambda} V(I_i), \quad D\left(\sum_{i \in \Lambda} I_i\right) = \bigcup_{i \in \Lambda} D(I_i).$$

3. trivially

$$V((0)) = \operatorname{Spec} A, \quad V(A) = \emptyset.$$

4. for ideals $I_1, I_2 \subseteq A$,

$$V(I_1) \subseteq V(I_2) \iff \sqrt{I_1} \supseteq \sqrt{I_2},$$

where

$$\sqrt{I} := \{a \in A \mid a^n \in I \text{ for some } n \geq 1\}.$$

Proof. For (i), since $I_1 I_2 \subseteq I_1$ and $I_1 I_2 \subseteq I_2$, we have

$$V(I_1 I_2) \supseteq V(I_1) \cup V(I_2).$$

Conversely, let $\mathfrak{p} \in V(I_1 I_2)$. If $\mathfrak{p} \notin V(I_1)$, then there exists

$$a \in I_1, \quad a \notin \mathfrak{p}.$$

For every $b \in I_2$, we have $ab \in I_1 I_2 \subseteq \mathfrak{p}$. Since $\mathfrak{p}$ is prime and $a \notin \mathfrak{p}$, it follows that $b \in \mathfrak{p}$. Hence $I_2 \subseteq \mathfrak{p}$, so $\mathfrak{p} \in V(I_2)$. Therefore

$$V(I_1 I_2) \subseteq V(I_1) \cup V(I_2),$$

and thus

$$V(I_1 I_2) = V(I_1) \cup V(I_2).$$

For (iv), first note that

$$V(I) = V(\sqrt{I})$$

for every ideal I , since a prime ideal $\mathfrak{p}$ contains I if and only if it contains $\sqrt{I}$. Hence if

$$\sqrt{I_1} \supseteq \sqrt{I_2},$$

then

$$V(I_1) = V(\sqrt{I_1}) \subseteq V(\sqrt{I_2}) = V(I_2).$$

Conversely, assume

$$V(I_1) \subseteq V(I_2).$$

To show

$$\sqrt{I_1} \supseteq \sqrt{I_2},$$

suppose not. Then there exists

$$f \in \sqrt{I_2}, \quad f \notin \sqrt{I_1}.$$

Consider the image of f in $A/\sqrt{I_1}$. Since $f \notin \sqrt{I_1}$, its image is nonzero, so

$$(A/\sqrt{I_1})_f \neq 0.$$

Hence there exists a prime ideal

$$\tilde{\mathfrak{p}} \in \text{Spec}((A/\sqrt{I_1})_f).$$

By the correspondence between prime ideals of a localization and prime ideals disjoint from the multiplicative set $\{1, f, f^2, \dots\}$, this gives a prime ideal $\mathfrak{p} \subseteq A$ such that

$$\mathfrak{p} \supseteq \sqrt{I_1} \quad \text{and} \quad f \notin \mathfrak{p}.$$

Thus

$$\mathfrak{p} \in V(I_1), \quad \mathfrak{p} \notin V(I_2),$$

because $f \in \sqrt{I_2} \subseteq \mathfrak{p}$ would be necessary for $\mathfrak{p} \in V(I_2)$. This contradicts

$$V(I_1) \subseteq V(I_2).$$

Therefore

$$\sqrt{I_1} \supseteq \sqrt{I_2}.$$

□

Lemma 2.1.3. *Let A be a ring. Then*

$$\bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p} = \sqrt{(0)},$$

where

$$\sqrt{(0)} = \{ a \in A \mid a^n = 0 \text{ for some } n \geq 1 \}$$

is the nilradical of A .

Consequently, if $f \in A$ is not nilpotent, then there exists a prime ideal

$$\mathfrak{p} \in \operatorname{Spec} A$$

such that $f \notin \mathfrak{p}$. Equivalently,

$$\operatorname{Spec}(A_f) \neq \emptyset.$$

Proof. First we prove

$$\bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p} = \sqrt{(0)}.$$

If $a \in \sqrt{(0)}$, then $a^n = 0$ for some $n \geq 1$. Hence for every prime ideal $\mathfrak{p} \subseteq A$, we have $a^n \in \mathfrak{p}$, so by primality,

$$a \in \mathfrak{p}.$$

Therefore

$$a \in \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

Thus

$$\sqrt{(0)} \subseteq \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

Conversely, let $a \notin \sqrt{(0)}$. Then the image of a in

$$A/\sqrt{(0)}$$

is not nilpotent, hence in particular is nonzero. Localizing at the powers of a , we get

$$(A/\sqrt{(0)})_a \neq 0.$$

Every nonzero ring has a maximal ideal, hence a prime ideal. So there exists

$$\tilde{\mathfrak{p}} \in \operatorname{Spec}((A/\sqrt{(0)})_a).$$

By the correspondence between prime ideals of a localization and prime ideals disjoint from the multiplicative set

$$\{1, a, a^2, \dots\},$$

this gives a prime ideal $\mathfrak{p} \subseteq A$ such that

$$\sqrt{(0)} \subseteq \mathfrak{p} \quad \text{and} \quad a \notin \mathfrak{p}.$$

Hence

$$a \notin \bigcap_{\mathfrak{p} \in \text{Spec } A} \mathfrak{p}.$$

This proves

$$\bigcap_{\mathfrak{p} \in \text{Spec } A} \mathfrak{p} \subseteq \sqrt{(0)}.$$

Therefore

$$\bigcap_{\mathfrak{p} \in \text{Spec } A} \mathfrak{p} = \sqrt{(0)}.$$

Now if $f \in A$ is not nilpotent, then

$$f \notin \sqrt{(0)} = \bigcap_{\mathfrak{p} \in \text{Spec } A} \mathfrak{p}.$$

So there exists a prime ideal $\mathfrak{p} \subseteq A$ with

$$f \notin \mathfrak{p}.$$

Again by the localization correspondence, this is equivalent to the existence of a prime ideal of A_f . Hence

$$\text{Spec}(A_f) \neq \emptyset.$$

□

Proposition 2.1.4. 1. For $f \in A$,

$$D(f) := D(fA) = D((f)).$$

Moreover, the family

$$\{D(f)\}_{f \in A}$$

is a basis for the topology on $\text{Spec } A$.

2. Closed points of $\text{Spec } A$ correspond to maximal ideals of A .

3. Generic points of $\text{Spec } A$ correspond to minimal prime ideals of A .

More precisely, $\eta \in \text{Spec } A$ is a generic point if there is no

$$y \in \text{Spec } A, \quad y \neq \eta,$$

such that

$$\overline{\{y\}} \supsetneq \eta.$$

4. $\text{Spec } A$ and each $D(f)$ are quasi-compact.

Proof. 1. Let $\mathfrak{p} \in \text{Spec } A$. Then

$$\mathfrak{p} \in D(I) \iff \mathfrak{p} \not\supseteq I.$$

Hence there exists $f \in I$ such that $f \notin \mathfrak{p}$, i.e.

$$\mathfrak{p} \in D(f).$$

Since $fA \subseteq I$, we have

$$D(f) = D(fA) \subseteq D(I).$$

Therefore every point of $D(I)$ lies in some basic open set $D(f) \subseteq D(I)$, so the family

$$\{D(f)\}_{f \in A}$$

is a basis for the topology on $\text{Spec } A$.

2. Let $\mathfrak{m} \subseteq A$ be a maximal ideal. Then

$$V(\mathfrak{m}) = \{\mathfrak{m}\},$$

since if $\mathfrak{p} \in V(\mathfrak{m})$, then $\mathfrak{p} \supseteq \mathfrak{m}$, and by maximality of $\mathfrak{m}$ we get $\mathfrak{p} = \mathfrak{m}$. Hence $\mathfrak{m}$ is a closed point.

Conversely, let $\mathfrak{p} \in \text{Spec } A$ be a closed point. Then

$$\{\mathfrak{p}\} = V(I)$$

for some ideal $I \subseteq A$. Since $\mathfrak{p} \in V(I)$, we have $I \subseteq \mathfrak{p}$, hence

$$V(\mathfrak{p}) \subseteq V(I) = \{\mathfrak{p}\}.$$

But always $\mathfrak{p} \in V(\mathfrak{p})$, so

$$V(\mathfrak{p}) = \{\mathfrak{p}\}.$$

If $\mathfrak{p}$ were not maximal, there would exist a prime ideal $\mathfrak{p}' \supsetneq \mathfrak{p}$, and then

$$\mathfrak{p}' \in V(\mathfrak{p}) \setminus \{\mathfrak{p}\},$$

a contradiction. Thus $\mathfrak{p}$ is maximal.

3. For $\mathfrak{p} \in \text{Spec } A$, one has

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

Indeed, the closed subsets containing $\mathfrak{p}$ are precisely the sets $V(I)$ with $I \subseteq \mathfrak{p}$. Therefore

$$\overline{\{\mathfrak{p}\}} = \bigcap_{I \subseteq \mathfrak{p}} V(I) = V\left(\sum_{I \subseteq \mathfrak{p}} I\right) = V(\mathfrak{p}).$$

Hence $\mathfrak{p}$ is a generic point of $V(\mathfrak{p})$. In particular, generic points correspond to minimal prime ideals.

4. Let

$$\operatorname{Spec} A = \bigcup_{i \in \Lambda} U_i$$

be an open cover. By (1), for each $x \in \operatorname{Spec} A$, there exists $f_x \in A$ such that

$$x \in D(f_x) \subseteq U_i$$

for some i . Thus

$$\operatorname{Spec} A = \bigcup_{x \in \operatorname{Spec} A} D(f_x).$$

Equivalently,

$$\operatorname{Spec} A = D\left(\sum_x f_x A\right).$$

Hence

$$V\left(\sum_x f_x A\right) = \emptyset,$$

so

$$\sqrt{\sum_x f_x A} = A.$$

Therefore

$$1_A \in \sum_x f_x A,$$

and so there exist $x_1, \dots, x_n$ such that

$$1_A \in \sum_{i=1}^n f_{x_i} A.$$

It follows that

$$\operatorname{Spec} A = \bigcup_{i=1}^n D(f_{x_i}),$$

and therefore

$$\operatorname{Spec} A = \bigcup_{j=1}^n U_{i_j}$$

for suitable $i_1, \dots, i_n$. Hence $\text{Spec } A$ is quasi-compact.

Since for any $f \in A$,

$$D(f) \cong \text{Spec}(A_f),$$

the same argument shows that each $D(f)$ is quasi-compact. □

Proposition 2.1.5. *For $f \in A$, the natural map*

$$\text{Spec}(A_f) \longrightarrow D(f)$$

is a homeomorphism.

Example 2.1.6. Let

$$R := \prod_{p \text{ prime}} \mathbb{F}_p.$$

Then

$$\text{Spec}(R) \neq \prod_{p \text{ prime}} \text{Spec}(\mathbb{F}_p).$$

Indeed, for each prime p_0 , define

$$\mathfrak{m}_{p_0} := \{(x_p) \in R \mid x_{p_0} = 0\}.$$

Then $\mathfrak{m}_{p_0}$ is a maximal ideal of R , and

$$R/\mathfrak{m}_{p_0} \cong \mathbb{F}_{p_0}.$$

However, R has prime ideals other than these. For example, let

$$I := \bigoplus_{p \text{ prime}} \mathbb{F}_p = \{(x_p) \in R \mid x_p = 0 \text{ for all but finitely many } p\}.$$

Then $I \subsetneq R$, so there exists a maximal ideal

$$\mathfrak{m} \subseteq R$$

such that

$$I \subseteq \mathfrak{m}.$$

Since every maximal ideal is prime, $\mathfrak{m} \in \text{Spec}(R)$.

We claim that

$$\mathfrak{m} \neq \mathfrak{m}_{p_0} \quad \text{for every prime } p_0.$$

Indeed, let

$$e_{p_0} := (\delta_{p,p_0})_p \in I,$$

where $\delta_{p,p_0} = 1$ if $p = p_0$ and 0 otherwise. Since $I \subseteq \mathfrak{m}$, we have

$$e_{p_0} \in \mathfrak{m}.$$

But

$$e_{p_0} \notin \mathfrak{m}_{p_0},$$

because its p_0 -th coordinate is 1. Hence $\mathfrak{m} \neq \mathfrak{m}_{p_0}$.

Therefore $\text{Spec}(R)$ contains points not coming from any factor $\text{Spec}(\mathbb{F}_p)$, so

$$\text{Spec}\left(\prod_p \mathbb{F}_p\right) \neq \coprod_p \text{Spec}(\mathbb{F}_p).$$

Example 2.1.7. One has

$$\text{Spec } \mathbb{Z} = \{(0)\} \cup \{(p) \mid p \text{ prime}\}.$$

Indeed, the prime ideals of $\mathbb{Z}$ are exactly (0) and the ideals (p) for prime numbers p .

The closed subsets of $\text{Spec } \mathbb{Z}$ are

$$\emptyset, \quad \text{Spec } \mathbb{Z}, \quad \{(p_1), \dots, (p_n)\},$$

where $p_1, \dots, p_n$ are prime numbers.

Example 2.1.8. Let k be a field. Then

$$\text{Spec}(k[t]) = \{(0)\} \cup \{(f(t)) \mid f(t) \in k[t] \text{ irreducible}\}.$$

Indeed, since $k[t]$ is a principal ideal domain, its nonzero prime ideals are exactly the ideals generated by irreducible polynomials.

The closed subsets of $\text{Spec}(k[t])$ are

$$\emptyset, \quad \text{Spec}(k[t]), \quad \{(f_1(t)), \dots, (f_n(t))\},$$

where $f_1(t), \dots, f_n(t) \in k[t]$ are irreducible polynomials.

Let $f : A \rightarrow B$ be a ring homomorphism. Then f induces a map

$$\varphi_f : \text{Spec } B \longrightarrow \text{Spec } A, \quad \mathfrak{p} \longmapsto f^{-1}(\mathfrak{p}).$$

This map is continuous, since for every ideal $I \subseteq A$,

$$\varphi_f^{-1}(V(I)) = \{\mathfrak{p} \in \text{Spec } B \mid f^{-1}(\mathfrak{p}) \supseteq I\} = \{\mathfrak{p} \in \text{Spec } B \mid \mathfrak{p} \supseteq f(I)\} = V(f(I)).$$

Example 2.1.9. Let $a \in A$. The localization map

$$A \longrightarrow A_a$$

induces a map

$$\mathrm{Spec}(A_a) \longrightarrow \mathrm{Spec} A, \quad \mathfrak{p} \longmapsto f^{-1}(\mathfrak{p}),$$

whose image is exactly

$$D(a) = \{\mathfrak{q} \in \mathrm{Spec} A \mid a \notin \mathfrak{q}\}.$$

Indeed, by the correspondence for localization, prime ideals of A_a are in bijection with prime ideals

$$\mathfrak{q} \subseteq A \quad \text{such that} \quad a \notin \mathfrak{q}.$$

Hence

$$\mathrm{Spec}(A_a) \xrightarrow{\sim} D(a)$$

is a homeomorphism.

More generally, if

$$\frac{t}{a^n} \in A_a,$$

then under the homeomorphism $\mathrm{Spec}(A_a) \cong D(a)$, one has

$$D\left(\frac{t}{a^n}\right) \longleftrightarrow D(a) \cap D(t) = D(at).$$

Indeed, for $\mathfrak{p} \in \mathrm{Spec}(A_a)$ corresponding to $\mathfrak{q} = f^{-1}(\mathfrak{p}) \in D(a)$,

$$\frac{t}{a^n} \notin \mathfrak{p} \iff t \notin \mathfrak{q},$$

since $a \notin \mathfrak{q}$. Therefore

$$D\left(\frac{t}{a^n}\right) \cong D(a) \cap D(t).$$

Example 2.1.10. Let $I \subseteq A$ be an ideal. The quotient map

$$A \longrightarrow A/I$$

induces a map

$$\mathrm{Spec}(A/I) \longrightarrow \mathrm{Spec} A.$$

By the correspondence for quotient rings, prime ideals of A/I are in bijection with prime ideals

$$\mathfrak{q} \subseteq A \quad \text{such that} \quad \mathfrak{q} \supseteq I.$$

Hence the image is exactly

$$V(I) = \{\mathfrak{q} \in \operatorname{Spec} A \mid \mathfrak{q} \supseteq I\},$$

and

$$\operatorname{Spec}(A/I) \xrightarrow{\sim} V(I)$$

is a homeomorphism.

Moreover, for any ideal $\tilde{I} \subseteq A/I$ corresponding to an ideal $J \subseteq A$ with $J \supseteq I$, one has

$$V(\tilde{I}) \subseteq \operatorname{Spec}(A/I) \iff V(J) \subseteq V(I),$$

and in particular

$$V(\tilde{I}) = V(J/I) \text{ corresponds to } V(J) \cap V(I) = V(J).$$

Equivalently, if $J \supseteq I$, then

$$V_{A/I}(J/I) \cong V_A(J).$$

Example 2.1.11. There is a natural homeomorphism

$$\operatorname{Spec}(A \times B) \cong \operatorname{Spec} A \sqcup \operatorname{Spec} B.$$

Indeed, let

$$x := (1, 0), \quad y := (0, 1) \in A \times B.$$

Then

$$xy = 0, \quad D(x) \cup D(y) = \operatorname{Spec}(A \times B), \quad D(x) \cap D(y) = D(xy) = \emptyset.$$

Moreover,

$$(A \times B)_x \cong A, \quad (A \times B)_y \cong B.$$

Hence

$$D(x) \cong \operatorname{Spec} A, \quad D(y) \cong \operatorname{Spec} B,$$

and therefore

$$\operatorname{Spec}(A \times B) \cong \operatorname{Spec} A \sqcup \operatorname{Spec} B.$$

Example 2.1.12. The closed points of

$$\operatorname{Spec} \mathbb{C}[x_1, \dots, x_n]$$

are in bijection with $\mathbb{C}^n$.

Indeed, by Hilbert's Nullstellensatz, the maximal ideals of $\mathbb{C}[x_1, \dots, x_n]$ are exactly the ideals of the form

$$(x_1 - a_1, \dots, x_n - a_n), \quad (a_1, \dots, a_n) \in \mathbb{C}^n.$$

Since closed points of $\text{Spec } \mathbb{C}[x_1, \dots, x_n]$ correspond to maximal ideals, we obtain a bijection

$$\{\text{closed points of } \text{Spec } \mathbb{C}[x_1, \dots, x_n]\} \longleftrightarrow \mathbb{C}^n.$$

More generally, let

$$I = (f_1, \dots, f_m) \subseteq \mathbb{C}[x_1, \dots, x_n].$$

Then the closed points of

$$\text{Spec}(\mathbb{C}[x_1, \dots, x_n]/I)$$

are in bijection with the set

$$\{(a_1, \dots, a_n) \in \mathbb{C}^n \mid f_i(a_1, \dots, a_n) = 0, 1 \leq i \leq m\}.$$

Equivalently, the closed points correspond to maximal ideals

$$(x_1 - a_1, \dots, x_n - a_n) \supseteq I,$$

that is,

$$f_i(a_1, \dots, a_n) = 0 \quad \text{for all } 1 \leq i \leq m.$$

2.1.1 Structure sheaf

Let $X := \text{Spec } A$. For an open subset $U \subseteq X$, define

$$\mathcal{O}_X(U) = \left\{ s : U \rightarrow \prod_{\mathfrak{q} \in U} A_{\mathfrak{q}} \mid \begin{array}{l} s(\mathfrak{q}) \in A_{\mathfrak{q}} \text{ for all } \mathfrak{q} \in U, \\ \text{and } s \text{ is locally a fraction} \end{array} \right\}.$$

Here “locally a fraction” means that for every $\mathfrak{p} \in U$, there exist

$$\mathfrak{p} \in D(t) \subseteq U, \quad a \in A_t,$$

such that for every $\mathfrak{q} \in D(t)$,

$$s(\mathfrak{q}) = a|_{A_{\mathfrak{q}}}.$$

Equivalently, if $a = \frac{b}{t^n} \in A_t$, then for every $\mathfrak{q} \in D(t)$,

$$s(\mathfrak{q}) = \frac{b}{t^n} \in A_{\mathfrak{q}}.$$

Proposition 2.1.13. *For every $\mathfrak{p} \in X$, there is a natural isomorphism*

$$\theta : \varinjlim_{\mathfrak{p} \in U} \mathcal{O}_X(U) \longrightarrow A_{\mathfrak{p}}, \quad [s, U] \longmapsto s(\mathfrak{p}).$$

In other words,

$$(\mathcal{O}_X)_{\mathfrak{p}} \cong A_{\mathfrak{p}}.$$

Proof. The map θ is well-defined, since if two representatives agree on some neighborhood of $\mathfrak{p}$, then they have the same value in $A_{\mathfrak{p}}$.

To prove surjectivity, let

$$\frac{a}{t} \in A_{\mathfrak{p}}, \quad t \notin \mathfrak{p}.$$

Then $\mathfrak{p} \in D(t)$, and we define a section

$$s \in \mathcal{O}_X(D(t))$$

by

$$s(\mathfrak{q}) = \frac{a}{t} \in A_{\mathfrak{q}}, \quad \mathfrak{q} \in D(t).$$

This is well-defined because $t \notin \mathfrak{q}$ for every $\mathfrak{q} \in D(t)$. Hence

$$\theta([s, D(t)]) = \frac{a}{t},$$

so θ is surjective.

To prove injectivity, let $[s, U] \in \varinjlim_{\mathfrak{p} \in U} \mathcal{O}_X(U)$ satisfy

$$\theta([s, U]) = s(\mathfrak{p}) = 0 \in A_{\mathfrak{p}}.$$

Choose a basic open neighborhood

$$\mathfrak{p} \in D(t) \subseteq U$$

such that

$$s|_{D(t)} = \frac{a}{t^n}$$

for some $a \in A$. Since

$$\frac{a}{t^n} = 0 \in A_{\mathfrak{p}},$$

there exists $r \in A \setminus \mathfrak{p}$ such that

$$ra = 0.$$

Hence on the smaller neighborhood

$$\mathfrak{p} \in D(tr) \subseteq D(t),$$

we have

$$\frac{a}{t^n} = 0$$

in every $A_{\mathfrak{q}}$ with $\mathfrak{q} \in D(tr)$, because $r \notin \mathfrak{q}$. Therefore

$$s|_{D(tr)} = 0,$$

so $[s, U] = 0$ in the direct limit. Thus θ is injective.

Therefore θ is an isomorphism. □

Remark 2.1.14. For a basic open set $D(t) \subseteq \operatorname{Spec} A$, one has

$$\mathcal{O}_X(D(t)) \cong A_t.$$

Under this identification, an element $\frac{a}{t^n} \in A_t$ corresponds to the section

$$\mathfrak{q} \longmapsto \frac{a}{t^n} \in A_{\mathfrak{q}}, \quad \mathfrak{q} \in D(t).$$

2.1.2 The structure sheaf on $\operatorname{Spec} A$

Let A be a ring, and let $X = \operatorname{Spec} A$. Denote by $\mathcal{O}_X$ the structure sheaf on X .

Proposition 2.1.15. *The structure sheaf $\mathcal{O}_X$ satisfies the following properties.*

1. $\mathcal{O}_X$ is a sheaf on $\operatorname{Spec} A$.
2. For every prime ideal $\mathfrak{p} \in \operatorname{Spec} A$, there is a canonical isomorphism

$$(\mathcal{O}_X)_{\mathfrak{p}} \cong A_{\mathfrak{p}}.$$

3. For every $f \in A$, there is a canonical isomorphism

$$\mathcal{O}_X(D(f)) \cong A_f.$$

4. As a special case of (3) with $f = 1$, one has

$$\mathcal{O}_X(\operatorname{Spec} A) \cong A.$$

Remark 2.1.16. Recall the following standard fact about principal open sets:

$$D(t) \subseteq D(s) \iff \sqrt{(t)} \subseteq \sqrt{(s)}.$$

Equivalently,

$$D(t) \subseteq D(s) \iff \exists n \geq 1, \exists a \in A \text{ such that } t^n = sa.$$

In that case one gets a canonical map

$$A_s \longrightarrow A_{sa} = A_{t^n} \cong A_t.$$

Also, for any $s, a \in A$, one has

$$(A_s)_a \cong A_{sa}.$$

Proof of (3). We construct a canonical homomorphism

$$\psi: A_t \longrightarrow \mathcal{O}_X(D(t))$$

by sending

$$\frac{a}{t^n} \longmapsto s_{a/t^n},$$

where s_{a/t^n} is the section defined by

$$s_{a/t^n}(\mathfrak{p}) = \frac{a}{t^n} \Big|_{A_{\mathfrak{p}}} \in A_{\mathfrak{p}}, \quad \mathfrak{p} \in D(t).$$

This is well-defined because $t \notin \mathfrak{p}$ for every $\mathfrak{p} \in D(t)$, so there is a canonical localization map

$$A_t \longrightarrow A_{\mathfrak{p}}.$$

We prove that ψ is bijective.

Step 1: ψ is injective.

Suppose

$$\psi\left(\frac{a}{t^n}\right) = 0.$$

Then for every $\mathfrak{p} \in D(t)$,

$$\frac{a}{t^n} = 0 \quad \text{in } A_{\mathfrak{p}}.$$

By the criterion for equality in a localization, this means that for each $\mathfrak{p} \in D(t)$ there exists

$$h_{\mathfrak{p}} \in A \setminus \mathfrak{p}$$

such that

$$h_{\mathfrak{p}} a = 0.$$

Let

$$I := \text{Ann}(a) = \{ b \in A \mid ba = 0 \}.$$

Then for every $\mathfrak{p} \in D(t)$ we have $I \not\subseteq \mathfrak{p}$, because $h_{\mathfrak{p}} \in I \setminus \mathfrak{p}$. Hence

$$\mathfrak{p} \in D(I) \quad \text{for all } \mathfrak{p} \in D(t),$$

so

$$D(t) \subseteq D(I).$$

Passing to complements gives

$$V(I) \subseteq V(t),$$

therefore

$$t \in \sqrt{I}.$$

Hence there exists $\ell \geq 1$ such that

$$t^\ell \in I,$$

i.e.

$$t^\ell a = 0.$$

It follows that

$$\frac{a}{t^n} = 0 \quad \text{in } A_t.$$

Thus ψ is injective.

Step 2: ψ is surjective.

Take any section

$$s \in \mathcal{O}_X(D(t)).$$

By definition of the structure sheaf, s is locally represented by fractions. Thus for every $\mathfrak{p} \in D(t)$ there exists a principal open neighborhood

$$\mathfrak{p} \in D(t_i) \subseteq D(t)$$

and an element

$$\frac{a_i}{t_i^{n_i}} \in A_{t_i}$$

such that

$$s|_{D(t_i)} = \psi\left(\frac{a_i}{t_i^{n_i}}\right).$$

Since $D(t)$ is quasi-compact, we may choose finitely many such opens:

$$D(t) = \bigcup_{i=1}^r D(t_i).$$

On overlaps

$$D_{ij} := D(t_i) \cap D(t_j) = D(t_i t_j),$$

the two local expressions agree:

$$\psi\left(\frac{a_i}{t_i^{n_i}}\right)\Big|_{D_{ij}} = \psi\left(\frac{a_j}{t_j^{n_j}}\right)\Big|_{D_{ij}}.$$

Since the analogous map

$$A_{t_i t_j} \longrightarrow \mathcal{O}_X(D_{ij})$$

is injective by the same argument as in Step 1, this implies

$$\frac{a_i}{t_i^{n_i}} = \frac{a_j}{t_j^{n_j}} \quad \text{in } A_{t_i t_j}.$$

Again by the criterion for equality in a localization, for each pair (i, j) there exists $N_{ij} \geq 0$ such that

$$(t_i t_j)^{N_{ij}} (t_j^{n_j} a_i - t_i^{n_i} a_j) = 0 \quad \text{in } A.$$

Since there are only finitely many pairs, choose N large enough so that for all i, j ,

$$(t_i t_j)^N (t_j^{n_j} a_i - t_i^{n_i} a_j) = 0.$$

Replacing each fraction by the equal fraction

$$\frac{a_i}{t_i^{n_i}} = \frac{a_i t_i^N}{t_i^{n_i + N}},$$

we may assume from the start that

$$t_j^{n_j} a_i = t_i^{n_i} a_j \quad \text{for all } i, j.$$

Now from

$$D(t) = \bigcup_{i=1}^r D(t_i)$$

we get

$$V(t) \supseteq \bigcap_{i=1}^r V(t_i),$$

hence

$$\sqrt{(t)} \subseteq \sqrt{(t_1, \dots, t_r)}.$$

Therefore

$$t \in \sqrt{(t_1, \dots, t_r)}.$$

So there exists $m \geq 1$ such that

$$t^m \in (t_1, \dots, t_r).$$

After replacing the t_i by suitable powers if necessary, we may write

$$t^m = \sum_{i=1}^r b_i t_i^{n_i} \quad \text{for some } b_i \in A.$$

Define

$$a := \sum_{i=1}^r b_i a_i \in A.$$

We claim that

$$\psi\left(\frac{a}{t^m}\right) = s.$$

Indeed, fix i . In A_{t_i} one has

$$a = \sum_{j=1}^r b_j a_j = \sum_{j=1}^r b_j \frac{t_j^{n_j}}{t_i^{n_i}} a_i = \frac{\sum_{j=1}^r b_j t_j^{n_j}}{t_i^{n_i}} a_i = \frac{t^m}{t_i^{n_i}} a_i.$$

Hence in A_{t_i} ,

$$\frac{a}{t^m} = \frac{a_i}{t_i^{n_i}}.$$

Therefore

$$\psi\left(\frac{a}{t^m}\right) \Big|_{D(t_i)} = \psi\left(\frac{a_i}{t_i^{n_i}}\right) \Big|_{D(t_i)} = s|_{D(t_i)}.$$

Since the opens $D(t_i)$ cover $D(t)$, we conclude

$$\psi\left(\frac{a}{t^m}\right) = s.$$

So ψ is surjective.

Combining injectivity and surjectivity, ψ is an isomorphism:

$$\mathcal{O}_X(D(t)) \cong A_t.$$

This proves (3). □

Corollary 2.1.17. *Taking $t = 1$, we obtain*

$$\mathcal{O}_X(\text{Spec } A) = \mathcal{O}_X(D(1)) \cong A_1 \cong A.$$

Remark 2.1.18. The statement

$$(\mathcal{O}_X)_{\mathfrak{p}} \cong A_{\mathfrak{p}}$$

means that the stalk of the structure sheaf at $\mathfrak{p}$ is the local ring $A_{\mathfrak{p}}$.

Example 2.1.19. Using the identifications

$$\mathcal{O}_X(D(t)) \cong A_t, \quad \mathcal{O}_X(D(s)) \cong A_s, \quad \mathcal{O}_X(D(t_i)) \cong A_{t_i},$$

the sheaf axiom on a principal open set

$$D(t) = \bigcup_i D(t_i)$$

gives an exact sequence

$$0 \longrightarrow \mathcal{O}_X(D(t)) \longrightarrow \prod_i \mathcal{O}_X(D(t_i)) \longrightarrow \prod_{i,j} \mathcal{O}_X(D(t_i t_j)).$$

Under the above identifications, this becomes

$$0 \longrightarrow A_t \longrightarrow \prod_i A_{t_i} \longrightarrow \prod_{i,j} A_{t_i t_j},$$

where the second map is the usual difference of restrictions.

Example 2.1.20. Let $X = \operatorname{Spec} A$. Then the canonical map

$$A = \mathcal{O}_X(X) \longrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} A_{\mathfrak{p}}$$

is injective.

Indeed, this is the natural map from global sections to the product of all stalks:

$$\mathcal{O}_X(X) \longrightarrow \prod_{\mathfrak{p} \in X} (\mathcal{O}_X)_{\mathfrak{p}}.$$

For any sheaf, a section is determined by its germs. Hence if a global section maps to zero in every stalk, then it is the zero section. Therefore the map is injective.

2.1.3 Ringed Spaces and Locally Ringed Spaces

Goal.

$$\text{代数对象} \iff \text{几何对象}$$

and one should think of elements such as

$$e \in C^0([0, 1], \mathbb{R})$$

as functions on a geometric object.

Definition 2.1.21. A *ringed space* is a pair

$$(X, \mathcal{O}_X),$$

where X is a topological space and $\mathcal{O}_X$ is a sheaf of rings on X .

Definition 2.1.22. A *locally ringed space* is a ringed space $(X, \mathcal{O}_X)$ such that for every $x \in X$, the stalk

$$\mathcal{O}_{X,x}$$

is a local ring.

Notation 2.1.23. For a locally ringed space $(X, \mathcal{O}_X)$ and a point $x \in X$, denote by

$$\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$$

the maximal ideal. The quotient field

$$k(x) := \mathcal{O}_{X,x} / \mathfrak{m}_x$$

is called the residue field.

Definition 2.1.24. A morphism

$$(f, f^\#): (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

of locally ringed spaces consists of:

1. a continuous map

$$f: X \longrightarrow Y,$$

2. a morphism of sheaves of rings

$$f^\#: \mathcal{O}_Y \longrightarrow f_* \mathcal{O}_X,$$

such that for every $x \in X$, if $y = f(x)$, the induced map on stalks

$$f_x^\#: \mathcal{O}_{Y,y} \longrightarrow \mathcal{O}_{X,x}$$

is a local homomorphism, i.e.

$$(f_x^\#)^{-1}(\mathfrak{m}_x) = \mathfrak{m}_y.$$

Remark 2.1.25. For any sheaf $\mathcal{F}$ on X and any continuous map $f: X \rightarrow Y$, one has

$$(f_* \mathcal{F})_{f(x)} \cong \varinjlim_{f(x) \in V} \mathcal{F}(f^{-1}(V)).$$

Since the neighborhoods of x form a cofinal system among the sets $f^{-1}(V)$ with $f(x) \in V$, it follows that

$$(f_* \mathcal{F})_{f(x)} \cong \mathcal{F}_x.$$

Remark 2.1.26. Terminology is not completely uniform in the literature. Some books use “ringed space” for what others call “locally ringed space”.

Proposition 2.1.27. For any ring A , the pair

$$(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$$

is a ringed space.

Proposition 2.1.28. *Any ring homomorphism*

$$\varphi: A \rightarrow B$$

induces a morphism of ringed spaces

$$(f, f^\#): (\operatorname{Spec} B, \mathcal{O}_{\operatorname{Spec} B}) \longrightarrow (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A}).$$

Moreover, on global sections, the induced map

$$f^\#_{\operatorname{Spec} A}: \mathcal{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) \longrightarrow \mathcal{O}_{\operatorname{Spec} B}(\operatorname{Spec} B)$$

identifies with

$$\varphi: A \rightarrow B$$

under the canonical identifications

$$\mathcal{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) \cong A, \quad \mathcal{O}_{\operatorname{Spec} B}(\operatorname{Spec} B) \cong B.$$

Remark 2.1.29. In this sense, morphisms

$$(\operatorname{Spec} B, \mathcal{O}_{\operatorname{Spec} B}) \rightarrow (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$$

come from ring homomorphisms

$$A \rightarrow B.$$

Lemma 2.1.30. *For any section $s \in \mathcal{O}_X(X)$, the subset*

$$X_s := \{x \in X \mid r_x(s) \notin \mathfrak{m}_x\}$$

is open in X .

Lemma 2.1.31. *For each $t \in A$, there exists a unique ring homomorphism*

$$\varphi_t: A_t \longrightarrow \mathcal{O}_X(X_{\varphi(t)})$$

such that

$$\varphi_t\left(\frac{a}{t^n}\right) = \varphi(a) \cdot \varphi(t)^{-n}.$$

Theorem 2.1.32. *Let $(X, \mathcal{O}_X)$ be a locally ringed space, and let A be a ring. Then there is a natural bijection*

$$\operatorname{Hom}_{\operatorname{Ringed Spaces}}((X, \mathcal{O}_X), (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})) \xrightarrow{\sim} \operatorname{Hom}_{\operatorname{Ring}}(A, \mathcal{O}_X(X)),$$

given by

$$(f, f^\#) \longmapsto f^\#_{\operatorname{Spec} A}.$$

Equivalently, the functor

$$F: \mathbf{RingedSpaces} \rightarrow (\mathbf{Ring})^{\text{op}}, \quad (X, \mathcal{O}_X) \mapsto \mathcal{O}_X(X)$$

has left adjoint

$$G: (\mathbf{Ring})^{\text{op}} \rightarrow \mathbf{RingedSpaces}, \quad A \mapsto (\text{Spec } A, \mathcal{O}_{\text{Spec } A}).$$

Proof. We prove surjectivity and injectivity separately.

Surjectivity. Start with a ring homomorphism

$$\varphi: A \longrightarrow \mathcal{O}_X(X).$$

We construct a morphism of ringed spaces

$$(f, f^\#): (X, \mathcal{O}_X) \longrightarrow (\text{Spec } A, \mathcal{O}_{\text{Spec } A})$$

such that

$$f^\#_{\text{Spec } A} = \varphi.$$

Step 1: define the underlying map $f: X \rightarrow \text{Spec } A$.

For each $x \in X$, let

$$r_x: \mathcal{O}_X(X) \longrightarrow \mathcal{O}_{X,x}$$

be the canonical map to the stalk, and let $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ be the maximal ideal. Define

$$f(x) := (r_x \circ \varphi)^{-1}(\mathfrak{m}_x).$$

Since $r_x \circ \varphi: A \rightarrow \mathcal{O}_{X,x}$ is a ring homomorphism into a local ring, the inverse image of $\mathfrak{m}_x$ is a prime ideal of A . Hence

$$f(x) \in \text{Spec } A,$$

so f is well-defined.

Step 2: prove that f is continuous.

For $s \in \mathcal{O}_X(X)$, define

$$X_s := \{x \in X \mid r_x(s) \notin \mathfrak{m}_x\}.$$

Lemma 2.1.33. *For every $s \in \mathcal{O}_X(X)$, the subset X_s is open in X .*

Proof. Let $x \in X_s$. Then $r_x(s) \notin \mathfrak{m}_x$, so $r_x(s)$ is a unit in the local ring $\mathcal{O}_{X,x}$. Hence there exists $\sigma \in \mathcal{O}_{X,x}$ such that

$$\sigma r_x(s) = 1.$$

Choose an open neighborhood $U \ni x$ and a section $u \in \mathcal{O}_X(U)$ whose germ at x is σ . After shrinking U , we may assume

$$u \cdot s|_U = 1 \quad \text{in } \mathcal{O}_X(U).$$

Then for every $y \in U$,

$$u_y s_y = 1 \quad \text{in } \mathcal{O}_{X,y},$$

so $s_y \notin \mathfrak{m}_y$. Thus $U \subseteq X_s$, proving that X_s is open. $\square$

Now let $t \in A$. Then

$$x \in f^{-1}(D(t)) \iff t \notin f(x) \iff (r_x \circ \varphi)(t) \notin \mathfrak{m}_x \iff r_x(\varphi(t)) \notin \mathfrak{m}_x.$$

Therefore

$$f^{-1}(D(t)) = X_{\varphi(t)},$$

which is open by the lemma. Since the principal opens form a basis of $\text{Spec } A$, f is continuous.

Step 3: localize φ on principal opens.

Fix $t \in A$. Since

$$f^{-1}(D(t)) = X_{\varphi(t)},$$

the section $\varphi(t) \in \mathcal{O}_X(X)$ becomes invertible on $X_{\varphi(t)}$. Hence, by the universal property of localization, there exists a unique ring homomorphism

$$\varphi_t: A_t \longrightarrow \mathcal{O}_X(X_{\varphi(t)})$$

such that the diagram

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & \mathcal{O}_X(X) \\ \downarrow & & \downarrow \\ A_t & \xrightarrow{\varphi_t} & \mathcal{O}_X(X_{\varphi(t)}) \end{array}$$

commutes.

Equivalently,

$$\varphi_t\left(\frac{a}{t^n}\right) = \varphi(a) \cdot \varphi(t)^{-n} \quad \text{on } X_{\varphi(t)}.$$

Step 4: define $f_V^\#$ for an arbitrary open $V \subseteq \text{Spec } A$.

Let $V \subseteq \operatorname{Spec} A$ be open. We want to construct

$$f_V^\# : \mathcal{O}_{\operatorname{Spec} A}(V) \longrightarrow \mathcal{O}_X(f^{-1}(V)).$$

Fix $x \in f^{-1}(V)$ and set

$$p = f(x) \in V.$$

Choose a principal open neighborhood

$$p \in D(t) \subseteq V.$$

Then there is a commutative diagram

$$\begin{array}{ccc}
 A & \xrightarrow{\varphi} & \mathcal{O}_X(X) \\
 \downarrow & & \downarrow \\
 \mathcal{O}_{\operatorname{Spec} A}(V) & \dashrightarrow & \mathcal{O}_X(f^{-1}(V)) \\
 \downarrow & & \downarrow \\
 A_t & \xrightarrow{\varphi_t} & \mathcal{O}_X(X_{\varphi(t)}) \\
 \downarrow & & \downarrow \\
 A_p & \dashrightarrow^{\varphi_x} & \mathcal{O}_{X,x}
 \end{array}$$

where the vertical maps are the canonical restriction/localization maps.

Since

$$p \in D(t) \subseteq V,$$

the canonical map

$$\mathcal{O}_{\operatorname{Spec} A}(V) \longrightarrow A_p$$

is the germ map at p , and similarly

$$\mathcal{O}_X(f^{-1}(V)) \longrightarrow \mathcal{O}_{X,x}$$

is the germ map at x .

The map

$$\varphi_x : A_p \longrightarrow \mathcal{O}_{X,x}$$

is uniquely determined from φ_t by localization, because

$$A_p \cong (A_t)_p.$$

Thus, for each $x \in f^{-1}(V)$, we obtain a prescribed map on stalks

$$(f_V^\#)_x: \mathcal{O}_{\text{Spec } A, p} \longrightarrow \mathcal{O}_{X, x}.$$

Step 5: existence and uniqueness of $f_V^\#$.

We claim that there exists a unique homomorphism

$$f_V^\#: \mathcal{O}_{\text{Spec } A}(V) \longrightarrow \mathcal{O}_X(f^{-1}(V))$$

whose induced map on the stalk at each $x \in f^{-1}(V)$ is the above map

$$\varphi_x: A_{f(x)} \rightarrow \mathcal{O}_{X, x}.$$

Indeed, sections of a sheaf are determined by their germs. So uniqueness is immediate once the stalk maps are fixed.

For existence, choose a principal open cover

$$V = \bigcup_i D(t_i).$$

On each $D(t_i)$ we already have

$$\varphi_{t_i}: A_{t_i} \rightarrow \mathcal{O}_X(f^{-1}(D(t_i))).$$

These are compatible on overlaps, because on

$$D(t_i) \cap D(t_j) = D(t_i t_j)$$

both induced maps localize further to the same map

$$\varphi_{t_i t_j}: A_{t_i t_j} \rightarrow \mathcal{O}_X(f^{-1}(D(t_i t_j))).$$

Hence they glue to a unique map

$$f_V^\#: \mathcal{O}_{\text{Spec } A}(V) \rightarrow \mathcal{O}_X(f^{-1}(V)).$$

Equivalently, using the sheaf equalizer diagrams, we have the large commutative diagram

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathcal{O}_{\text{Spec } A}(V) & \longrightarrow & \prod_i A_{t_i} & \rightrightarrows & \prod_{i,j} A_{t_i t_j} \\ & & \downarrow f_V^\# & & \downarrow \prod_i \varphi_{t_i} & & \downarrow \prod_{i,j} \varphi_{t_i t_j} \\ 0 & \rightarrow & \mathcal{O}_X(f^{-1}(V)) & \rightarrow & \prod_i \mathcal{O}_X(f^{-1}(D(t_i))) & \rightrightarrows & \prod_{i,j} \mathcal{O}_X(f^{-1}(D(t_i t_j))) \end{array}$$

and $f_V^\#$ is the unique map induced by the universal property of the equalizer.

Step 6: the maps $f_V^\#$ define a morphism of sheaves.

If $V_1 \subseteq V_2 \subseteq \operatorname{Spec} A$, then both

$$f_{V_1}^\# \quad \text{and} \quad f_{V_2}^\#|_{V_1}$$

induce the same maps on all stalks, hence they are equal. Therefore the family

$$\{f_V^\#\}_{V \subseteq \operatorname{Spec} A}$$

is compatible with restriction and defines a morphism of sheaves of rings

$$f^\# : \mathcal{O}_{\operatorname{Spec} A} \longrightarrow f_* \mathcal{O}_X.$$

Step 7: check the local condition on stalks.

For each $x \in X$, write

$$p = f(x).$$

Then the induced map on stalks is

$$f_x^\# : \mathcal{O}_{\operatorname{Spec} A, p} \cong A_p \longrightarrow \mathcal{O}_{X, x},$$

and by construction this is precisely φ_x .

Moreover, since

$$p = (r_x \circ \varphi)^{-1}(\mathfrak{m}_x),$$

we have

$$(f_x^\#)^{-1}(\mathfrak{m}_x) = pA_p,$$

which is the maximal ideal of the local ring A_p . Thus

$$f_x^\#$$

is a local homomorphism. Hence

$$(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$$

is a morphism of ringed spaces.

Step 8: recover φ .

By construction,

$$f_{\operatorname{Spec} A}^\# : \mathcal{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) \longrightarrow \mathcal{O}_X(X)$$

is exactly the original homomorphism φ , because

$$\operatorname{Spec} A = D(1), \quad \mathcal{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) = \mathcal{O}_{\operatorname{Spec} A}(D(1)) \cong A_1 = A.$$

So every ring homomorphism

$$\varphi: A \rightarrow \mathcal{O}_X(X)$$

comes from a morphism

$$(f, f^\#): (X, \mathcal{O}_X) \rightarrow (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A}).$$

Injectivity.

Suppose

$$(f_1, f_1^\#), (f_2, f_2^\#): (X, \mathcal{O}_X) \rightarrow (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$$

have the same image in

$$\operatorname{Hom}_{\operatorname{Ring}}(A, \mathcal{O}_X(X)),$$

that is,

$$f_{1, \operatorname{Spec} A}^\# = f_{2, \operatorname{Spec} A}^\# =: \varphi: A \rightarrow \mathcal{O}_X(X).$$

Fix $x \in X$. For $i = 1, 2$, let

$$p_i := f_i(x) \in \operatorname{Spec} A.$$

Then the induced map on stalks gives a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & \mathcal{O}_X(X) \\ \downarrow & & \downarrow r_x \\ A_{p_i} & \xrightarrow{f_{i,x}^\#} & \mathcal{O}_{X,x} \end{array}$$

where $f_{i,x}^\#$ is local. Hence

$$(f_{i,x}^\#)^{-1}(\mathfrak{m}_x) = p_i A_{p_i}.$$

Pulling back along

$$A \rightarrow A_{p_i},$$

we get

$$(r_x \circ \varphi)^{-1}(\mathfrak{m}_x) = p_i.$$

Therefore

$$p_1 = p_2,$$

so

$$f_1(x) = f_2(x) \quad \text{for all } x \in X.$$

Hence

$$f_1 = f_2 =: f.$$

Now for each $x \in X$, both maps

$$f_{1,x}^\#, f_{2,x}^\#: A_{f(x)} \rightarrow \mathcal{O}_{X,x}$$

are the unique localizations of the same map

$$\varphi: A \rightarrow \mathcal{O}_X(X),$$

so

$$f_{1,x}^\# = f_{2,x}^\# \quad \text{for all } x \in X.$$

Since a morphism of sheaves is determined by its maps on stalks, it follows that

$$f_1^\# = f_2^\#.$$

Thus

$$(f_1, f_1^\#) = (f_2, f_2^\#).$$

This proves injectivity.

Therefore the map

$$(f, f^\#) \longmapsto f_{\text{Spec } A}^\#$$

is bijective. □

Definition 2.1.34. An *affine scheme* is a ringed space $(X, \mathcal{O}_X)$ such that there exists a ring A with

$$(X, \mathcal{O}_X) \cong (\text{Spec } A, \mathcal{O}_{\text{Spec } A}).$$

Definition 2.1.35. A *scheme* is a ringed space $(X, \mathcal{O}_X)$ such that there exists an open cover

$$X = \bigcup_i U_i$$

for which each restricted ringed space

$$(U_i, \mathcal{O}_X|_{U_i})$$

is an affine scheme.

Example 2.1.36. If k is a field, then

$$\text{Spec } k = \{*\}.$$

Hence $(\text{Spec } k, \mathcal{O}_{\text{Spec } k})$ is an affine scheme consisting of one point, and its structure sheaf is k .

Example 2.1.37. If T is a discrete valuation ring, then $\operatorname{Spec} T$ has two points:

$$\operatorname{Spec} T = \{\eta, s\},$$

where $\eta = (0)$ is the generic point and $s = \mathfrak{m}$ is the closed point.

Proposition 2.1.38. If $(X, \mathcal{O}_X)$ is a scheme and $U \subseteq X$ is open, then

$$(U, \mathcal{O}_X|_U)$$

is again a scheme.

Proof. Choose an affine open cover

$$X = \bigcup_i U_i, \quad (U_i, \mathcal{O}_X|_{U_i}) \cong (\operatorname{Spec} A_i, \mathcal{O}_{\operatorname{Spec} A_i}).$$

Then

$$U = \bigcup_i (U \cap U_i).$$

So it is enough to show that for each i , the open subspace

$$(U \cap U_i, \mathcal{O}_X|_{U \cap U_i})$$

is a scheme. Since $U_i \cong \operatorname{Spec} A_i$, this reduces to showing that every open subset of an affine scheme is covered by principal opens, and each principal open is affine.

Thus it remains to check that for every $f \in A$,

$$(D(f), \mathcal{O}_{\operatorname{Spec} A}|_{D(f)})$$

is an affine scheme. □

Proposition 2.1.39. For any ring A and any $f \in A$, there is an isomorphism of ringed spaces

$$(D(f), \mathcal{O}_{\operatorname{Spec} A}|_{D(f)}) \cong (\operatorname{Spec} A_f, \mathcal{O}_{\operatorname{Spec} A_f}).$$

Proof. Let

$$\varphi: A \rightarrow A_f$$

be the localization map. It induces a morphism of ringed spaces

$$(g, g^\#): (\operatorname{Spec} A_f, \mathcal{O}_{\operatorname{Spec} A_f}) \longrightarrow (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A}).$$

On underlying spaces, g identifies $\operatorname{Spec} A_f$ with $D(f) \subseteq \operatorname{Spec} A$, so g is a homeomorphism

$$\operatorname{Spec} A_f \xrightarrow{\sim} D(f).$$

It remains to show that the sheaf morphism is an isomorphism over $D(f)$. It is enough to check on stalks. If

$$\mathfrak{p} \in D(f),$$

then the corresponding prime of A_f is $\mathfrak{p}A_f$, and the induced map on stalks is

$$g_{\mathfrak{p}}^{\#}: \mathcal{O}_{\mathrm{Spec} A, \mathfrak{p}} \longrightarrow \mathcal{O}_{\mathrm{Spec} A_f, \mathfrak{p}A_f}.$$

Using the standard identifications

$$\mathcal{O}_{\mathrm{Spec} A, \mathfrak{p}} \cong A_{\mathfrak{p}}, \quad \mathcal{O}_{\mathrm{Spec} A_f, \mathfrak{p}A_f} \cong (A_f)_{\mathfrak{p}A_f},$$

this becomes the canonical map

$$A_{\mathfrak{p}} \longrightarrow (A_f)_{\mathfrak{p}A_f},$$

which is an isomorphism because $f \notin \mathfrak{p}$. Hence $g^{\#}$ is an isomorphism on all stalks, so

$$(D(f), \mathcal{O}_{\mathrm{Spec} A}|_{D(f)}) \cong (\mathrm{Spec} A_f, \mathcal{O}_{\mathrm{Spec} A_f}).$$

□

Corollary 2.1.40. *Every open subscheme of a scheme is a scheme.*

Definition 2.1.41. *A closed immersion*

$$(f, f^{\#}): (Z, \mathcal{O}_Z) \longrightarrow (X, \mathcal{O}_X)$$

is a morphism of schemes such that

1. the map of topological spaces

$$f: Z \hookrightarrow X$$

identifies Z with a closed subspace of X ;

2. for every $z \in Z$, the induced map on stalks

$$f_z^{\#}: \mathcal{O}_{X, f(z)} \longrightarrow \mathcal{O}_{Z, z}$$

is surjective.

Remark 2.1.42. If

$$(f, f^{\#}): (Z, \mathcal{O}_Z) \longrightarrow (X, \mathcal{O}_X)$$

is a closed immersion, one also says that $(Z, \mathcal{O}_Z)$ is a *closed subscheme* of $(X, \mathcal{O}_X)$.

Example 2.1.43. For an ideal $I \subseteq A$, the canonical morphism

$$\mathrm{Spec}(A/I) \longrightarrow \mathrm{Spec} A$$

is a closed immersion. Thus $\mathrm{Spec}(A/I)$ is a closed subscheme of $\mathrm{Spec} A$.

Remark 2.1.44. See Hartshorne, Exercise II.3.11.

Remark 2.1.45 (Gluing schemes). Let X_1 and X_2 be schemes, and let

$$U_1 \subseteq X_1, \quad U_2 \subseteq X_2$$

be open subschemes. Suppose there is an isomorphism

$$(U_1, \mathcal{O}_{X_1}|_{U_1}) \xrightarrow{\sim} (U_2, \mathcal{O}_{X_2}|_{U_2}).$$

Then one may glue X_1 and X_2 along U_1 and U_2 to obtain a scheme

$$X = X_1 \cup X_2.$$

See Hartshorne, Exercise I.2.12 and Exercise II.1.22.

Example 2.1.46. Let k be a field. Set

$$X_1 = \mathbb{A}_k^1 = \mathrm{Spec} k[x], \quad X_2 = \mathbb{A}_k^1 = \mathrm{Spec} k[x].$$

Let

$$U_1 = X_1 \setminus \{(x)\} = D(x) \cong \mathrm{Spec} k \left[x, \frac{1}{x} \right],$$

and similarly

$$U_2 = X_2 \setminus \{(x)\} = D(x) \cong \mathrm{Spec} k \left[x, \frac{1}{x} \right].$$

Identifying U_1 and U_2 via the identity on

$$\mathrm{Spec} k \left[x, \frac{1}{x} \right],$$

one can glue X_1 and X_2 along $U_1 \cong U_2$.

Remark 2.1.47. If

$$A = k[x_1, \dots, x_n]/(f_1, \dots, f_m)$$

for a field k , then $\mathrm{Spec} A$ is an affine scheme. This is the scheme-theoretic affine variety associated to the quotient algebra A .

Remark 2.1.48. Assume $k = \bar{k}$ is algebraically closed. Then by Hilbert's Nullstellensatz, the closed points of

$$\operatorname{Spec} A, \quad A = k[x_1, \dots, x_n]/(f_1, \dots, f_m),$$

correspond to maximal ideals, hence to common zeros of

$$f_1, \dots, f_m$$

in k^n .

Remark 2.1.49. Let k be a field, let A be a k -algebra, and let B be another k -algebra. Then

$$\operatorname{Hom}_{k\text{-Sch}}(\operatorname{Spec} B, \operatorname{Spec} A) \cong \operatorname{Hom}_{k\text{-alg}}(A, B).$$

If

$$A = k[x_1, \dots, x_n]/(f_1, \dots, f_m),$$

then

$$\operatorname{Hom}_{k\text{-alg}}(A, B) \cong \{(b_1, \dots, b_n) \in B^n \mid f_1(b_1, \dots, b_n) = \dots = f_m(b_1, \dots, b_n) = 0\}.$$

These are called the B -points of $\operatorname{Spec} A$.

Definition 2.1.50. For any scheme X and any scheme T , define

$$X(T) := \operatorname{Hom}_{\operatorname{Sch}}(T, X).$$

Elements of $X(T)$ are called the T -points of X .

Remark 2.1.51. Two important special cases are:

1. when X is affine;
2. when $T = \operatorname{Spec} k$ for a field k .

Example 2.1.52. Define

$$SL_{n,\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[x_{ij}]/(\det(x_{ij}) - 1).$$

Also define

$$GL_{n,\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[x_{ij}, t]/(t \det(x_{ij}) - 1).$$

Then for any ring R ,

$$SL_{n,\mathbb{Z}}(R) = \operatorname{Hom}_{\operatorname{Sch}}(\operatorname{Spec} R, SL_{n,\mathbb{Z}}) \cong \{M \in M_n(R) \mid \det(M) = 1\},$$

and

$$GL_{n,\mathbb{Z}}(R) = \operatorname{Hom}_{\operatorname{Sch}}(\operatorname{Spec} R, GL_{n,\mathbb{Z}}) \cong GL_n(R).$$

In particular,

$$GL_{n,\mathbb{Z}}(R) = \{M \in M_n(R) \mid \det(M) \in R^\times\}.$$

Example 2.1.53. Let

$$X = \operatorname{Spec} \frac{\mathbb{Z}[x_1, \dots, x_n]}{(f_1, \dots, f_m)}.$$

Then for any commutative ring R , we have a natural bijection

$$\operatorname{Hom}(\operatorname{Spec} R, X) \cong \operatorname{Hom}_{\mathbf{Ring}} \left(\frac{\mathbb{Z}[x_1, \dots, x_n]}{(f_1, \dots, f_m)}, R \right).$$

By the universal property of the quotient, this is further identified with

$$\{\varphi : \mathbb{Z}[x_1, \dots, x_n] \rightarrow R \mid \varphi(f_1) = \dots = \varphi(f_m) = 0\}.$$

Such a ring homomorphism φ is uniquely determined by the images

$$a_i := \varphi(x_i) \in R, \quad i = 1, \dots, n.$$

Hence

$$\operatorname{Hom}(\operatorname{Spec} R, X) \cong \{(a_1, \dots, a_n) \in R^n \mid f_1(a_1, \dots, a_n) = \dots = f_m(a_1, \dots, a_n) = 0\}.$$

In other words, the set of R -valued points (R -值点, R -valued points) of X is exactly the set of solutions in R^n of the equations

$$f_1 = \dots = f_m = 0.$$

Example 2.1.54. Consider

$$GL_n = \operatorname{Spec} \frac{\mathbb{Z}[\{x_{ij}\}_{1 \leq i, j \leq n}, t]}{(\det(x_{ij})t - 1)}.$$

Then for any commutative ring R , we have

$$GL_n(R) := \operatorname{Hom}(\operatorname{Spec} R, GL_n).$$

By the description of R -valued points (R -值点, R -valued points) of an affine scheme, this set is naturally identified with

$$\left\{ (a_{ij}, s) \in R^{n^2} \times R \mid \det(a_{ij})s = 1 \right\}.$$

Equivalently,

$$GL_n(R) \cong \{(a_{ij}) \in M_n(R) \mid \det(a_{ij}) \in R^\times\}.$$

Thus $GL_n(R)$ is the group of invertible $n \times n$ matrices over R .

In particular, if $R = \mathbb{Z}$, then

$$\mathbb{Z}^\times = \{\pm 1\},$$

so

$$GL_n(\mathbb{Z}) = \{(a_{ij}) \in M_n(\mathbb{Z}) \mid \det(a_{ij}) = \pm 1\}.$$

Similarly,

$$SL_n = \operatorname{Spec} \frac{\mathbb{Z}[\{x_{ij}\}_{1 \leq i, j \leq n}]}{(\det(x_{ij}) - 1)},$$

and for any commutative ring R ,

$$SL_n(R) = \operatorname{Hom}(\operatorname{Spec} R, SL_n) \cong \{(a_{ij}) \in M_n(R) \mid \det(a_{ij}) = 1\}.$$

Remark 2.1.55. The notation GL_n above is not only a scheme (概形, scheme), but in fact an algebraic group (代数群, algebraic group).

Roughly speaking, an algebraic group (代数群, algebraic group) is a group object in the category of schemes (概形范畴中的群对象, group object in the category of schemes). This means that the group law, the identity element, and the inverse map are all given by morphisms of schemes:

$$m : GL_n \times GL_n \rightarrow GL_n, \quad e : \operatorname{Spec} \mathbb{Z} \rightarrow GL_n, \quad i : GL_n \rightarrow GL_n.$$

On R -valued points (R -值点, R -valued points), these morphisms induce the usual operations

$$(A, B) \mapsto AB, \quad I_n, \quad A \mapsto A^{-1}.$$

So the phrase “called algebraic group” written on the board means that GL_n carries not only the structure of an affine scheme (仿射概形, affine scheme), but also a compatible group structure. The same is true for SL_n .

Example 2.1.56. A basic example of a non-affine scheme (非仿射概形, non-affine scheme) is the punctured affine plane (去心仿射平面, punctured affine plane)

$$U = \mathbb{A}_k^2 \setminus V((x, y)),$$

where

$$\mathbb{A}_k^2 = \operatorname{Spec} k[x, y].$$

Here $V((x, y))$ is the closed subset corresponding to the maximal ideal (x, y) , namely the origin.

Then U is an open subscheme (开子概形, open subscheme) of $\mathbb{A}_k^2$, but U is not affine (仿射的, affine).

Remark 2.1.57. The key point is that

$$\mathcal{O}_U(U) \cong k[x, y].$$

Indeed, since

$$U = D(x) \cup D(y),$$

a regular function (正则函数, regular function) on U is given by sections on $D(x)$ and $D(y)$ that agree on

$$D(x) \cap D(y) = D(xy).$$

Now

$$\mathcal{O}_{\mathbb{A}_k^2}(D(x)) = k[x, y]_x, \quad \mathcal{O}_{\mathbb{A}_k^2}(D(y)) = k[x, y]_y,$$

and

$$\mathcal{O}_{\mathbb{A}_k^2}(D(xy)) = k[x, y]_{xy}.$$

Hence

$$\mathcal{O}_U(U) = k[x, y]_x \cap k[x, y]_y \subset k[x, y]_{xy}.$$

But one checks that

$$k[x, y]_x \cap k[x, y]_y = k[x, y].$$

Therefore

$$\mathcal{O}_U(U) \cong k[x, y].$$

If U were affine, then by the affine reconstruction theorem (仿射重构定理, affine reconstruction theorem) we would have

$$U \cong \operatorname{Spec} \mathcal{O}_U(U) \cong \operatorname{Spec} k[x, y] = \mathbb{A}_k^2.$$

This is impossible, since $U = \mathbb{A}_k^2 \setminus \{(x, y)\}$ is a proper open subset of $\mathbb{A}_k^2$. Thus U is not affine.

2.2 Projective scheme

2.2.1 Motivation from projective varieties

Our goal is to find a scheme-theoretic counterpart of projective varieties (射影簇, projective varieties).

For affine varieties (仿射簇, affine varieties), we have a correspondence

$$V(f_1, \dots, f_m) \subseteq \mathbb{A}^n \quad \longleftrightarrow \quad \text{closed points of } \operatorname{Spec} \frac{\mathbb{C}[x_1, \dots, x_n]}{(f_1, \dots, f_m)}.$$

For projective varieties (射影簇, projective varieties), we consider

$$V(f_1, \dots, f_m) \subseteq \mathbb{P}^n,$$

where

$$f_i \in \mathbb{C}[x_0, \dots, x_n]$$

are homogeneous polynomials (齐次多项式, homogeneous polynomials) .

Recall that a polynomial is called homogeneous (齐次的, homogeneous) if all of its terms have the same degree. For example,

$$x^2 + xy + y^2$$

is homogeneous of degree 2, while

$$x^2 + 1, \quad x^2 + y$$

are not homogeneous.

2.2.2 Graded rings

The abstract framework for projective schemes is given by graded rings (分次环, graded rings) .

Definition 2.2.1. A ring S is called a graded ring (分次环, graded ring) if

$$S = \bigoplus_{d \geq 0} S_d$$

as an abelian group (阿贝尔群, abelian group) , and the multiplication satisfies

$$S_m \cdot S_n \subseteq S_{m+n} \quad \text{for all } m, n \geq 0.$$

Notation 2.2.2. We write

$$S_+ := \bigoplus_{d \geq 1} S_d.$$

This is an ideal (理想, ideal) of S .

Definition 2.2.3. An element $f \in S$ is called homogeneous of degree d (d 次齐次元, homogeneous of degree d) if

$$f \in S_d.$$

Definition 2.2.4. An ideal $I \subseteq S$ is called homogeneous (齐次理想, homogeneous ideal) if it is generated by homogeneous elements.

Remark 2.2.5. Equivalently, an ideal $I \subseteq S$ is homogeneous if and only if

$$I = \bigoplus_{d \geq 0} (I \cap S_d).$$

Example 2.2.6. The polynomial ring

$$S = k[x_0, \dots, x_n]$$

is naturally a graded ring with

$$S = \bigoplus_{d \geq 0} S_d,$$

where S_d is the set of homogeneous polynomials of degree d .

2.2.3 The construction of $\text{Proj } S$

Definition 2.2.7. Let $S = \bigoplus_{d \geq 0} S_d$ be a graded ring. Define

$$\text{Proj } S := \{\mathfrak{p} \subseteq S \mid \mathfrak{p} \text{ is a homogeneous prime ideal and } \mathfrak{p} \not\supseteq S_+\}.$$

Remark 2.2.8. Thus $\text{Proj } S$ consists of homogeneous prime ideals (齐次素理想, homogeneous prime ideals) that do not contain the irrelevant ideal (无关理想, irrelevant ideal) S_+ .

2.2.4 Closed sets in $\text{Proj } S$

For every homogeneous ideal (齐次理想, homogeneous ideal) $I \subseteq S$, define

$$V_+(I) := \{\mathfrak{p} \in \text{Proj } S \mid \mathfrak{p} \supseteq I\}.$$

Also define

$$D_+(I) := \{\mathfrak{p} \in \text{Proj } S \mid \mathfrak{p} \not\supseteq I\}.$$

In particular, for a homogeneous element $f \in S$, we write

$$D_+(f) := D_+((f)).$$

2.2.5 Topology on $\text{Proj } S$

Lemma 2.2.9. *The sets*

$$V_+(I) = \{\mathfrak{p} \in \text{Proj } S \mid \mathfrak{p} \supseteq I\},$$

where $I \subseteq S$ is a homogeneous ideal (齐次理想, homogeneous ideal), define the closed sets of a topology on $\text{Proj } S$.

Proof. We verify the axioms for closed sets.

(1) For any family of homogeneous ideals $\{I_\lambda\}_{\lambda \in \Lambda}$,

$$\bigcap_{\lambda \in \Lambda} V_+(I_\lambda) = V_+\left(\sum_{\lambda \in \Lambda} I_\lambda\right).$$

Indeed, for $\mathfrak{p} \in \text{Proj } S$,

$$\mathfrak{p} \in \bigcap_{\lambda} V_+(I_\lambda) \iff I_\lambda \subseteq \mathfrak{p} \text{ for all } \lambda \iff \sum_{\lambda} I_\lambda \subseteq \mathfrak{p} \iff \mathfrak{p} \in V_+\left(\sum_{\lambda} I_\lambda\right).$$

(2) If $I, J \subseteq S$ are homogeneous ideals, then

$$V_+(I) \cup V_+(J) = V_+(I \cap J).$$

Indeed, since every $\mathfrak{p} \in \text{Proj } S$ is prime (素理想, prime ideal),

$$I \cap J \subseteq \mathfrak{p} \iff IJ \subseteq \mathfrak{p} \iff I \subseteq \mathfrak{p} \text{ or } J \subseteq \mathfrak{p}.$$

Hence

$$\mathfrak{p} \in V_+(I \cap J) \iff \mathfrak{p} \in V_+(I) \cup V_+(J).$$

(3) We have

$$V_+(0) = \text{Proj } S$$

and

$$V_+(S) = V_+(S_+) = \emptyset.$$

Indeed, every homogeneous prime ideal contains 0, so $V_+(0) = \text{Proj } S$. On the other hand, no prime ideal can contain S , and by definition every $\mathfrak{p} \in \text{Proj } S$ satisfies $\mathfrak{p} \not\supseteq S_+$, so

$$V_+(S) = V_+(S_+) = \emptyset.$$

Therefore the sets $V_+(I)$ satisfy the axioms for closed sets. $\square$

Remark 2.2.10. Some books, including Hartshorne, write $V(I)$ instead of $V_+(I)$ in the projective setting. Here we use the notation $V_+(I)$ to emphasize that we are working in $\text{Proj } S$, not in $\text{Spec } S$.

Proposition 2.2.11. *The sets*

$$D_+(f) := \text{Proj } S \setminus V_+((f)), \quad f \in S_+ \text{ homogeneous},$$

form a topological basis (拓扑基, topological basis) for $\text{Proj } S$.

Proof. First, we show that these open sets cover $\text{Proj } S$. Since

$$\bigcap_{f \in S_+} V_+(f) = V_+(S_+) = \emptyset,$$

we get

$$\bigcup_{f \in S_+} D_+(f) = \text{Proj } S.$$

Now let $\mathfrak{p} \in D_+(I)$ for some homogeneous ideal $I \subseteq S$. Then

$$\mathfrak{p} \not\supseteq I.$$

Hence there exists some $g \in I \setminus \mathfrak{p}$. Write

$$g = g_{d_1} + \cdots + g_{d_r}$$

as a sum of homogeneous components. Since $\mathfrak{p}$ is homogeneous (齐次的, homogeneous), if every $g_{d_i} \in \mathfrak{p}$, then $g \in \mathfrak{p}$, contradiction. Thus there exists a homogeneous component $f := g_{d_i} \in I \setminus \mathfrak{p}$. In particular $f \in S_+$ if I is relevant here, and

$$\mathfrak{p} \in D_+(f) \subseteq D_+(I).$$

Therefore the sets $D_+(f)$, for homogeneous $f \in S_+$, form a basis of open sets. $\square$

2.2.6 The structure sheaf on $\text{Proj } S$

Our goal is to endow $\text{Proj } S$ with a scheme structure (概形结构, scheme structure).

Definition 2.2.12. Let $\mathfrak{p} \in \text{Proj } S$. Define

$$T(\mathfrak{p}) := \{\text{homogeneous elements of } S \setminus \mathfrak{p}\}.$$

Consider the localization (局部化, localization)

$$T(\mathfrak{p})^{-1}S.$$

We define

$$S_{(\mathfrak{p})} := (T(\mathfrak{p})^{-1}S)_0,$$

the degree 0 part of $T(\mathfrak{p})^{-1}S$.

Remark 2.2.13. An element of $S_{(\mathfrak{p})}$ is represented by a fraction

$$\frac{a}{t},$$

where a, t are homogeneous, $t \notin \mathfrak{p}$, and

$$\deg a = \deg t.$$

Thus $S_{(\mathfrak{p})}$ is a ring.

Definition 2.2.14. For any open subset $U \subseteq \text{Proj } S$, define

$$\mathcal{O}_{\text{Proj } S}(U) := \left\{ s : U \rightarrow \coprod_{\mathfrak{p} \in U} S_{(\mathfrak{p})} \mid s(\mathfrak{p}) \in S_{(\mathfrak{p})} \text{ for all } \mathfrak{p} \in U, \text{ and } s \text{ is locally a quotient } \frac{a}{f} \right\}.$$

More precisely, s satisfies the following local condition. For every $\mathfrak{p} \in U$, there exist an open neighborhood $V \subseteq U$. There also exist homogeneous elements $a, f \in S$ such that

$$V \subseteq D_+(f), \quad \deg a = \deg f,$$

and for every $\mathfrak{q} \in V$,

$$s(\mathfrak{q}) = \frac{a}{f} \in S_{(\mathfrak{q})}.$$

Definition 2.2.15. If $f \in S$ is homogeneous, let

$$S_{(f)} := (S_f)_0,$$

the degree 0 part of the localization S_f . Equivalently,

$$S_{(f)} = \left\{ \frac{a}{f^n} \in S_f \mid a \text{ homogeneous and } \deg a = n \deg f \right\}.$$

Lemma 2.2.16. *The assignment*

$$U \longmapsto \mathcal{O}_{\text{Proj } S}(U)$$

defines a sheaf (层, sheaf) on $\text{Proj } S$. Moreover, for every $\mathfrak{p} \in \text{Proj } S$, the stalk (茎, stalk) satisfies

$$\mathcal{O}_{\text{Proj } S, \mathfrak{p}} \cong S_{(\mathfrak{p})}.$$

Proof. The proof is the same as in the affine case. The sheaf axiom follows from the local nature of the definition, and the identification of the stalk with $S_{(\mathfrak{p})}$ is obtained by passing to germs of locally represented fractions. $\square$

2.2.7 A lemma on graded localization

Lemma 2.2.17. *Let S be a graded ring (分次环, graded ring), and let*

$$T$$

be a multiplicative set (乘法集, multiplicative set) consisting of homogeneous elements. Then there is a natural bijection

$$\left\{ \mathfrak{p} \subseteq S \mid \begin{array}{l} \mathfrak{p} \text{ is a homogeneous prime ideal and} \\ \mathfrak{p} \cap T = \emptyset \end{array} \right\} \longleftrightarrow \left\{ \mathfrak{q} \subseteq T^{-1}S \mid \mathfrak{q} \text{ is a homogeneous prime ideal} \right\},$$

given by

$$\mathfrak{p} \longmapsto T^{-1}\mathfrak{p}.$$

Proof. This is the graded version of the usual correspondence between prime ideals under localization (局部化下素理想对应, correspondence of prime ideals under localization) .

If $\mathfrak{p} \subseteq S$ is a homogeneous prime ideal with $\mathfrak{p} \cap T = \emptyset$, then $T^{-1}\mathfrak{p}$ is a prime ideal of $T^{-1}S$. It is homogeneous because $\mathfrak{p}$ is homogeneous.

Conversely, if $\mathfrak{q} \subseteq T^{-1}S$ is a homogeneous prime ideal, then its contraction

$$\mathfrak{q}^c := \{a \in S \mid a/1 \in \mathfrak{q}\}$$

is a homogeneous prime ideal of S , and $\mathfrak{q}^c \cap T = \emptyset$.

These two constructions are inverse to each other. □

Lemma 2.2.18. *Let S be a graded ring, and let T be a multiplicative set consisting of homogeneous elements of nonzero degree. Set*

$$S_{(T)} := (T^{-1}S)_0,$$

the degree 0 part of $T^{-1}S$. Then there is a natural bijection

$$\{\mathfrak{p} \subseteq T^{-1}S \mid \mathfrak{p} \text{ is a homogeneous prime ideal}\} \longleftrightarrow \operatorname{Spec} S_{(T)},$$

given by contraction

$$\mathfrak{p} \longmapsto \mathfrak{p} \cap S_{(T)}.$$

Proof. Let $R = T^{-1}S$, which is still a graded ring. We must show that homogeneous prime ideals of R correspond bijectively to prime ideals of $R_0 = S_{(T)}$.

Define

$$\Phi : \{\text{homogeneous prime ideals of } R\} \rightarrow \operatorname{Spec} R_0, \quad \mathfrak{p} \mapsto \mathfrak{p} \cap R_0.$$

Since $R_0 \subseteq R$, the contraction of a prime ideal is prime, so Φ is well-defined.

Conversely, if $\mathfrak{r} \in \operatorname{Spec} R_0$, define

$$\Psi(\mathfrak{r}) := \bigoplus_{d \in \mathbb{Z}} \{x \in R_d \mid x^n \in \mathfrak{r}R \text{ for some } n \geq 1\}.$$

Equivalently, one may define $\Psi(\mathfrak{r})$ as the radical of the ideal generated by $\mathfrak{r}$ in R :

$$\Psi(\mathfrak{r}) = \sqrt{\mathfrak{r}R}.$$

This is a homogeneous prime ideal of R .

We now check that Φ and Ψ are inverse to each other.

First, let $\mathfrak{p} \subseteq R$ be a homogeneous prime ideal. We claim that

$$\Psi(\Phi(\mathfrak{p})) = \sqrt{(\mathfrak{p} \cap R_0)R} = \mathfrak{p}.$$

Indeed, if $a \in \mathfrak{p}$ is homogeneous of degree d , choose any homogeneous $f \in T$. Since $\deg f \neq 0$, the element

$$\frac{a^{\deg f}}{f^{\deg a}}$$

has degree 0, hence belongs to R_0 . Moreover it lies in $\mathfrak{p} \cap R_0$, so $a \in \sqrt{(\mathfrak{p} \cap R_0)R}$. Thus every homogeneous element of $\mathfrak{p}$ belongs to $\sqrt{(\mathfrak{p} \cap R_0)R}$, and therefore

$$\mathfrak{p} \subseteq \sqrt{(\mathfrak{p} \cap R_0)R}.$$

The reverse inclusion is obvious because $(\mathfrak{p} \cap R_0)R \subseteq \mathfrak{p}$ and $\mathfrak{p}$ is radical.

Next, let $\mathfrak{r} \in \text{Spec } R_0$. Then

$$\Phi(\Psi(\mathfrak{r})) = \sqrt{\mathfrak{r}R} \cap R_0 = \mathfrak{r}.$$

Hence Φ is a bijection. □

Notation 2.2.19. If $f \in S$ is homogeneous, we write

$$S_{(f)} := (S_f)_0.$$

More generally, if T is a multiplicative set consisting of homogeneous elements, we write

$$S_{(T)} := (T^{-1}S)_0.$$

Proposition 2.2.20. *Let S be a graded ring and let $f \in S_+$ be homogeneous. Then*

$$D_+(f) \cong \text{Spec } S_{(f)}.$$

2.2.8 The affine charts of $\text{Proj } S$

Proposition 2.2.21. *Let S be a graded ring (分次环, graded ring), and let $f \in S_+$ be homogeneous. Then:*

(1) *There is an isomorphism of ringed spaces (环化空间同构, isomorphism of ringed spaces)*

$$(D_+(f), \mathcal{O}_{\text{Proj } S}|_{D_+(f)}) \cong \text{Spec } S_{(f)}.$$

(2) *In particular, $\text{Proj } S$ is a scheme (概形, scheme).*

Proof. It is clear that (1) $\Rightarrow$ (2), since the open sets $D_+(f)$, for homogeneous $f \in S_+$, form a basis of $\text{Proj } S$, and each of them is affine (仿射的, affine) by (1).

So it remains to prove (1).

Step 1: Construction of the canonical map. By definition,

$$\mathcal{O}_{\text{Proj } S}(D_+(f)) = \left\{ g : D_+(f) \rightarrow \coprod_{\mathfrak{p} \in D_+(f)} S_{(\mathfrak{p})} \mid g(\mathfrak{p}) \in S_{(\mathfrak{p})} \text{ and } g \text{ is locally a quotient} \right\}.$$

There is a canonical ring homomorphism

$$\psi : S_{(f)} \longrightarrow \mathcal{O}_{\text{Proj } S}(D_+(f))$$

defined as follows. If

$$\frac{a}{f^n} \in S_{(f)}, \quad a \text{ homogeneous, } \deg a = n \deg f,$$

then set

$$\psi\left(\frac{a}{f^n}\right)(\mathfrak{p}) := \frac{a}{f^n} \in S_{(\mathfrak{p})} \quad (\mathfrak{p} \in D_+(f)).$$

This is well-defined because $f \notin \mathfrak{p}$ for every $\mathfrak{p} \in D_+(f)$, so f becomes invertible in $S_{(\mathfrak{p})}$.

Step 2: The induced map on underlying topological spaces. By Step 1 we obtain a morphism of locally ringed spaces

$$\Phi : D_+(f) \longrightarrow \text{Spec } S_{(f)}.$$

On points, Φ is given by

$$\Phi(\mathfrak{p}) = \psi^{-1}(\mathfrak{p} \mathcal{O}_{\text{Proj } S, \mathfrak{p}}).$$

Using the description

$$\mathcal{O}_{\text{Proj } S, \mathfrak{p}} \cong S_{(\mathfrak{p})},$$

this becomes

$$\Phi(\mathfrak{p}) = (\mathfrak{p} S_f) \cap S_{(f)}.$$

We claim that Φ is a homeomorphism (同胚, homeomorphism) .

Indeed, let

$$T = \{\text{homogeneous elements of } S \setminus \mathfrak{p}\}.$$

Then

$$S_{(\mathfrak{p})} = (T^{-1}S)_0.$$

By the lemma on graded localization (分次局局部化引理, lemma on graded localization), prime ideals (素理想, prime ideals) of $S_{(f)} = (S_f)_0$ correspond to homogeneous prime ideals of S_f , and these in turn correspond to homogeneous prime ideals $\mathfrak{p} \subseteq S$ such that $f \notin \mathfrak{p}$. Hence

$$D_+(f) \longleftrightarrow \text{Spec } S_{(f)}$$

bijectively.

More explicitly, if $\mathfrak{q} \in \operatorname{Spec} S_{(f)}$, then the corresponding point of $D_+(f)$ is the contraction to S of the homogeneous prime ideal of S_f lying over $\mathfrak{q}$.

Step 3: Compatibility with the topology. Let $\mathfrak{a} \subseteq S$ be a homogeneous ideal (齐次理想, homogeneous ideal). Then

$$\mathfrak{p} \in V_+(\mathfrak{a}) \cap D_+(f) \iff f \notin \mathfrak{p} \text{ and } \mathfrak{p} \supseteq \mathfrak{a}.$$

Under the above bijection, this is equivalent to

$$\Phi(\mathfrak{p}) \supseteq (\mathfrak{a}S_f) \cap S_{(f)}.$$

Therefore

$$\Phi(V_+(\mathfrak{a}) \cap D_+(f)) = V((\mathfrak{a}S_f) \cap S_{(f)}).$$

So Φ is a homeomorphism.

Step 4: Isomorphism on stalks. Let $\mathfrak{p} \in D_+(f)$, and let

$$\mathfrak{q} = \Phi(\mathfrak{p}) \in \operatorname{Spec} S_{(f)}.$$

We must show that the induced local homomorphism

$$\mathcal{O}_{\operatorname{Spec} S_{(f)}, \mathfrak{q}} \longrightarrow \mathcal{O}_{\operatorname{Proj} S, \mathfrak{p}}$$

is an isomorphism.

Now

$$\mathcal{O}_{\operatorname{Spec} S_{(f)}, \mathfrak{q}} = (S_{(f)})_{\mathfrak{q}}, \quad \mathcal{O}_{\operatorname{Proj} S, \mathfrak{p}} \cong S_{(\mathfrak{p})}.$$

So it is enough to prove

$$(S_{(f)})_{\mathfrak{q}} \cong S_{(\mathfrak{p})}.$$

Let

$$T = \{\text{homogeneous elements of } S \setminus \mathfrak{p}\}.$$

Since $f \notin \mathfrak{p}$, we have $f \in T$. Consider the set

$$\tilde{T} = \left\{ \frac{a}{f^n} \in S_{(f)} \mid a \text{ homogeneous, } \deg a = n \deg f, a \in T \right\}.$$

Then $\tilde{T}$ is exactly the complement of $\mathfrak{q}$ in $S_{(f)}$, and

$$(S_{(f)})_{\mathfrak{q}} = \tilde{T}^{-1} S_{(f)}.$$

On the other hand, every element of $S_{(\mathfrak{p})}$ may be written as

$$\frac{a}{t}, \quad a, t \text{ homogeneous, } t \in T, \deg a = \deg t.$$

Since $f \in T$, after multiplying numerator and denominator by suitable powers of f , we may rewrite such an element in the form

$$\frac{af^m}{tf^m},$$

with numerator and denominator representing an element of $\tilde{T}^{-1}S_{(f)}$. Hence

$$\tilde{T}^{-1}S_{(f)} \cong S_{(\mathfrak{p})}.$$

Therefore

$$(S_{(f)})_{\mathfrak{q}} \cong S_{(\mathfrak{p})} \cong \mathcal{O}_{\text{Proj } S, \mathfrak{p}}.$$

Thus Φ is an isomorphism of locally ringed spaces, and we conclude

$$(D_+(f), \mathcal{O}_{\text{Proj } S}|_{D_+(f)}) \cong \text{Spec } S_{(f)}.$$

This proves (1), and hence also (2). □

Remark 2.2.22. There is also a direct proof that the induced map on stalks

$$(S_{(f)})_{\mathfrak{q}} \rightarrow S_{(\mathfrak{p})}$$

is an isomorphism, by checking surjectivity and injectivity explicitly with homogeneous fractions.

Step 4 (Method 2): direct proof on stalks. Let

$$P \in D_+(f), \quad \mathfrak{q} = \psi^{-1}(P) \in \text{Spec } S_{(f)}.$$

We want to show that the induced local homomorphism

$$h : (S_{(f)})_{\mathfrak{q}} \longrightarrow S_{(P)}$$

is an isomorphism.

Recall that

$$S_{(P)} = (T^{-1}S)_0, \quad T = \{\text{homogeneous elements of } S \setminus P\}.$$

Since $f \notin P$, we have $f \in T$, so there is a natural map

$$h : (S_{(f)})_{\mathfrak{q}} \rightarrow S_{(P)}.$$

Surjectivity. Take any element

$$\frac{a}{b} \in S_{(P)},$$

where a, b are homogeneous, $b \notin P$, and

$$\deg a = \deg b.$$

We claim that it comes from $(S_{(f)})_{\mathfrak{q}}$.

Consider

$$\frac{a b^{\deg f - 1} / f^{\deg b}}{b^{\deg f} / f^{\deg b}} \in (S_{(f)})_{\mathfrak{q}}.$$

Indeed, both numerator and denominator are homogeneous of degree 0 in S_f :

$$\deg(a b^{\deg f - 1}) = \deg a + (\deg f - 1) \deg b = \deg f \deg b,$$

and similarly

$$\deg(b^{\deg f}) = \deg f \deg b.$$

Moreover,

$$\frac{b^{\deg f}}{f^{\deg b}} \notin \mathfrak{q},$$

since $b \notin P$. Under the map h , this element is sent to

$$\frac{a b^{\deg f - 1} / f^{\deg b}}{b^{\deg f} / f^{\deg b}} = \frac{a}{b}.$$

Hence h is surjective.

Injectivity. Suppose an element

$$\frac{a/f^{n_1}}{b/f^{n_2}} \in (S_{(f)})_{\mathfrak{q}}$$

maps to 0 in $S_{(P)}$, where a, b are homogeneous, $b \notin P$, and

$$\deg a = n_1 \deg f, \quad \deg b = n_2 \deg f.$$

Its image in $S_{(P)}$ is

$$\frac{a f^{n_2}}{b f^{n_1}}.$$

If this is 0 in $S_{(P)}$, then by the definition of localization there exists a homogeneous element

$$c \in T$$

such that

$$c a f^{n_2} = 0 \quad \text{in } S.$$

Multiplying by a suitable power of c if necessary, we may assume the multiplier has compatible degree. Then in $S_{(f)}$ we get

$$\frac{c^{\deg f}}{f^{\deg c}} \cdot \frac{a}{f^{n_1}} = 0.$$

Now

$$\frac{c^{\deg f}}{f^{\deg c}} \in S_{(f)}$$

is homogeneous of degree 0, and since $c \notin P$, it does not belong to $\mathfrak{q}$. Therefore

$$\frac{a/f^{n_1}}{b/f^{n_2}} = 0 \quad \text{in } (S_{(f)})_{\mathfrak{q}}.$$

So h is injective.

Therefore h is an isomorphism:

$$(S_{(f)})_{\mathfrak{q}} \cong S_{(P)}.$$

This proves that the morphism

$$(D_+(f), \mathcal{O}_{\text{Proj } S}|_{D_+(f)}) \rightarrow \text{Spec } S_{(f)}$$

induces an isomorphism on stalks.

2.2.9 Projective space

Example 2.2.23. Let A be a ring. The projective n -space (n 维射影空间, projective n -space) over A is defined by

$$\mathbb{P}_A^n := \text{Proj } A[x_0, \dots, x_n],$$

where the grading is the standard grading (标准分次, standard grading)

$$\deg x_i = 1, \quad i = 0, \dots, n.$$

In this case,

$$A[x_0, \dots, x_n]_+ = (x_0, \dots, x_n).$$

Proposition 2.2.24. We have

$$\mathbb{P}_A^n = \bigcup_{i=0}^n D_+(x_i).$$

Moreover, for each i ,

$$D_+(x_i) \cong \text{Spec } A\left[\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i}\right] \cong \mathbb{A}_A^n.$$

Proof. Since

$$A[x_0, \dots, x_n]_+ = (x_0, \dots, x_n),$$

we have

$$V_+((x_0, \dots, x_n)) = \emptyset.$$

Hence

$$\mathbb{P}_A^n = \text{Proj } A[x_0, \dots, x_n] = \bigcup_{i=0}^n D_+(x_i).$$

Now fix i . By the general result on affine charts of Proj,

$$D_+(x_i) \cong \text{Spec } A[x_0, \dots, x_n]_{(x_i)}.$$

So it remains to identify the ring $A[x_0, \dots, x_n]_{(x_i)}$.

An element of $A[x_0, \dots, x_n]_{(x_i)}$ is a homogeneous fraction (齐次分式, homogeneous fraction) of degree 0, hence can be written in the form

$$\frac{f}{x_i^m},$$

where f is homogeneous of degree m . Such an element is generated by the degree 0 elements

$$\frac{x_j}{x_i}, \quad j \neq i.$$

Therefore

$$A[x_0, \dots, x_n]_{(x_i)} = A\left[\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i}\right].$$

This is a polynomial ring in n variables over A , so

$$D_+(x_i) \cong \mathbb{A}_A^n.$$

□

Example 2.2.25. For $i = 0$, we have

$$D_+(x_0) \cong \text{Spec } A[x_0, \dots, x_n]_{(x_0)} \cong \text{Spec } A\left[\frac{x_1}{x_0}, \frac{x_2}{x_0}, \dots, \frac{x_n}{x_0}\right] \cong \mathbb{A}_A^n.$$

Remark 2.2.26. For a general ring R , the set

$$\mathbb{P}_A^n(R) := \text{Hom}(\text{Spec } R, \mathbb{P}_A^n)$$

is not simply the union

$$\bigcup_{i=0}^n \text{Hom}(\text{Spec } R, D_+(x_i)).$$

One should be careful: a morphism $\text{Spec } R \rightarrow \mathbb{P}_A^n$ is more subtle than choosing a point in one affine chart.

However, if K is a field (域, field), then $\text{Spec } K$ has only one point, and one recovers the classical picture:

$$\mathbb{P}_K^n(K) = \bigcup_{i=0}^n D_+(x_i)(K),$$

with each chart $D_+(x_i)(K) \cong \mathbb{A}_K^n(K) = K^n$. Thus $\mathbb{P}_K^n(K)$ agrees with the usual projective space in classical algebraic geometry.

Remark 2.2.27. When K is a field, the K -valued points (K -值点, K -valued points) of $\mathbb{P}_K^n$ are in bijection with the usual projective points

$$[a_0 : \cdots : a_n], \quad (a_0, \dots, a_n) \in K^{n+1} \setminus \{0\},$$

modulo the equivalence relation

$$(a_0, \dots, a_n) \sim (\lambda a_0, \dots, \lambda a_n), \quad \lambda \in K^\times.$$

Remark 2.2.28. If K is a field (域, field), then

$$\mathbb{P}_A^n(K) = \text{Hom}(\text{Spec } K, \mathbb{P}_A^n) = \bigcup_{i=0}^n \text{Hom}(\text{Spec } K, D_+(x_i)).$$

Indeed, $\text{Spec } K$ has only one point, so every morphism $\text{Spec } K \rightarrow \mathbb{P}_A^n$ factors through one of the open sets $D_+(x_i)$.

More generally, the same holds if R is a local ring (局部环, local ring) :

$$\mathbb{P}_A^n(R) = \text{Hom}(\text{Spec } R, \mathbb{P}_A^n) = \bigcup_{i=0}^n \text{Hom}(\text{Spec } R, D_+(x_i)).$$

Proof. Let R be a local ring, and let

$$f : \text{Spec } R \rightarrow \mathbb{P}_A^n$$

be a morphism. Let $\mathfrak{m} \in \text{Spec } R$ be the unique closed point (唯一闭点, unique closed point), i.e. the unique maximal ideal.

Since

$$\mathbb{P}_A^n = \bigcup_{i=0}^n D_+(x_i),$$

there exists some i such that

$$f(\mathfrak{m}) \in D_+(x_i).$$

Hence

$$\mathfrak{m} \in f^{-1}(D_+(x_i)).$$

Now $f^{-1}(D_+(x_i))$ is an open subset of $\text{Spec } R$ containing $\mathfrak{m}$. We claim that any open subset $U \subseteq \text{Spec } R$ containing $\mathfrak{m}$ must be all of $\text{Spec } R$.

Indeed, choose $D(t) \subseteq U$ with

$$\mathfrak{m} \in D(t).$$

Then $t \notin \mathfrak{m}$. Since $\mathfrak{m}$ is the unique maximal ideal of R , it follows that t is a unit (単位, unit), i.e.

$$t \in R^\times.$$

Therefore

$$D(t) = \text{Spec } R.$$

So $U = \text{Spec } R$.

Applying this to $U = f^{-1}(D_+(x_i))$, we get

$$f^{-1}(D_+(x_i)) = \text{Spec } R,$$

hence

$$f(\text{Spec } R) \subseteq D_+(x_i).$$

Therefore f factors through $D_+(x_i)$, and so

$$\text{Hom}(\text{Spec } R, \mathbb{P}_A^n) = \bigcup_{i=0}^n \text{Hom}(\text{Spec } R, D_+(x_i)).$$

□

Lemma 2.2.29. *Let R be a local ring with maximal ideal $\mathfrak{m}$. If $U \subseteq \text{Spec } R$ is an open subset containing $\mathfrak{m}$, then*

$$U = \text{Spec } R.$$

Proof. Choose $D(t) \subseteq U$ such that

$$\mathfrak{m} \in D(t).$$

Then $t \notin \mathfrak{m}$, so $t \in R^\times$. Hence

$$D(t) = \text{Spec } R.$$

Therefore $U = \text{Spec } R$.

□

Proposition 2.2.30. *Let $A \rightarrow R$ be an A -algebra, and assume that R is a local ring (局部环, local ring) with maximal ideal $\mathfrak{m}$. Then there is a natural bijection*

$$\mathbb{P}_A^n(R) = \operatorname{Hom}_A(\operatorname{Spec} R, \mathbb{P}_A^n) \cong \{(a_0, \dots, a_n) \in R^{n+1} \mid \exists i, a_i \notin \mathfrak{m}\} / R^\times,$$

where $R^\times$ acts by simultaneous scaling

$$u \cdot (a_0, \dots, a_n) = (ua_0, \dots, ua_n), \quad u \in R^\times.$$

Proof. Since R is local, every morphism

$$f : \operatorname{Spec} R \rightarrow \mathbb{P}_A^n$$

factors through some standard open subset $D_+(x_i)$. Indeed,

$$\mathbb{P}_A^n = \bigcup_{i=0}^n D_+(x_i),$$

and if $\mathfrak{m} \in \operatorname{Spec} R$ is the unique closed point, then $f(\mathfrak{m}) \in D_+(x_i)$ for some i , hence

$$f(\operatorname{Spec} R) \subseteq D_+(x_i).$$

Therefore

$$\mathbb{P}_A^n(R) = \bigcup_{i=0}^n D_+(x_i)(R).$$

Now

$$D_+(x_i) \cong \operatorname{Spec} A \left[\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i} \right].$$

Hence an A -morphism

$$f : \operatorname{Spec} R \rightarrow D_+(x_i)$$

is equivalent to an A -algebra homomorphism

$$A \left[\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i} \right] \rightarrow R.$$

Such a homomorphism is uniquely determined by the images

$$b_{0,i}, \dots, b_{i-1,i}, b_{i+1,i}, \dots, b_{n,i} \in R,$$

where

$$b_{j,i} := f^\# \left(\frac{x_j}{x_i} \right).$$

It is convenient to insert $b_{i,i} = 1$, and write the resulting tuple as

$$(b_{0,i}, \dots, b_{i,i} = 1, \dots, b_{n,i}) \in R^{n+1}.$$

This suggests defining

$$\Phi := \{(a_0, \dots, a_n) \in R^{n+1} \mid \exists i, a_i \notin \mathfrak{m}\} / R^\times.$$

We now define a map

$$\Psi : \mathbb{P}_A^n(R) \rightarrow \Phi.$$

Given $f : \text{Spec } R \rightarrow \mathbb{P}_A^n$, choose i such that f factors through $D_+(x_i)$. Then define

$$\Psi(f) := [(b_{0,i}, \dots, b_{i,i} = 1, \dots, b_{n,i})].$$

We must check that this is well-defined, i.e. independent of the chosen index i .

Suppose f factors through both $D_+(x_i)$ and $D_+(x_j)$. Let

$$(b_{0,i}, \dots, b_{n,i}) \quad \text{and} \quad (c_{0,j}, \dots, c_{n,j})$$

be the tuples obtained from the i -th and j -th charts respectively, with

$$b_{i,i} = 1, \quad c_{j,j} = 1.$$

On the overlap $D_+(x_i) \cap D_+(x_j)$, the coordinate change is given by

$$\frac{x_\ell}{x_i} = \frac{x_\ell/x_j}{x_i/x_j}.$$

Applying $f^\sharp$, we obtain

$$b_{\ell,i} = \frac{c_{\ell,j}}{c_{i,j}} \quad \text{for all } \ell,$$

where $c_{i,j} \in R^\times$, since on the overlap $\frac{x_i}{x_j}$ is invertible. Therefore

$$(b_{0,i}, \dots, b_{n,i}) = c_{i,j}^{-1}(c_{0,j}, \dots, c_{n,j}),$$

so the two tuples define the same class in Φ . Thus Ψ is well-defined.

Next we show that Ψ is injective. Suppose

$$\Psi(f) = \Psi(g).$$

Choose a chart $D_+(x_i)$ through which both f and g factor. Then the corresponding tuples differ by multiplication by a unit $u \in R^\times$. After normalizing the i -th coordinate

to be 1, this scalar must be $u = 1$. Hence the affine coordinates on $D_+(x_i)$ agree, so the corresponding morphisms

$$\mathrm{Spec} R \rightarrow D_+(x_i)$$

are equal. Therefore $f = g$.

Finally we show that Ψ is surjective. Let

$$[(a_0, \dots, a_n)] \in \Phi,$$

so for some i we have $a_i \notin \mathfrak{m}$, hence $a_i \in R^\times$. Define an A -algebra homomorphism

$$A \left[\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i} \right] \rightarrow R$$

by sending

$$\frac{x_j}{x_i} \mapsto \frac{a_j}{a_i}.$$

This defines a morphism

$$f : \mathrm{Spec} R \rightarrow D_+(x_i) \subseteq \mathbb{P}_A^n,$$

and by construction

$$\Psi(f) = [(a_0, \dots, a_n)].$$

So Ψ is surjective.

Therefore Ψ is a bijection, and we obtain

$$\mathbb{P}_A^n(R) \cong \{(a_0, \dots, a_n) \in R^{n+1} \mid \exists i, a_i \notin \mathfrak{m}\} / R^\times.$$

□

Corollary 2.2.31. *If K is a field (域, field), then*

$$\mathbb{P}_A^n(K) \cong (K^{n+1} \setminus \{0\}) / K^\times.$$

In other words, the K -valued points (K -值点, K -valued points) of $\mathbb{P}_A^n$ are exactly the usual projective points

$$[a_0 : \dots : a_n], \quad (a_0, \dots, a_n) \in K^{n+1} \setminus \{0\}.$$

2.3 Properties of Schemes

2.3.1 Connected, irreducible, reduced, and integral

Definition 2.3.1. Let X be a scheme (概形, scheme).

- (1) We say that X is connected (连通的, connected) if the underlying topological space (底层拓扑空间, underlying topological space) of X is connected.
- (2) We say that X is irreducible (不可约的, irreducible) if the underlying topological space of X is irreducible. Equivalently, for any closed subsets (闭子集, closed subsets) $Z_1, Z_2 \subseteq X$, if

$$Z_1 \cup Z_2 = X,$$

then

$$Z_1 = X \quad \text{or} \quad Z_2 = X.$$

- (3) We say that X is reduced (约化的, reduced) if for every open subset (开子集, open subset) $U \subseteq X$, the ring

$$\mathcal{O}_X(U)$$

has no nonzero nilpotent element (非零幂零元, nonzero nilpotent element) .

- (4) We say that X is integral (整的, integral) if for every open subset $U \subseteq X$, the ring

$$\mathcal{O}_X(U)$$

is an integral domain (整环, integral domain) .

Remark 2.3.2. The irreducibility of X is equivalent to the following condition: for any two nonempty open subsets (非空开子集, nonempty open subsets) $U_1, U_2 \subseteq X$, one has

$$U_1 \cap U_2 \neq \emptyset.$$

Remark 2.3.3. The scheme X is reduced if and only if for every point $p \in X$, the stalk (茎, stalk)

$$\mathcal{O}_{X,p}$$

has no nonzero nilpotent element.

Example 2.3.4. Let

$$X = \operatorname{Spec} A.$$

Then the geometric properties of X can be translated into algebraic properties of A .

- (1) **Connectedness.**

If X is not connected, then

$$X = V(I) \sqcup V(J)$$

for some ideals $I, J \subseteq A$, where both $V(I)$ and $V(J)$ are nonempty and open-and-closed. Since

$$\mathcal{O}_X(X) \cong \mathcal{O}_X(V(I)) \times \mathcal{O}_X(V(J)),$$

we get

$$A \cong A/I \times A/J.$$

Hence A contains a nontrivial idempotent (幂等元, idempotent) :

$$e = (1, 0), \quad e^2 = e, \quad e \neq 0, 1.$$

Conversely, if A has a nontrivial idempotent e , then

$$A \cong A/(e) \times A/(1 - e),$$

so $\text{Spec } A$ is disconnected. Therefore

$$X \text{ is connected} \iff A \text{ has no nontrivial idempotent.}$$

(2) Irreducibility.

The space X is irreducible if and only if for any ideals $I, J \subseteq A$,

$$V(I) \cup V(J) = X \implies V(I) = X \text{ or } V(J) = X.$$

Since

$$V(I) \cup V(J) = V(IJ) = V(I \cap J),$$

this is equivalent to saying that for all ideals $I, J \subseteq A$,

$$I \cap J \subseteq \sqrt{(0)} \implies I \subseteq \sqrt{(0)} \text{ or } J \subseteq \sqrt{(0)}.$$

In other words,

$$X \text{ is irreducible} \iff \sqrt{(0)} = \text{nil}(A) \text{ is a prime ideal.}$$

(3) Reducedness.

The scheme X is reduced if and only if

$$\text{nil}(A) = 0.$$

One implication is immediate from the definition, since

$$\mathcal{O}_X(X) = A.$$

Conversely, assume $\text{nil}(A) = 0$. We show that every stalk $\mathcal{O}_{X,\mathfrak{p}} = A_{\mathfrak{p}}$ is reduced. Take

$$\frac{a}{f} \in \text{nil}(A_{\mathfrak{p}}), \quad f \notin \mathfrak{p}.$$

Then

$$\left(\frac{a}{f}\right)^n = 0$$

for some n , so there exists $b \notin \mathfrak{p}$ such that

$$ba^n = 0.$$

Hence

$$(ba)^n = 0.$$

Since $\text{nil}(A) = 0$, we get

$$ba = 0.$$

Therefore

$$\frac{a}{f} = 0 \quad \text{in } A_{\mathfrak{p}}.$$

So $A_{\mathfrak{p}}$ is reduced for every $\mathfrak{p}$, and thus X is reduced.

(4) Integrality.

If X is integral, then by definition every ring $\mathcal{O}_X(U)$ is an integral domain (整环, integral domain) ; in particular,

$$A = \mathcal{O}_X(X)$$

is an integral domain.

Conversely, assume that A is an integral domain. Let $U \subseteq X = \text{Spec } A$ be a nonempty open subset. We show that

$$\mathcal{O}_X(U)$$

is also an integral domain.

First note that if $U = D(f)$, then

$$\mathcal{O}_X(U) = A_f,$$

and since A is a domain, A_f is also a domain.

For a general nonempty open subset U , choose a cover

$$U = \bigcup_i D(f_i).$$

Since $U \neq \emptyset$, at least one $D(f_i)$ is nonempty. The restriction maps give an injective map

$$\mathcal{O}_X(U) \hookrightarrow \prod_i \mathcal{O}_X(D(f_i)) = \prod_i A_{f_i}.$$

Each A_{f_i} embeds into the fraction field (分式域, fraction field)

$$\text{Frac}(A),$$

so $\mathcal{O}_X(U)$ embeds into $\text{Frac}(A)$. Therefore $\mathcal{O}_X(U)$ is a subring of a field, hence an integral domain.

Thus

$$X \text{ is integral} \iff A \text{ is an integral domain.}$$

Proposition 2.3.5. *A scheme X is integral (整的, integral) if and only if X is reduced (约化的, reduced) and irreducible (不可约的, irreducible).*

Proof. Assume first that X is integral.

For every open subset $U \subseteq X$, the ring

$$\mathcal{O}_X(U)$$

is an integral domain (整环, integral domain). Hence $\mathcal{O}_X(U)$ has no nonzero nilpotent elements, so X is reduced.

We now show that X is irreducible. Suppose not. Then there exist two nonempty open subsets

$$U_1, U_2 \subseteq X$$

such that

$$U_1 \cap U_2 = \emptyset.$$

Then

$$\mathcal{O}_X(U_1 \cup U_2) \cong \mathcal{O}_X(U_1) \times \mathcal{O}_X(U_2),$$

because U_1 and U_2 are disjoint. Since U_1 and U_2 are both nonempty, the two rings on the right are both nonzero. Therefore

$$\mathcal{O}_X(U_1 \cup U_2)$$

is not an integral domain, contradiction. Thus X is irreducible.

Conversely, assume that X is reduced and irreducible. We must show that for every open subset $U \subseteq X$, the ring

$$\mathcal{O}_X(U)$$

is an integral domain.

Let $U \subseteq X$ be open, and let

$$f, g \in \mathcal{O}_X(U)$$

satisfy

$$fg = 0.$$

Define

$$Y := \{x \in U \mid f_x \in \mathfrak{m}_x\}, \quad Z := \{x \in U \mid g_x \in \mathfrak{m}_x\},$$

where $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is the maximal ideal of the local ring (局部环, local ring) $\mathcal{O}_{X,x}$.

We claim that Y and Z are closed subsets of U . Indeed,

$$U \setminus Y = \{x \in U \mid f_x \notin \mathfrak{m}_x\} = \{x \in U \mid f_x \text{ is invertible in } \mathcal{O}_{X,x}\},$$

which is open; similarly $U \setminus Z$ is open.

Moreover,

$$Y \cup Z = U.$$

Indeed, for every $x \in U$, since

$$f_x g_x = 0 \quad \text{in } \mathcal{O}_{X,x},$$

and $\mathcal{O}_{X,x}$ is a local ring, at least one of f_x, g_x must lie in the maximal ideal $\mathfrak{m}_x$. Hence $x \in Y \cup Z$.

Since X is irreducible, every nonempty open subset of X is also irreducible, so U is irreducible. Therefore

$$Y = U \quad \text{or} \quad Z = U.$$

Assume for instance that

$$Y = U.$$

Then for every $x \in U$, we have

$$f_x \in \mathfrak{m}_x.$$

Now choose an affine open cover

$$U = \bigcup_i \text{Spec } A_i.$$

For each i , the restriction $f|_{\text{Spec } A_i}$ corresponds to an element of A_i , still denoted by f . The condition

$$f_x \in \mathfrak{m}_x \quad \text{for all } x \in \text{Spec } A_i$$

means that f belongs to every prime ideal of A_i . Hence

$$f \in \bigcap_{\mathfrak{p} \in \text{Spec } A_i} \mathfrak{p} = \text{nil}(A_i).$$

Since X is reduced, each A_i is reduced, so

$$\text{nil}(A_i) = 0.$$

Therefore

$$f|_{\text{Spec } A_i} = 0 \quad \text{for all } i.$$

Thus $f = 0$ in $\mathcal{O}_X(U)$.

We have shown that whenever $fg = 0$, one has $f = 0$ or $g = 0$. Hence

$$\mathcal{O}_X(U)$$

is an integral domain. Therefore X is integral. $\square$

Lemma 2.3.6. *Let A be an integral domain, and let $X = \text{Spec } A$. For every nonempty open subset $U \subseteq X$, there is a natural injective map*

$$\mathcal{O}_X(U) \hookrightarrow \text{Frac}(A).$$

In particular, $\mathcal{O}_X(U)$ is an integral domain.

Proof. If $U = D(f)$, then

$$\mathcal{O}_X(U) = A_f \subseteq \text{Frac}(A).$$

For a general nonempty open subset U , choose a cover

$$U = \bigcup_i D(f_i).$$

Then

$$\mathcal{O}_X(U) \hookrightarrow \prod_i \mathcal{O}_X(D(f_i)) = \prod_i A_{f_i}.$$

Each A_{f_i} embeds into $\text{Frac}(A)$, hence we obtain an induced injective map

$$\mathcal{O}_X(U) \hookrightarrow \prod_i \text{Frac}(A).$$

By compatibility of restrictions on overlaps, the image lies in the diagonal copy of $\text{Frac}(A)$. Therefore

$$\mathcal{O}_X(U) \subseteq \text{Frac}(A).$$

$\square$

$\text{spec } A$ is integral iff A integral domain

2.4 Locally Noetherian Schemes

2.4.1 Definitions

Definition 2.4.1 (Locally Noetherian / 局部诺特). X is *locally Noetherian* if there exists an affine cover $\bigcup_i \text{Spec } A_i = X$ such that each A_i is a Noetherian ring.

Definition 2.4.2 (Noetherian / 诺特). X is *Noetherian* if it is locally Noetherian and quasi-compact.

Corollary 2.4.3. X is Noetherian $\iff$ there exists a finite affine cover $\bigcup_{i \in J} \text{Spec } A_i = X$ with $|J| < \infty$ and each A_i Noetherian.

2.4.2 Noetherianness of the underlying topological space

Definition 2.4.4 (Noetherian topological space / 诺特拓扑空间). A topological space X is *Noetherian* if for every descending sequence of closed subsets

$$X \supseteq X_1 \supseteq X_2 \supseteq \cdots,$$

there exists r such that $X_r = X_{r+1} = \cdots$.

Proposition 2.4.5. If a scheme X is Noetherian, then its underlying topological space is Noetherian.

Proof. Write $X = \bigcup_{i \in J} \text{Spec } A_i$ with $|J| < \infty$ and each A_i Noetherian. Given a descending chain of closed subsets

$$X \supseteq X_1 \supseteq X_2 \supseteq \cdots,$$

intersecting with each affine piece gives

$$\text{Spec } A_i \supseteq \text{Spec } A_i \cap X_1 \supseteq \text{Spec } A_i \cap X_2 \supseteq \cdots,$$

where $\text{Spec } A_i \cap X_n = V(I_{n,i})$ for some ideal $I_{n,i} \subseteq A_i$. This corresponds to an ascending chain of ideals

$$I_{1,i} \subseteq I_{2,i} \subseteq \cdots$$

in the Noetherian ring A_i , so there exists r_i such that $I_{r_i,i} = I_{r_i+1,i} = \cdots$. Since $|J| < \infty$, take $r = \max_i r_i$; then

$$X_r \cap \text{Spec } A_i = X_{r+1} \cap \text{Spec } A_i = \cdots \quad \forall i \in J,$$

hence $X_r = X_{r+1} = \cdots$ as topological spaces. □

2.4.3 Noetherianness is affine-local

Proposition 2.4.6. X is locally Noetherian $\iff$ for every affine open $\text{Spec } A \subseteq X$, the ring A is Noetherian.

Corollary 2.4.7. $X = \text{Spec } A$ is Noetherian $\iff A$ is a Noetherian ring.

Remark 2.4.8 (Warning / 警告). If $U \subseteq X$ is not affine, it is possible that $\mathcal{O}_X(U)$ is not Noetherian, even when X itself is Noetherian.

Proof of Proposition 2.4.6. “ $\Leftarrow$ ” Trivial: apply the hypothesis to an affine cover.

“ $\Rightarrow$ ” Write $X = \bigcup_i \text{Spec } A_i$ with each A_i Noetherian.

Claim (intersection of two affine opens). Let $U = \text{Spec } A \subseteq X \supseteq \text{Spec } B = V$. Then

$$U \cap V = \bigcup_i W_i, \quad \text{with } W_i = D(a_i) \subseteq U \text{ and } W_i = D(b_i) \subseteq V,$$

for suitable $a_i \in A, b_i \in B$.

Proof of claim. Cover $U \cap V = \bigcup_j D(\tilde{a}_j)$ with $\tilde{a}_j \in A$, and for each j further refine

$$D(\tilde{a}_j) = \bigcup_k D(\widetilde{b_{jk}}), \quad \widetilde{b_{jk}} \in B.$$

Then $D(\widetilde{b_{jk}}) = \text{Spec } B_{\widetilde{b_{jk}}}$. On the other hand, $D(\tilde{a}_j) \subseteq \text{Spec } B$ induces a ring map

$$\begin{array}{ccc} \mathcal{O}_X(\text{Spec } B) & \xrightarrow{\phi} & \mathcal{O}_X(D(\tilde{a}_j)) \\ \parallel & & \parallel \\ B & & A_{\tilde{a}_j} \end{array}$$

so

$$D(\widetilde{b_{jk}}) \subseteq \text{Spec } A_{\tilde{a}_j} \text{ becomes } D(\phi(\widetilde{b_{jk}})) \subseteq \text{Spec } A_{\tilde{a}_j},$$

i.e.

$$D(\widetilde{b_{jk}}) \cong \text{Spec}((A_{\tilde{a}_j})_{\phi(\widetilde{b_{jk}})}), \quad \phi(\widetilde{b_{jk}}) = \frac{t}{\widetilde{a_j}^n}, \quad t \in A.$$

So $(A_{\tilde{a}_j})_{\phi(\widetilde{b_{jk}})} = A_{t \cdot \tilde{a_j}}$, giving the desired form $D(a') \subseteq U$ and $D(b') \subseteq V$ simultaneously. □_{claim}

Recall: a ring map $B \xrightarrow{\phi} A$ induces $\text{Spec } A \rightarrow \text{Spec } B$, with $D(f) \xleftarrow{\phi} D(\phi(f))$.

Main step. Let $U = \text{Spec } A \subseteq X$ be any affine open; we show A is Noetherian. By the claim,

$$U = \text{Spec } A = \bigcup_i (U \cap \text{Spec } A_i) = \bigcup_i \bigcup_j \text{Spec } A_{a_{ij}},$$

where $\text{Spec } A_{a_{ij}} = \text{Spec}(A_i)_{b_j}$. Since A_i is Noetherian, so is $(A_i)_{b_j}$, hence each $A_{a_{ij}}$ is Noetherian.

Since $\text{Spec } A$ is quasi-compact, finitely many $f_1, \dots, f_m \in A$ suffice:

$$\text{Spec } A = \bigcup_{i=1}^m D(f_i), \quad A_{f_i} \text{ Noetherian.}$$

It remains to show that A itself is Noetherian. For any ascending chain $I_1 \subseteq I_2 \subseteq \dots$ in A , since each A_{f_i} is Noetherian, there exists a common r such that

$$I_r \cdot A_{f_i} = I_{r+1} \cdot A_{f_i} = \dots, \quad \forall 1 \leq i \leq m.$$

That is, for any $x \in I_{r+M}$, there exists $y_i \in I_r$ with

$$\frac{y_i}{f_i^{m_i}} = x \quad \text{in } A_{f_i},$$

i.e. in A ,

$$f_i^{\tilde{m}_i} \cdot x = f_i^{\tilde{m}_i - m_i} \cdot y_i \quad (\text{since } I_r A_{f_i} = I_{r+M} A_{f_i}).$$

Since $\text{Spec } A = \bigcup_i D(f_i) = \bigcup_i D(f_i^{\tilde{m}_i})$, we get

$$1_A \in (f_1^{\tilde{m}_1}, \dots, f_m^{\tilde{m}_m}),$$

i.e. $1_A = \sum_i b_i f_i^{\tilde{m}_i}$ for some $b_i \in A$. Therefore

$$x = 1_A \cdot x = \sum_{i=1}^m b_i f_i^{\tilde{m}_i} x = \sum_{i=1}^m b_i \underbrace{f_i^{\tilde{m}_i - m_i} y_i}_{\in I_r} \in I_r.$$

Hence $I_{r+M} = I_r$ for all $M > 0$, so A is Noetherian. $\square$

2.4.4 Examples

Example 2.4.9.

1. $A = k[x_1, \dots, x_n]/I$ is a Noetherian ring (by Hilbert's basis theorem / 希尔伯特基定理), so $\text{Spec } A$ is a Noetherian scheme.
2. Let $S = k[x_0, \dots, x_n]/I$ with I a homogeneous ideal (齐次理想). Then

$$\text{Proj } S = \bigcup_{i=0}^n \underbrace{D_+(x_i)}_{\text{Spec } S_{(x_i)} \text{ Noetherian}},$$

since $(x_0, \dots, x_n) = S_+$ (the irrelevant ideal / 无关理想). Hence $\text{Proj } S$ is a Noetherian scheme.

2.5 Morphisms of Finite Type

2.5.1 Definitions

Definition 2.5.1. Let $f: X \rightarrow Y$ be a morphism of schemes (概形态射).

1. f is *locally of finite type* / 局部有限型 if there exists an affine cover $Y = \bigcup_i V_i$ with $V_i = \text{Spec } B_i$, such that $f^{-1}(V_i) = \bigcup_j U_{ij}$ with $U_{ij} = \text{Spec } A_{ij}$, where each A_{ij} is a finitely generated B_i -algebra (有限生成代数), i.e. there exists N and a surjection

$$B_i[x_1, \dots, x_N] \twoheadrightarrow A_{ij}.$$

2. f is *of finite type* / 有限型 if it is locally of finite type and each $f^{-1}(V_i)$ is quasi-compact.
3. f is *affine* / 仿射态射 if there exists an affine cover $Y = \bigcup_i V_i$, $V_i = \text{Spec } B_i$, such that $f^{-1}(V_i)$ is affine.
4. f is *finite* / 有限态射 if it is affine and $f^{-1}(V_i) = \text{Spec } A_i$ with A_i a finitely generated B_i -module (有限生成模), i.e. there exists N and a surjection

$$B_i^{\oplus N} \twoheadrightarrow A_i \quad \text{as } B_i\text{-modules.}$$

2.5.2 Affine-local nature of each property

Remark 2.5.2. As with local Noetherianness, each of the four properties in Definition 2.5.1 (locally of finite type, finite type, affine, finite) satisfies

$$\iff \text{for every affine open } V \subseteq Y, f^{-1}(V) \text{ has the same property.}$$

(See Hartshorne, Exercises 3.1–3.4.)

2.5.3 Examples

Example 2.5.3 (Transfer of finiteness to the source). Let $f: X \rightarrow Y$ with Y Noetherian. Then

$$\begin{aligned} f \text{ locally of finite type} &\implies X \text{ locally Noetherian,} \\ f \text{ of finite type} &\implies X \text{ Noetherian.} \end{aligned}$$

Example 2.5.4 (Finite type $\not\Rightarrow$ finite). Take $f: X = \mathbb{A}_k^1 = \text{Spec } k[t] \rightarrow \text{Spec } k$: of finite type but not finite.

At any $p \in X$, the morphism

$$\mathrm{Spec} \mathcal{O}_{X,p} \longrightarrow \mathrm{Spec} k$$

is *not* locally of finite type. (E.g. at $p = (t)$, this is $\mathrm{Spec} k[t]_{(t)} \rightarrow \mathrm{Spec} k$.)

Example 2.5.5. $\mathrm{Spec} A[x_1, \dots, x_n]/I \longrightarrow \mathrm{Spec} A$ is of finite type.

2.6 Open and Closed Subschemes; Dimension

2.6.1 Open subschemes and open immersions

Definition 2.6.1. An *open subscheme* / 开子概形 of X is $(U, \mathcal{O}_X|_U) \subseteq (X, \mathcal{O}_X)$ where $U \subseteq X$ is open.

An *open immersion* / 开浸入 is a morphism $f: Y \rightarrow X$ such that there exists an open $U \subseteq X$ for which f induces an isomorphism

$$(Y, \mathcal{O}_Y) \xrightarrow[f]{} (U, \mathcal{O}_X|_U).$$

2.6.2 Closed subschemes and closed immersions

Definition 2.6.2. A *closed immersion* / 闭浸入 is a morphism $f: X \rightarrow Y$ such that

- (i) f is a homeomorphism onto a closed subset of Y (closed immersion of topological spaces);
- (ii) $f^\#: \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ is surjective.

A *closed subscheme* / 闭子概形 $X \subseteq Y$ is the data of a closed immersion $f: X \hookrightarrow Y$.

The basic example: $\mathrm{Spec} A/I \hookrightarrow \mathrm{Spec} A$

Example 2.6.3. Let $Y = \mathrm{Spec} A$, $X = \mathrm{Spec} A/I$ for an ideal $I \subseteq A$, and $f: X \rightarrow Y$ the natural map.

Then f is a closed immersion of topological spaces with image $V(I)$. For the sheaf condition, we need

$$\mathcal{O}_{Y,y} \longrightarrow (f_*\mathcal{O}_X)_y \quad \text{surjective for all } y \in Y,$$

and it suffices to check $y \in X \subseteq Y$ (outside the image the stalk of $f_*\mathcal{O}_X$ is zero).

For $\mathfrak{p} \in X$ (so $I \subseteq \mathfrak{p}$), the stalk map is

$$A_{\mathfrak{p}} \twoheadrightarrow (A/I)_{\mathfrak{p}},$$

which is surjective.

Every closed immersion into an affine is affine

Example 2.6.4 (Hartshorne, Ex. 3.11). Closed immersion $\implies$ affine: if $Y \subseteq \operatorname{Spec} A$ is a closed subscheme, then there exists an ideal $I \subseteq A$ such that

$$Y \cong \operatorname{Spec} A/I \subseteq \operatorname{Spec} A.$$

Non-uniqueness of closed subscheme structures

Example 2.6.5. Let $Z \subseteq Y$ be a closed subset. There can be many closed subscheme structures on Z .

For example, $\{0\} \subseteq \mathbb{A}^1 = \operatorname{Spec} k[t]$ admits the closed subscheme structures

$$\operatorname{Spec} k[t]/(t^n), \quad n \geq 1.$$

However, the *reduced* / 既约 closed subscheme structure on Z is unique — making Z a reduced scheme (既约概形). (Hartshorne, Ex. 3.11.)

2.6.3 Dimension**Definition and Krull dimension**

Definition 2.6.6 (Dimension / 维数). Let X be a scheme. Define

$$\dim X := \sup_n \{ Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \subseteq X \},$$

where each Z_i is an irreducible closed subset (不可约闭子集) of X .

Example 2.6.7. For $X = \operatorname{Spec} A$,

$$\dim X = \text{Krull dimension of } A \text{ (克鲁尔维数)} := \sup_n \{ \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n \subseteq A \},$$

where each $\mathfrak{p}_i$ is a prime ideal.

Theorem 2.6.8. Let k be a field. Then $\operatorname{Krull dim} k[x_1, \dots, x_n] = n$, and hence

$$\dim \mathbb{A}_k^n = n.$$

Dimension of fibers of a dominant morphism

Theorem 2.6.9 (Hartshorne, Ex. 3.22). Let $f: X \rightarrow Y$ be a dominant morphism (优势态射) with X, Y integral and of finite type over a field k . Then Y has a unique generic point (一般点) $\eta_Y \in Y$, and $f^{-1}(\eta_Y) \neq \emptyset$. Moreover,

$$\dim X \leq \dim Y + \dim f^{-1}(y)$$

for every y with $f^{-1}(y) \neq \emptyset$, and

$$\dim X = \dim Y + \dim f^{-1}(\eta_Y).$$

Recall: Y is *integral* / 整 $\iff$

- (1) any two nonempty opens $U, V \subseteq Y$ satisfy $U \cap V \neq \emptyset$ (irreducibility);
- (2) for every affine open $V = \operatorname{Spec} A \subseteq Y$, A is an integral domain, with $(0) \rightsquigarrow \eta_Y \in Y$ the generic point.

2.7 Fibre Product

Definition 2.7.1 (Fibre Product 纤维积). Let X, Y, S be schemes with morphisms $\pi_1 : X \rightarrow S$ and $\pi_2 : Y \rightarrow S$. The *fibre product* $X \times_S Y$ is the fibred product in the category of schemes, i.e. a scheme $X \times_S Y$ together with morphisms p_1, p_2 such that

1. $\pi_1 \circ p_1 = \pi_2 \circ p_2$;
2. for any scheme Z with morphisms $q_1 : Z \rightarrow X, q_2 : Z \rightarrow Y$ satisfying $\pi_1 \circ q_1 = \pi_2 \circ q_2$, there exists a unique $q : Z \rightarrow X \times_S Y$ such that $p_1 \circ q = q_1$ and $p_2 \circ q = q_2$.

$$\begin{array}{ccccc}
 & & & q_2 & \\
 & & & \searrow & \\
 Z & & & & Y \\
 & \searrow \exists! q & & & \uparrow p_2 \\
 & & X \times_S Y & \xrightarrow{\quad} & Y \\
 & & \downarrow p_1 & & \downarrow \pi_2 \\
 & & X & \xrightarrow{\quad \pi_1 \quad} & S \\
 & \swarrow q_1 & & &
 \end{array}$$

Remark 2.7.2 (Functor of points 函子观点). Define $h : \mathbf{Sch} \rightarrow \mathbf{Set}$ by

$$h(Z) := \operatorname{Hom}(Z, X) \times_{\operatorname{Hom}(Z, S)} \operatorname{Hom}(Z, Y) = \left\{ (q_1, q_2) \left| \begin{array}{l} q_1 \in \operatorname{Hom}(Z, X) \\ q_2 \in \operatorname{Hom}(Z, Y) \end{array} \right. , \pi_1 \circ q_1 = \pi_2 \circ q_2 \right\}.$$

The fibre product exists $\iff h$ is representable, i.e. there exists $X \times_S Y$ together with a natural isomorphism

$$\operatorname{Hom}(Z, X \times_S Y) \xrightarrow{\sim} h(Z) \quad \forall Z.$$

Theorem 2.7.3. $X \times_S Y$ exists and is unique.

Proof. Uniqueness. Suppose T also satisfies the same universal property. Then by the universal property applied both ways, there exist unique morphisms

$$p : T \rightarrow X \times_S Y, \quad q : X \times_S Y \rightarrow T$$

compatible with the projections. Then $q \circ p : T \rightarrow T$ and $p \circ q : X \times_S Y \rightarrow X \times_S Y$ both satisfy

$$(q \circ p) \circ (\text{proj}) = (\text{proj}), \quad (p \circ q) \circ p_1 = p_1, \quad (p \circ q) \circ p_2 = p_2,$$

so by uniqueness in the universal property, $q \circ p = \text{id}_T$ and $p \circ q = \text{id}_{X \times_S Y}$.

Existence.

Step 1. X, Y, S are affine. Write $X = \text{Spec } A$, $Y = \text{Spec } B$, $S = \text{Spec } R$, with ring maps $R \rightarrow A$ and $R \rightarrow B$. For any scheme Z ,

$$\begin{aligned} h(Z) &:= \text{Hom}(Z, X) \times_{\text{Hom}(Z, S)} \text{Hom}(Z, Y) \\ &\cong \text{Hom}_{\text{Ring}}(A, \mathcal{O}_Z(Z)) \times_{\text{Hom}_{\text{Ring}}(R, \mathcal{O}_Z(Z))} \text{Hom}_{\text{Ring}}(B, \mathcal{O}_Z(Z)) \\ &\stackrel{\text{check}}{=} \text{Hom}_{\text{Ring}}(A \otimes_R B, \mathcal{O}_Z(Z)) \\ &= \text{Hom}(Z, \text{Spec}(A \otimes_R B)). \end{aligned}$$

Hence

$$X \times_S Y = \text{Spec}(A \otimes_R B).$$

Check ($\Leftarrow$). Given $f : A \rightarrow \mathcal{O}_Z(Z)$ and $g : B \rightarrow \mathcal{O}_Z(Z)$ agreeing on R , the map

$$f \cdot g : A \times B \rightarrow \mathcal{O}_Z(Z), \quad (a, b) \mapsto f(a)g(b)$$

is bilinear, hence induces $A \otimes_R B \rightarrow \mathcal{O}_Z(Z)$.

Check ($\Rightarrow$). Given $q^\# : A \otimes_R B \rightarrow \mathcal{O}_Z(Z)$, the diagram

$$\begin{array}{ccc} R & \longrightarrow & A \\ \downarrow & & \downarrow p_1^\# \\ B & \xrightarrow[p_2^\#]{} & A \otimes_R B \end{array} \quad \begin{array}{c} a \\ \downarrow \\ a \otimes 1 \end{array}$$

$$b \longmapsto 1 \otimes b$$

gives induced maps

$$q^\# \circ p_1^\# : A \rightarrow \mathcal{O}_Z(Z), \quad q^\# \circ p_2^\# : B \rightarrow \mathcal{O}_Z(Z),$$

which agree on R . These two constructions are mutually inverse, establishing the bijection.

Step 3. Gluing morphisms. Let X, Y be schemes, $X = \bigcup_i U_i$ an open cover, and $f_i : U_i \rightarrow Y$ morphisms with

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}.$$

Then there exists a unique morphism $f : X \rightarrow Y$ such that $f|_{U_i} = f_i$.

Proof. (Exercise: gluing of sheaves; gluing of schemes —Hartshorne, Ex. 2.12.) $\square$

Step 4. Let $X \rightarrow S \leftarrow Y$ and $U \subseteq X$ open. If $X \times_S Y$ exists, then $U \times_S Y$ exists, and

$$U \times_S Y = p_1^{-1}(U) \subseteq_{\text{open}} X \times_S Y.$$

Proof.

$$\begin{array}{ccccc}
 & & & q_2 & \\
 & & & \curvearrowright & \\
 Z & & & & \\
 & \searrow & & & \\
 & p_1^{-1}(U) & \hookrightarrow & X \times_S Y & \xrightarrow{p_2} & Y \\
 & \downarrow & & \downarrow p_1 & & \downarrow \pi_2 \\
 & U & \xrightarrow{\text{open}} & X & \xrightarrow{\pi_1} & S \\
 & \nearrow q_1 & & & &
 \end{array}$$

We only need to show the universal property of $p_1^{-1}(U)$. For any Z with $q_1 : Z \rightarrow U$, $q_2 : Z \rightarrow Y$ satisfying $q_2 \circ \pi_2 = q_1 \circ \pi_1$ (after composing with $U \hookrightarrow X$), the universal property of $X \times_S Y$ gives a unique $q : Z \rightarrow X \times_S Y$ with $p_1 \circ q = q_1$, $p_2 \circ q = q_2$. Since $\text{Im}(p_1 \circ q) \subseteq U$, we get $\text{Im}(q) \subseteq p_1^{-1}(U)$, i.e. there exists a unique $\tilde{q} : Z \rightarrow p_1^{-1}(U)$ with $i \circ \tilde{q} = q$, where $i : p_1^{-1}(U) \hookrightarrow X \times_S Y$.

Topologically: $\text{Im } q \subseteq p_1^{-1}(U)$, so there exists $\tilde{q} : Z \rightarrow p_1^{-1}(U)$ as a continuous map.

On sheaves: $q^\# : q^{-1}\mathcal{O}_{X \times_S Y} \rightarrow \mathcal{O}_Z$ becomes

$$\tilde{q}^\# : \tilde{q}^{-1}(\mathcal{O}_{X \times_S Y}|_{p_1^{-1}(U)}) \rightarrow \mathcal{O}_Z. \quad \square$$

Step 5. Let $X \rightarrow S \leftarrow Y$ with $X = \bigcup_i U_i$ an open cover. If $U_i \times_S Y$ exists for all i , then $X \times_S Y$ exists.

Proof. Let $U_{ij} := U_i \cap U_j$. By Step 4, inside $U_i \times_S Y$ we have an open subscheme

$$W_{ij} := p_1^{-1}(U_{ij}) \cong U_{ij} \times_S Y \subseteq U_i \times_S Y.$$

Since $U_{ij} = U_{ji}$, uniqueness of fibre products gives a canonical isomorphism

$$\begin{array}{ccc}
 \phi_{ij} : W_{ij} & \xrightarrow{\sim} & W_{ji} \\
 \parallel & & \parallel \\
 U_{ij} \times_S Y & & U_{ji} \times_S Y
 \end{array}$$

Uniqueness also forces the cocycle condition $\phi_{jk} \circ \phi_{ij} = \phi_{ik}$ on triple overlaps. By gluing of schemes we obtain

$$X \times_S Y := \bigcup_i U_i \times_S Y.$$

Universal property. For any Z with morphisms $f : Z \rightarrow X, g : Z \rightarrow Y$ over S , let $Z_i := f^{-1}(U_i)$. Then $f|_{Z_i} = f_i : Z_i \rightarrow U_i$ and $g|_{Z_i} : Z_i \rightarrow Y$ give, by the universal property of $U_i \times_S Y$, a unique

$$\theta_i : Z_i \rightarrow U_i \times_S Y.$$

On overlaps $Z_i \cap Z_j$, both $\theta_i|_{Z_i \cap Z_j}$ and $\theta_j|_{Z_i \cap Z_j}$ map into $(U_i \cap U_j) \times_S Y$ and satisfy the same universal property, so by uniqueness

$$\theta_i|_{Z_i \cap Z_j} = \theta_j|_{Z_i \cap Z_j}.$$

By gluing morphisms (Step 3), the θ_i glue to a unique morphism $\theta : Z \rightarrow X \times_S Y$. $\square$

$\square$

Example 2.7.4. 设 $f : X \rightarrow S$ 为态射, $U \subseteq S$ 为开子概形 (open subscheme), 记 $\phi : U \hookrightarrow S$ 为开浸入. 则

$$X \times_S U = f^{-1}(U) \subseteq X.$$

Proof. 验证 $f^{-1}(U)$ 满足 $X \times_S U$ 的泛性质. 设 Z 为任一概形, 配有态射 $g : Z \rightarrow X$ 与 $h : Z \rightarrow U$, 使得下图交换:

$$\begin{array}{ccccc} & & h & & \\ & \searrow & \curvearrowright & \searrow & \\ Z & \dashrightarrow & f^{-1}(U) & \longrightarrow & U \\ & \searrow g & \downarrow & & \downarrow \phi \\ & & X & \xrightarrow{f} & S \end{array}$$

由 $f \circ g = \phi \circ h$ 知 $\text{Im}(f \circ g) \subseteq U$, 故

$$\text{Im}(g) \subseteq f^{-1}(U).$$

因此 g 唯一地分解过 $f^{-1}(U)$, 这正是纤维积的泛性质. $\square$

Definition 2.7.5 (纤维 Fibre). 设 $f : X \rightarrow Y$ 为概形态射. 对 $y \in Y$, 定义其剩余域 (residue field)

$$k(y) := \mathcal{O}_{Y,y}/\mathfrak{m}_y.$$

自然态射

$$y: \operatorname{Spec} k(y) \longrightarrow \operatorname{Spec} \mathcal{O}_{Y,y} \longrightarrow Y$$

将 $\operatorname{Spec} k(y)$ 视为 Y 中“点 y ”的概形实现. 局部地, 取 y 的仿射邻域 $V = \operatorname{Spec} A \subseteq Y$, 对应素理想 $\mathfrak{p} \subseteq A$, 上述态射对应于

$$\operatorname{Spec} A_{\mathfrak{p}} \longrightarrow \operatorname{Spec} A, \quad A_{\mathfrak{p}} \longleftarrow A, \quad y \leftrightarrow \mathfrak{p}.$$

f 在 y 处的纤维 (fibre of f over y) 定义为

$$X_y := X \times_Y \operatorname{Spec} k(y).$$

Remark 2.7.6. 作为拓扑空间, 有典范同胚 (homeomorphism)

$$X_y \xrightarrow{\sim} f^{-1}(y).$$

Example 2.7.7 (A family whose fibres change dimension / 纤维维数变化的族). Consider the morphism

$$f: X = \operatorname{Spec} \frac{k[x, y, t]}{(ty - x^2)} \longrightarrow Y = \operatorname{Spec} k[t], \quad t \longmapsto t.$$

Write $A = k[x, y, t]/(ty - x^2)$. Geometrically, X is the family of conics $ty = x^2$ parametrised by t , and f records the parameter.

Before computing fibres, recall two elementary tensor-product identities that will be used repeatedly:

$$A \otimes_R B/I \simeq (A \otimes_R B)/(I), \quad A \otimes_k k[y] \simeq A[y].$$

(i) The special fibre over $t = 0$. The residue field at the closed point $t = 0 \in Y$ is $k(t = 0) = k[t]/(t) = k$, so

$$\begin{aligned} X_{(t=0)} &\simeq \operatorname{Spec}(A \otimes_{k[t]} k[t]/(t)) = \operatorname{Spec}(A/(t)) \\ &= \operatorname{Spec} \frac{k[x, y, t]}{(ty - x^2, t)} = \operatorname{Spec} \frac{k[x, y]}{(x^2)} \\ &\simeq \operatorname{Spec} k[y] \times_k \operatorname{Spec} k[x]/(x^2). \end{aligned}$$

This fibre is *non-reduced*: the nilpotent $\bar{x}$ satisfies $\bar{x}^2 = 0$. Set-theoretically it is the line $\{x = 0\}$, but it carries a “thickening” in the x -direction—a *double line* (双线).

几何 $\leftrightarrow$ 代数

集合上, $t = 0$ 处的纤维只是 y 轴 $\{x = 0\}$; 但代数上方程是 $x^2 = 0$ 而非 $x = 0$ 。结构层 (structure sheaf) 中的 $\bar{x}$ 是 **幂零元 (nilpotent element)**: $\bar{x} \neq 0$ 但 $\bar{x}^2 = 0$ 。这条“看上去普通”的直线因此带 **重数 (multiplicity) 2**, 称为**双线 (double line)**——它“记得”自己是从一族抛物线退化而来的。

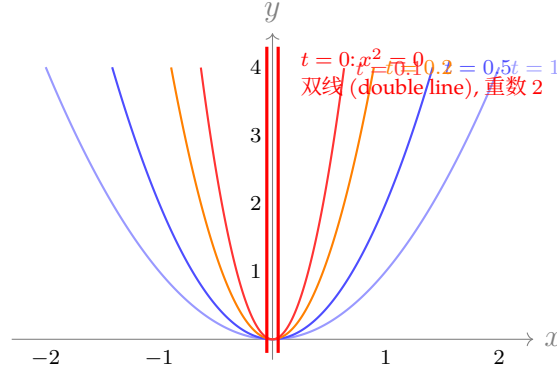


Figure 2.1: $X = \text{Spec } k[x, y, t]/(ty - x^2) \rightarrow \mathbb{A}_k^1$ 的纤维族: $t \neq 0$ 时是抛物线 $y = x^2/t$; $t \rightarrow 0$ 时抛物线“压扁”到 y 轴上, 得到带重数 2 的双线 (double line)。

(ii) **The fibre over a closed point $t = a \neq 0, a \in k$.** Here $k(t = a) = k[t]/(t - a) \simeq k$, and

$$\begin{aligned} X_{(t=a)} &\simeq \text{Spec}(A \otimes_{k[t]} k[t]/(t - a)) = \text{Spec} \frac{k[x, y, t]/(ty - x^2)}{(t - a)} \\ &= \text{Spec} \frac{k[x, y]}{(ay - x^2)} = \text{Spec } k[x] \cong \mathbb{A}_k^1, \end{aligned}$$

since $a \neq 0$ lets us solve $y = x^2/a$ and eliminate y . So every fibre over $a \neq 0$ is a smooth affine line.

(iii) **The fibre over the generic point (and other non-closed points).** Let $\mathfrak{p} \subseteq k[t]$ be a non-zero prime, corresponding to an irreducible polynomial $p(t) \in k[t]$. Then $k(\mathfrak{p}) = k[t]/(p(t))$ is a finite extension of k , and

$$X_{\mathfrak{p}} \cong \mathbb{A}_k^1 \times_{\text{Spec } k} \text{Spec } k(\mathfrak{p}) =: \mathbb{A}_k^1 \times_k k(\mathfrak{p}) =: \mathbb{A}_{k(\mathfrak{p})}^1.$$

(The generic point $\mathfrak{p} = (0)$ gives $k(\mathfrak{p}) = k(t)$ and $X_{\eta} = \mathbb{A}_{k(t)}^1$.)

Remark 2.7.8. This example illustrates why fibres must be defined via $\text{Spec } k(y)$ rather than just set-theoretic preimages: the special fibre $X_{(t=0)}$ and a nearby fibre $X_{(t=a)}$ are both 1-dimensional as topological spaces, but the scheme structure distinguishes them—one is the reduced affine line $\mathbb{A}^1$, the other is a non-reduced thickening. The nilpotents remember that $X_{(t=0)}$ is a *limit* of smooth conics degenerating onto a double line.

(iv) **The generic fibre over $\eta = (0) \in Y$.** The generic point $\eta = (0)$ has residue field $k(\eta) = k(t)$, the fraction field of $k[t]$. Then

$$\begin{aligned} X_{\eta} &= \text{Spec}(A \otimes_{k[t]} k(t)) = \text{Spec} \frac{k(t)[x, y]}{(ty - x^2)} \\ &= \text{Spec } k(t)[x] = \mathbb{A}_{k(t)}^1, \end{aligned}$$

because t is invertible in $k(t)$, so the relation $ty = x^2$ lets us eliminate $y = x^2/t$.

Remark 2.7.9 (Localization commutes with tensor product / 局部化与张量积交换). The computation above used the identity

$$S^{-1}A \otimes_R B \simeq S^{-1}(A \otimes_R B),$$

applied with $S = k[t] \setminus \{0\}$. Tensoring with the fraction field is the same as localizing afterwards.

Base Change / 基变换

The fibre construction $X_y = X \times_Y \operatorname{Spec} k(y)$ is a special case of a general operation.

Definition 2.7.10 (Base change / 基变换). Let $X \rightarrow S$ be a morphism and let $S' \rightarrow S$ be any morphism ("a change of base / 换底"). The *base change of X along $S' \rightarrow S$* , written

$$X_{S'} := X \times_S S',$$

is the fibre product. It fits into the cartesian square

$$\begin{array}{ccc} X_{S'} & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S. \end{array}$$

Example 2.7.11 (Base change by a field extension / 域扩张下的基变换). Let L/k be a field extension, and let

$$X = \operatorname{Spec} \frac{k[x_1, \dots, x_n]}{(f_1, \dots, f_m)}$$

be an affine scheme over k . Its base change to L is

$$X_L = X \times_{\operatorname{Spec} k} \operatorname{Spec} L = \operatorname{Spec} \frac{L[x_1, \dots, x_n]}{(f_1, \dots, f_m)}.$$

The computation proceeds in two steps:

$$\begin{aligned} k[x_1, \dots, x_n] \otimes_k L &= L[x_1, \dots, x_n], \\ \frac{k[x_1, \dots, x_n]}{(f_1, \dots, f_m)} \otimes_k L &= \frac{L[x_1, \dots, x_n]}{(f_1, \dots, f_m)}. \end{aligned}$$

Geometrically, X_L is "the same equations, now viewed over the bigger field L ."

Remark on example (iii): how $\text{Spec } k(P) \rightarrow Y$ is defined

Returning to case (iii) of the running example, where $P = (p(t)) \subset k[t]$ corresponds to an irreducible $p(t)$ with residue field $k(P) = k[u]/(p(u))$: the morphism $\text{Spec } k(P) \rightarrow Y = \text{Spec } k[t]$ is not quite “plug in a value of t ” —since $k(P)$ may be strictly larger than k , one must be careful.

The clean way to see it is via base change. Consider the diagram of schemes

$$\begin{array}{ccccc} \text{Spec } k(P) & \dashrightarrow^{\exists!} & Y_{k(P)} = \text{Spec } k(P)[t] & \longrightarrow & \text{Spec } k(P) \\ & \searrow & \downarrow & & \downarrow \\ & & Y = \text{Spec } k[t] & \longrightarrow & \text{Spec } k, \end{array}$$

corresponding, on rings, to

$$\begin{array}{ccccc} k & \longrightarrow & k(P) = k[u]/(p(u)) & & \\ \downarrow & & \downarrow & \searrow & \\ k[t] & \longrightarrow & (k[u]/(p(u)))[t] & \xrightarrow{t \mapsto u} & k[u]/(p(u)). \end{array}$$

The right-hand ring map sends $t \mapsto u$, realising $\text{Spec } k(P)$ as a $k(P)$ -rational point of the base change $Y_{k(P)}$. Composing with $Y_{k(P)} \rightarrow Y$ yields the desired morphism $\text{Spec } k(P) \rightarrow Y$, which picks out $P \in Y$ as a scheme-theoretic point.

Explicit construction of X_P via iterated base change

The full computation of X_P in case (iii) factors through two cartesian squares, making precise the intuition “change base to $k(P)$, then pick out the point”:

$$\begin{array}{ccccc} \mathbb{A}_{k(P)}^1 & \longrightarrow & X_{k(P)} & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow \\ \text{Spec } k(P) & \xrightarrow[\tilde{P}]{(t-u)} & Y_{k(P)} & \longrightarrow & Y \\ & & \downarrow & & \downarrow \\ & & \text{Spec } k(P) & \longrightarrow & \text{Spec } k. \end{array}$$

Reading bottom to top:

- The *bottom-right* square is cartesian and defines $Y_{k(P)} = Y \times_{\text{Spec } k} \text{Spec } k(P) = \text{Spec } k(P)[t]$.

- The $k(P)$ -point $\tilde{P}: \operatorname{Spec} k(P) \rightarrow Y_{k(P)}$ is given by the ring map $k(P)[t] \twoheadrightarrow k(P)[t]/(t-u) = k(P)$, i.e. it cuts out $t = u$ (recall $u = \bar{t} \in k(P)$).
- The *top-right* square is cartesian and defines $X_{k(P)} = X \times_Y Y_{k(P)}$.
- The *top-left* square is cartesian, and pulling $X_{k(P)}$ back along $\tilde{P}$ yields $X_P = \mathbb{A}_{k(P)}^1$.

The outer rectangle, being a composition of cartesian squares, is itself cartesian, recovering

$$X_P = X \times_Y \operatorname{Spec} k(P).$$

Example 2.7.12 (The hyperbola family $xy = t$ / 双曲线族). Consider

$$f: X = \operatorname{Spec} \frac{k[x, y, t]}{(xy - t)} \longrightarrow Y = \operatorname{Spec} k[t].$$

(i) **Generic behavior:** $t = a \neq 0$.

$$\begin{aligned} X_{(t=a)} &= \operatorname{Spec} \frac{k[x, y, t]}{(t - a, xy - t)} = \operatorname{Spec} \frac{k[x, y]}{(xy - a)} \\ &= \operatorname{Spec} k[x, \frac{1}{x}] = \operatorname{Spec} k[x]_x \\ &\cong D(x) \subseteq \mathbb{A}_k^1 \quad (\text{open, irreducible} / \text{不可约}). \end{aligned}$$

(Since $a \neq 0$, from $xy = a$ we get $x \neq 0$ and $y = a/x$; this identifies the fibre with $\mathbb{A}_k^1$ minus the origin.)

(ii) **Special fibre:** $t = 0$.

$$X_{(t=0)} = \operatorname{Spec} \frac{k[x, y]}{(xy)},$$

the union of the two coordinate axes $\{x = 0\} \cup \{y = 0\}$ —not irreducible / 不是不可约的.

Remark 2.7.13. This is the classical picture of a family of hyperbolas $xy = t$ degenerating, as $t \rightarrow 0$, into the union of their two asymptotes. Unlike the earlier parabola family (which degenerated into a *non-reduced* double line), here the special fibre is *reduced but reducible*: the two branches of the hyperbola pull apart into two distinct components. Together, the parabola and hyperbola examples illustrate the two basic ways a smooth family can degenerate:

non-reduced (multiplicity) vs. reducible (splitting into components).

2.8 Application: Ax–Grothendieck Theorem

Remark 2.8.1 (Reference). A. Grothendieck and J.-P. Serre, *How to use finite fields for problems concerning infinite fields*. The result below is usually called the Ax–Grothendieck theorem / Ax–Grothendieck 定理.

2.8.1 Statement and functor-of-points interpretation

Theorem 2.8.2 (Ax–Grothendieck for affine space). *Let $f: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ be a morphism of schemes over a field k , and consider the induced map on $\bar{k}$ -points*

$$f(\bar{k}): \mathbb{A}_k^n(\bar{k}) \longrightarrow \mathbb{A}_k^n(\bar{k}).$$

If $f(\bar{k})$ is injective, then $f(\bar{k})$ is surjective.

Remark 2.8.3 (Unpacking the functor-of-points / 函子观点下的解读). By definition

$$\mathbb{A}_k^n(\bar{k}) := \operatorname{Hom}_k(\operatorname{Spec} \bar{k}, \mathbb{A}_k^n) = \bar{k}^n,$$

and $f(\bar{k})$ is post-composition $\phi \mapsto f \circ \phi$:

$$\begin{array}{ccc} \mathbb{A}_k^n(\bar{k}) & \xrightarrow{f(\bar{k})} & \mathbb{A}_k^n(\bar{k}) \\ \parallel & & \parallel \\ \bar{k}^n & \xrightarrow{\quad} & \bar{k}^n \\ (a_1, \dots, a_n) & \longmapsto & (f_1(a_1, \dots, a_n), \dots, f_n(a_1, \dots, a_n)). \end{array}$$

On rings this is $k[x_1, \dots, x_n] \xleftarrow{f^\#} k[x_1, \dots, x_n], x_i \mapsto f_i$. So f is just an n -tuple of polynomials $(f_1, \dots, f_n)$ with coefficients in k , and Theorem 2.8.2 says: *an injective polynomial self-map of $\bar{k}^n$ is automatically surjective.*

Theorem 2.8.4 (Ax–Grothendieck, general form). *Let X be an algebraic variety (i.e. a scheme of finite type / 有限型) over k , and let $f: X \rightarrow X$ be a k -morphism. If $f(\bar{k})$ is injective, then $f(\bar{k})$ is surjective.*

2.8.2 Strategy: reduce to finite fields

Remark 2.8.5 (Strategy / 证明策略). The strategy is the motto in the title of Serre’s paper: *reduce from an infinite field to a finite field, where injective and surjective coincide for trivial cardinality reasons.* The reduction has three ingredients:

- (i) the finite-field case, by pigeonhole (Lemma 2.8.6);
- (ii) *spreading out* / 展开: replace k by a finitely generated $\mathbb{Z}$ -subalgebra $\Lambda \subseteq k$ carrying all relevant coefficients (Lemma 2.8.7);
- (iii) the fact that every maximal ideal $\mathfrak{m} \subset \Lambda$ has *finite* residue field (Lemma 2.8.8), so each specialization falls under (i).

The finite-field case

Lemma 2.8.6 (Finite-field case). *Let $\mathbb{F}_q$ be a finite field and let $f: \mathbb{A}_{\mathbb{F}_q}^n \rightarrow \mathbb{A}_{\mathbb{F}_q}^n$ be a morphism. If $f(\overline{\mathbb{F}_q})$ is injective, then it is surjective.*

Proof. Since f is defined over $\mathbb{F}_q$, its coefficients lie in $\mathbb{F}_q$, and hence in every finite subfield $\mathbb{F}_{q^d} \subseteq \overline{\mathbb{F}_q}$. Consequently f restricts to each finite layer:

$$\begin{array}{ccc} \overline{\mathbb{F}_q}^n & \xrightarrow{f(\overline{\mathbb{F}_q})} & \overline{\mathbb{F}_q}^n \\ \uparrow & & \uparrow \\ \mathbb{F}_{q^d}^n & \longrightarrow & \mathbb{F}_{q^d}^n. \end{array}$$

The bottom arrow is an injection between finite sets of equal cardinality q^{dn} , hence a bijection. Since

$$\overline{\mathbb{F}_q} = \bigcup_{d \geq 1} \mathbb{F}_{q^d},$$

surjectivity at every finite level forces $f(\overline{\mathbb{F}_q})$ itself to be surjective. $\square$

Spreading out the coefficients

Lemma 2.8.7 (Spreading out). *Let $f: \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ be a morphism. Then there exists a finitely generated $\mathbb{Z}$ -subalgebra $\Lambda \subseteq k$, with*

$$\mathbb{Z} \subseteq \Lambda \subseteq k \quad (\text{char } k = 0), \quad \mathbb{F}_p \subseteq \Lambda \subseteq k \quad (\text{char } k = p),$$

and a morphism $\hat{f}: \mathbb{A}_\Lambda^n \rightarrow \mathbb{A}_\Lambda^n$ such that the base change recovers f :

$$\hat{f} \times_\Lambda k = f.$$

Proof. On rings, f corresponds to

$$k[x_1, \dots, x_n] \xleftarrow{f^\#} k[x_1, \dots, x_n], \quad x_i \longmapsto f_i(x_1, \dots, x_n).$$

Let $\Lambda \subseteq k$ be the subring generated (over the prime subring $\mathbb{Z}$ or $\mathbb{F}_p$) by all coefficients of the finitely many polynomials $f_1, \dots, f_n$. Then Λ is finitely generated over $\mathbb{Z}$, and the same formulas define a ring map

$$\Lambda[x_1, \dots, x_n] \xleftarrow{\hat{f}^\#} \Lambda[x_1, \dots, x_n], \quad x_i \longmapsto f_i.$$

Tensoring $- \otimes_\Lambda k$ recovers the original map, which geometrically is the base change $\hat{f} \times_\Lambda k = f$. $\square$

Residue fields of finitely generated $\mathbb{Z}$ -algebras are finite

Lemma 2.8.8 (Residue field lemma / 剩余域引理). *Let Λ be a finitely generated $\mathbb{Z}$ -algebra and $\mathfrak{m} \subset \Lambda$ a maximal ideal. Then $\Lambda/\mathfrak{m}$ is a finite field.*

Proof. We split according to the characteristic.

Case (1): $\text{char}(\Lambda/\mathfrak{m}) = p > 0$. Then $\mathbb{F}_p \hookrightarrow \Lambda/\mathfrak{m}$, and $\Lambda/\mathfrak{m}$ is a finitely generated $\mathbb{F}_p$ -algebra which is also a field. By Zariski's lemma (the weak Nullstellensatz), a finitely generated algebra over a field which is itself a field must be a *finite field extension* / 有限域扩张 of the base. Hence $[\Lambda/\mathfrak{m} : \mathbb{F}_p] < \infty$, so $\Lambda/\mathfrak{m}$ is a finite field.

Case (2): $\text{char}(\Lambda/\mathfrak{m}) = 0$. Write $K = \Lambda/\mathfrak{m}$ and consider the structure map $\mathbb{Z} \rightarrow K$.

Subcase (2.1): $\mathbb{Z} \hookrightarrow K$. Then $K \supseteq \mathbb{Q} \supseteq \mathbb{Z}$, and K is a finitely generated $\mathbb{Z}$ -algebra. But $\mathbb{Q}$ is *not* finitely generated as a $\mathbb{Z}$ -algebra: inverting finitely many primes $p_1, \dots, p_r$ only produces $\mathbb{Z}[\frac{1}{p_1 \cdots p_r}] \subsetneq \mathbb{Q}$. Hence any $\mathbb{Z}$ -algebra containing $\mathbb{Q}$ cannot be finitely generated — contradiction (#). So this subcase does not occur.

Subcase (2.2): $\mathbb{Z} \rightarrow K$ has kernel (p) for some prime p . Then $K \supseteq \mathbb{F}_p$, and K is finitely generated as an $\mathbb{F}_p$ -algebra (since the $\mathbb{Z}$ -generators, read modulo p , also generate over $\mathbb{F}_p$). We are back in Case (1), so K is a finite field. $\square$

2.8.3 Algebraic reformulations via the Nullstellensatz

Remark 2.8.9 (Reduction to $\bar{k}$ / 归约到代数闭包). There is no loss of generality in assuming k is algebraically closed. Indeed, for any k -scheme X ,

$$X(\bar{k}) = (X \times_{\text{Spec } k} \text{Spec } \bar{k})(\bar{k}),$$

because a $\bar{k}$ -point of X is the same as a $\bar{k}$ -point of $X_{\bar{k}} := X \times_k \bar{k}$:

$$\begin{array}{ccccc} \text{Spec } \bar{k} & \dashrightarrow & X_{\bar{k}} & \xrightarrow{\text{id}} & \text{Spec } \bar{k} \\ & & \downarrow & \square & \downarrow \\ & \searrow & X & \longrightarrow & \text{Spec } k. \end{array}$$

So in the proof of Theorem 2.8.2 we may replace k by $\bar{k}$ and assume k algebraically closed, at which point the Nullstellensatz is available in its classical form.

From now on assume k is algebraically closed. Fix $f = (f_1, \dots, f_n): \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$.

Surjectivity as a Bézout identity

Lemma 2.8.10 (Surjectivity, algebraically). *For $P = (P_1, \dots, P_n) \in \mathbb{A}_k^n(k) = k^n$,*

$$P \notin \text{Im } f(k) \iff \exists q_1, \dots, q_n \in k[x_1, \dots, x_n] \text{ such that } \sum_{i=1}^n q_i(f_i - P_i) = 1.$$

Proof. $P \notin \text{Im } f(k)$ means the system $f_i(\vec{x}) = P_i$ has no solution in k^n , i.e.

$$V(f_1 - P_1, \dots, f_n - P_n) = \emptyset.$$

By the Nullstellensatz (Hilbert's Nullstellensatz / 希尔伯特零点定理), an ideal in $k[x_1, \dots, x_n]$ cuts out $\emptyset$ iff it is the unit ideal:

$$(f_1 - P_1, \dots, f_n - P_n) = (1),$$

which is precisely the Bézout-type identity $\sum q_i(f_i - P_i) = 1$. □

Injectivity via the Rabinowitsch trick

Lemma 2.8.11 (Injectivity, algebraically). *$f(k)$ is injective if and only if, for every $i_0 \in \{1, \dots, n\}$, there exist polynomials $\phi_0, \phi_1, \dots, \phi_n \in k[x_1, \dots, x_n, y_1, \dots, y_n, t]$ such that*

$$\phi_0(t(x_{i_0} - y_{i_0}) - 1) + \sum_{i=1}^n \phi_i(f_i(\vec{x}) - f_i(\vec{y})) = 1.$$

Proof. Unpack injectivity: $f(k)$ is injective iff for all $\vec{x}, \vec{y} \in k^n$ with $\vec{x} \neq \vec{y}$, there exists some i with $f_i(\vec{x}) \neq f_i(\vec{y})$. Negating, failure of injectivity means

$$\exists \vec{x}, \vec{y} \in k^n \text{ with } x_{i_0} \neq y_{i_0} \text{ for some } i_0, \text{ and } f_i(\vec{x}) = f_i(\vec{y}) \text{ for all } i.$$

So $f(k)$ is injective iff, for every i_0 , the system

$$\begin{cases} x_{i_0} \neq y_{i_0}, \\ f_i(\vec{x}) = f_i(\vec{y}) \quad \forall i, \end{cases}$$

has no solution in k^{2n} .

Now apply the *Rabinowitsch trick* / 拉比诺维奇技巧: the inequation $x_{i_0} \neq y_{i_0}$ is equivalent to the solvability of $t(x_{i_0} - y_{i_0}) - 1 = 0$ in a new variable t . The system above is thus unsolvable iff

$$V(t(x_{i_0} - y_{i_0}) - 1, f_1(\vec{x}) - f_1(\vec{y}), \dots, f_n(\vec{x}) - f_n(\vec{y})) = \emptyset$$

in $\mathbb{A}_k^{2n+1}$. By the Nullstellensatz this is equivalent to the Bézout identity

$$\phi_0(t(x_{i_0} - y_{i_0}) - 1) + \sum_{i=1}^n \phi_i(f_i(\vec{x}) - f_i(\vec{y})) = 1$$

for suitable $\phi_0, \dots, \phi_n \in k[\vec{x}, \vec{y}, t]$. □

Remark 2.8.12 (Why this reformulation matters). Lemmas 2.8.10 and 2.8.11 convert the set-theoretic statements “ $P \in \text{Im } f$ ” and “ f is injective” into purely algebraic identities in polynomial rings—Bézout-style equations involving only finitely many coefficients. In particular, each such identity *descends*: if it holds over k , then already over the finitely generated subring $\Lambda \subseteq k$ generated by the coefficients of f and of the witnesses ϕ_i (or q_i). Combined with Lemmas 2.8.8 and 2.8.6, this reduces Theorem 2.8.2 to the pigeonhole principle.

2.8.4 Completing the proof

Proof of Theorem 2.8.2. Argue by contradiction (反证 / *proof by contradiction*). By the remark above we may assume $k = \bar{k}$. Suppose $f(k)$ is injective but *not* surjective, and pick $P = (P_1, \dots, P_n) \in k^n$ with $P \notin \text{Im } f(k)$.

Step 1: collect all the algebraic witnesses. By Lemmas 2.8.10 and 2.8.11,

- $P \notin \text{Im } f(k)$ gives $q_1, \dots, q_n \in k[\vec{x}]$ with $\sum_i q_i(f_i - P_i) = 1$;
- $f(k)$ injective gives, for each i_0 , polynomials $\phi_0^{(i_0)}, \dots, \phi_n^{(i_0)} \in k[\vec{x}, \vec{y}, t]$ satisfying the Bézout identity of Lemma 2.8.11.

Step 2: spread out. Apply Lemma 2.8.7: take $\Lambda \subseteq k$ to be the subring generated over $\mathbb{Z}$ by the coefficients of

$$f_i, \quad P_i, \quad q_i, \quad \phi_j^{(i_0)} \quad (\text{for all } i, j, i_0).$$

This is a finitely generated $\mathbb{Z}$ -algebra, and $\Lambda \subseteq k$. The morphism f descends to $\hat{f}: \mathbb{A}_\Lambda^n \rightarrow \mathbb{A}_\Lambda^n$, the point P descends to a Λ -point $\hat{P} \in \mathbb{A}_\Lambda^n(\Lambda)$, and the Bézout identities hold *verbatim* over Λ .

Step 3: specialize to a finite field. Choose any maximal ideal $\mathfrak{m} \subset \Lambda$. By Lemma 2.8.8, $\Lambda/\mathfrak{m}$ is a finite field. Reduce all witnesses modulo $\mathfrak{m}$: write

$$\bar{f}_i, \quad \bar{P}_i, \quad \bar{q}_i, \quad \bar{\phi}_j^{(i_0)}$$

for the images in $(\Lambda/\mathfrak{m})[\vec{x}, \vec{y}, t]$, and set $\bar{f} = (\bar{f}_1, \dots, \bar{f}_n)$. The Bézout identities reduce to identities over $\Lambda/\mathfrak{m}$, hence also over its algebraic closure $\overline{\Lambda/\mathfrak{m}}$.

Applying Lemmas 2.8.10 and 2.8.11 *in reverse* now gives, over $\overline{\Lambda/\mathfrak{m}}$:

$$\bar{P} \notin \text{Im } \bar{f}(\overline{\Lambda/\mathfrak{m}}), \quad \bar{f}(\overline{\Lambda/\mathfrak{m}}) \text{ is injective.}$$

In other words, $\bar{f}$ is injective but not surjective on $\overline{\Lambda/\mathfrak{m}}$ -points.

Step 4: contradiction. But $\Lambda/\mathfrak{m}$ is a finite field, so Lemma 2.8.6 applies to $\bar{f}$ and forces $\bar{f}(\overline{\Lambda/\mathfrak{m}})$ to be surjective. Contradiction (矛盾). $\square$

Remark 2.8.13 (The geometric picture / 几何图像). The whole argument can be visualized as a tower of base changes, where Λ plays the role of a “bridge” between the finite and infinite worlds:

$$\begin{array}{ccccc}
 \mathbb{A}_{\Lambda/\mathfrak{m}}^n & \longleftrightarrow & \mathbb{A}_{\Lambda}^n & \longleftrightarrow & \mathbb{A}_k^n \\
 \downarrow \bar{f} & & \downarrow \hat{f} & & \downarrow f \\
 \mathbb{A}_{\Lambda/\mathfrak{m}}^n & \longleftrightarrow & \mathbb{A}_{\Lambda}^n & \longleftrightarrow & \mathbb{A}_k^n \\
 \downarrow & & \downarrow & & \downarrow \\
 \mathrm{Spec}(\Lambda/\mathfrak{m}) & \xrightarrow{\text{closed pt}} & \mathrm{Spec} \Lambda & \xleftarrow{\text{generic pt}} & \mathrm{Spec} k
 \end{array}$$

At the bottom, $\mathrm{Spec} \Lambda$ has two kinds of points: *closed points* / 闭点 $\mathfrak{m}$ with finite residue field, and the *generic point* / 一般点 (the zero ideal), whose residue field is the fraction field of Λ (a subfield of k). The morphism $\hat{f}$ over $\mathrm{Spec} \Lambda$ is a single family interpolating between f (at the generic point) and all the finite-field specializations $\bar{f}$ (at closed points). The properties “injective” and “not surjective” are expressed by the existence of polynomial witnesses (Bézout identities), and such identities propagate both ways along this tower—that is what makes the reduction to finite fields legitimate.

2.9 Non-isomorphism of SL_n and affine space

Proposition 2.9.1. *For $n \geq 3$,*

$$SL_{n, \mathbb{Q}} \not\cong \mathbb{A}_{\mathbb{Q}}^{n^2-1}$$

as schemes over $\mathbb{Q}$.

Both sides have dimension $n^2 - 1$, so dimension alone does not distinguish them. As another application of the *spread out, then reduce mod p and count points* philosophy, we give an entirely algebraic proof below (Method 2), after briefly contrasting it with the topological approach (Method 1).

2.9.1 Method 1: reduction to $\mathbb{C}$ and cohomology / 归约到 $\mathbb{C}$ 用上同调

$$\begin{aligned}
 SL_{n,\mathbb{Q}} &\cong \mathbb{A}_{\mathbb{Q}}^{n^2-1} \\
 &\Downarrow \text{base change to } \mathbb{C} \\
 SL_{n,\mathbb{C}} &\cong \mathbb{A}_{\mathbb{C}}^{n^2-1} \\
 &\Downarrow \text{analytification} \\
 SL_n(\mathbb{C}) &\cong \mathbb{C}^{n^2-1}.
 \end{aligned}$$

The last line would be a homeomorphism (in fact a biholomorphism) of complex manifolds, which can be ruled out by *cohomology computation* / 上同调计算: $\mathbb{C}^{n^2-1}$ is contractible, whereas $SL_n(\mathbb{C})$ is homotopy equivalent to $SU(n)$ and has nontrivial cohomology (e.g. $H^3(SL_n(\mathbb{C}), \mathbb{Z}) \neq 0$ for $n \geq 2$). This is powerful but uses heavy analytic/topological machinery (“need big Thm”).

2.9.2 Method 2: spread out and count points mod p / 展开并模 p 数点

Proof of Proposition 2.9.1. Argue by contradiction (反证). Assume we have mutually inverse $\mathbb{Q}$ -morphisms

$$SL_{n,\mathbb{Q}} \xrightleftharpoons[g]{f} \mathbb{A}_{\mathbb{Q}}^{n^2-1} \quad f \circ g = \text{id}, \quad g \circ f = \text{id}.$$

Write $f = (f_1, \dots, f_{n^2-1})$ with $f_i \in \mathbb{Q}[x_{ij}]$ (polynomials in the matrix entries) and $g = (g_1, \dots, g_{n^2}) \in \mathbb{Q}[y_1, \dots, y_{n^2-1}]$.

Step 1 (spread out / 展开). Only finitely many rational coefficients appear in f_i and g_j . Let $N \in \mathbb{Z}_{>0}$ be a common denominator for all of them. Then f and g are both well-defined over the localization

$$\mathbb{Z}[1/N] \subseteq \mathbb{Q},$$

and the two identities $f \circ g = \text{id}, g \circ f = \text{id}$ —being identities of polynomials in finitely many variables with coefficients in $\mathbb{Z}[1/N]$ —hold *verbatim* over $\mathbb{Z}[1/N]$.

Step 2 (reduce mod p / 模 p 约化). For any prime $p \nmid N$, the reduction $\mathbb{Z}[1/N] \rightarrow \mathbb{F}_p$ is defined. Base-changing f and g along $\text{Spec } \mathbb{F}_p \rightarrow \text{Spec } \mathbb{Z}[1/N]$ yields $\mathbb{F}_p$ -morphisms

$$SL_{n,\mathbb{F}_p} \xrightleftharpoons[\bar{g}]{\bar{f}} \mathbb{A}_{\mathbb{F}_p}^{n^2-1}$$

still satisfying $\bar{f} \circ \bar{g} = \text{id}, \bar{g} \circ \bar{f} = \text{id}$. So

$$SL_{n,\mathbb{F}_p} \cong \mathbb{A}_{\mathbb{F}_p}^{n^2-1} \quad \text{for every prime } p \nmid N. \quad (*)$$

Step 3 (count $\mathbb{F}_p$ -points / 数 $\mathbb{F}_p$ -点). An isomorphism of $\mathbb{F}_p$ -schemes induces a bijection of $\mathbb{F}_p$ -points. But

$$|\mathbb{A}_{\mathbb{F}_p}^{n^2-1}(\mathbb{F}_p)| = p^{n^2-1},$$

while the order of $SL_n(\mathbb{F}_p)$ is the classical formula

$$|SL_n(\mathbb{F}_p)| = \frac{|GL_n(\mathbb{F}_p)|}{p-1} = p^{\binom{n}{2}} \prod_{i=2}^n (p^i - 1).$$

This is *not* a pure power of p —it has the factor $(p^n - 1)$, which is coprime to p and > 1 . Hence

$$|SL_n(\mathbb{F}_p)| \neq p^{n^2-1},$$

contradicting (*). □

2.9.3 Further remarks

Remark 2.9.2 (Functoriality of base change / 基变换的函子性). The silent workhorse of Step 2 is that *base change preserves commutative diagrams*. Concretely, for any $S' \rightarrow S$, the functor $(-)\times_S S': \text{Sch}/_S \rightarrow \text{Sch}/_{S'}$ sends compositions to compositions and identities to identities, so the commutative diagrams encoding “ f and g are inverse”

$$\begin{array}{ccc} & X & \\ & \parallel & \\ X & & Y \\ & \parallel & \\ & X & \end{array} \quad \begin{array}{ccc} & Y & \\ & \parallel & \\ Y & & X \\ & \parallel & \\ & Y & \end{array}$$

$\begin{array}{ccc} & f & \\ & \searrow & \\ & & g \\ & \swarrow & \end{array}$ $\begin{array}{ccc} & g & \\ & \searrow & \\ & & f \\ & \swarrow & \end{array}$

survive base change verbatim, producing inverse morphisms $\bar{f}, \bar{g}$ over S' .

Remark 2.9.3 (A refinement via Hom-schemes / 通过 Hom-概形的精细化). A more sophisticated invariant available on both sides is the *Hom-scheme* / 同态概形

$$\underline{\text{Hom}}(C, X),$$

which, when C is a projective curve and X is suitable (e.g. projective or admitting a good compactification), is itself a scheme. Over $\mathbb{F}_p$ one can count its $\mathbb{F}_p$ -rational points (or its irreducible components), and one finds growth of the form

$$\#\underline{\text{Hom}}(C, X)(\mathbb{F}_p) \sim c p^d$$

for specific constants c, d depending on X and the genus of C . Comparing these counts for $X = SL_n$ and $X = \mathbb{A}^{n^2-1}$ (equivalently, comparing the spaces of maps $\mathbb{F}_p[C] \rightarrow \mathcal{O}(X)$) yields further obstructions to isomorphism. This perspective places Proposition 2.9.1 inside a much wider circle of ideas (Weil conjectures, arithmetic statistics).

Chapter 3

Separated and Proper Morphisms

In this chapter we introduce two fundamental finiteness conditions on morphisms of schemes. Both are algebro-geometric analogues of classical topological properties:

- *separatedness* corresponds to Hausdorffness;
- *properness* corresponds to the preimage-of-compact-is-compact condition.

Because Zariski topology is almost never Hausdorff, neither condition can be stated in its classical form. Both are instead expressed through categorical avatars—respectively the diagonal morphism and the universal-closedness condition.

3.1 Topological prototypes

We first recall how Hausdorffness can be reformulated in terms of the diagonal map. This reformulation is internal to the category of topological spaces and generalizes verbatim to any category with finite products, which is the bridge we shall use.

Proposition 3.1.1. *Let X be a topological space, and let*

$$\Delta_X : X \longrightarrow X \times X, \quad x \longmapsto (x, x)$$

be the diagonal map. The following are equivalent:

- (i) X is Hausdorff;
- (ii) $\text{Im}(\Delta_X)$ is closed in $X \times X$;
- (iii) Δ_X is a closed immersion, i.e. a homeomorphism onto a closed subspace.

Proof. (i) $\Rightarrow$ (ii). Let $(x, y) \notin \text{Im}(\Delta_X)$, so $x \neq y$. Hausdorffness yields opens $x \in U_x$, $y \in U_y$ with $U_x \cap U_y = \emptyset$. Then

$$(x, y) \in \underbrace{U_x \times U_y}_{\text{open in } X \times X}, \quad (U_x \times U_y) \cap \text{Im}(\Delta_X) = \emptyset,$$

since a common point $z \in U_x \cap U_y$ would give (z, z) in both. Hence $\text{Im}(\Delta_X)$ is closed.

(ii) $\Rightarrow$ (iii). The map Δ_X is visibly injective and, by hypothesis, has closed image. It remains to check that Δ_X is a homeomorphism onto its image, i.e. that for every open $U \subseteq X$ there is an open $W \subseteq X \times X$ with $W \cap \text{Im}(\Delta_X) = \Delta_X(U)$. Take $W = U \times U$.

(iii) $\Rightarrow$ (i). For any $x \neq y$, $(x, y) \notin \text{Im}(\Delta_X)$. Closedness of the image provides an open $W \subseteq X \times X$ with $(x, y) \in W$ and $W \cap \text{Im}(\Delta_X) = \emptyset$. The product topology has a basis of rectangles, so we may shrink to $(x, y) \in U_x \times U_y \subseteq W$. Then $U_x \cap U_y = \emptyset$: any common point z would give $(z, z) \in (U_x \times U_y) \cap \text{Im}(\Delta_X) \subseteq W \cap \text{Im}(\Delta_X) = \emptyset$, a contradiction. $\square$

Remark 3.1.2. 此命题将“点点分离”这一外在逐点条件，翻译成对对角映射 Δ_X 的一个内在条件。后者在任何具有有限积的范畴中都有意义——这正是 scheme 范畴中 separated morphism 定义的原型：一个 morphism $f : X \rightarrow Y$ 被称作 separated，若相对对角 $\Delta_{X/Y} : X \rightarrow X \times_Y X$ 是闭浸入。

We turn next to properness.

Definition 3.1.3. A continuous map $f : X \rightarrow Y$ between topological spaces is *proper* (紧合) if

1. f is a closed map, and
2. for every compact $Z \subseteq Y$, the preimage $f^{-1}(Z) \subseteq X$ is compact.

Example 3.1.4.

1. The projection $X \times S^1 \rightarrow X$, where $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, is proper: the fibre is the compact circle.
2. The projection $X \times \mathbb{R} \rightarrow X$ is *not* proper: the preimage of a point is $\mathbb{R}$, which is not compact.

Remark 3.1.5. 这两个例子体现了 proper 的几何直观：纤维不能“跑到无穷远”。 S^1 紧致（纤维被关住）， $\mathbb{R}$ 不紧致（纤维可逃逸）。在代数几何中，这对应到 $\mathbb{P}_k^n \rightarrow \text{Spec } k$ 是 proper 而 $\mathbb{A}_k^n \rightarrow \text{Spec } k$ 不是——射影空间正是把“无穷远点”补上后的紧化。

A classical source of proper maps is projection from a compact fibre. The following proposition captures the precise hypothesis needed; its proof illustrates the interplay of local compactness and Hausdorffness that, in the schematic setting, will be replaced by quasi-compactness and separatedness.

Proposition 3.1.6. *Let X be a Hausdorff locally compact space and Z a compact space. Then the projection $f : X \times Z \rightarrow X$ is proper.*

Proof. We verify the two conditions separately.

Preimage of compact is compact. For any compact $C \subseteq X$,

$$f^{-1}(C) = C \times Z,$$

which is compact as a product of two compact spaces.

f is closed. Let $Y \subseteq X \times Z$ be closed; we show $f(Y)$ is closed in X . By local compactness, choose an open cover $\{U_i\}_{i \in I}$ of X with each closure $\overline{U_i}$ compact. It suffices to show $f(Y) \cap U_i$ is closed in U_i for every i , since then

$$X \setminus f(Y) = \bigcup_{i \in I} (U_i \setminus (U_i \cap f(Y)))$$

is open in X .

Fix i . Using the identity $f(A) \cap B = f(A \cap f^{-1}(B))$,

$$f(Y) \cap \overline{U_i} = f(Y \cap f^{-1}(\overline{U_i})) = f(Y \cap (\overline{U_i} \times Z)).$$

Since $\overline{U_i} \times Z$ is compact, its closed subset $Y \cap (\overline{U_i} \times Z)$ is compact, hence so is its continuous image $f(Y) \cap \overline{U_i}$. A compact subset of the Hausdorff space X is closed, so

$$f(Y) \cap U_i = (f(Y) \cap \overline{U_i}) \cap U_i$$

is closed in U_i , as required. $\square$

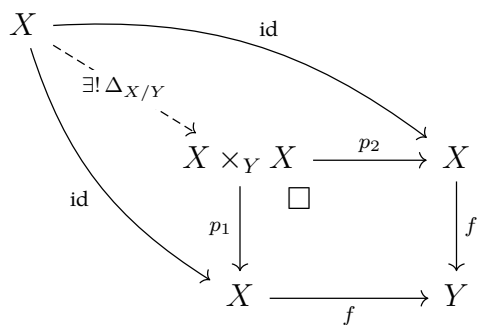
Remark 3.1.7. 证明里实际用到的不是“每点有紧邻域”，而是“每点有具紧闭包的开邻域”。在 Hausdorff 空间中这两者等价，所以“Hausdorff + locally compact”正是让论证跑通的最弱假设。代数几何里的“紧合态射”正是把这一图像——纤维紧、闭映射——搬进 scheme 范畴。由于 Zariski 拓扑几乎从不 Hausdorff，定义须改用“万有闭”来表达。

3.2 Separated morphisms of schemes

Let $f : X \rightarrow Y$ be a morphism of schemes. The pair of morphisms $(\text{id}_X, \text{id}_X) : X \rightarrow X \times X$ compatible with f factors through the fibre product, producing the unique morphism

$$\Delta_{X/Y} : X \rightarrow X \times_Y X$$

characterised by $p_1 \circ \Delta_{X/Y} = p_2 \circ \Delta_{X/Y} = \text{id}_X$:



Definition 3.2.2. A morphism of schemes $f : X \rightarrow Y$ is *separated* (分离) if the diagonal morphism $\Delta_{X/Y} : X \rightarrow X \times_Y X$ is a closed immersion. In this case we say:

1. X is *separated over* Y (在 Y 上分离) ;
2. when $Y = \operatorname{Spec} \mathbb{Z}$, we simply call X a *separated scheme* (分离概形), i.e. the structural morphism $X \rightarrow \operatorname{Spec} \mathbb{Z}$ is separated.

Remark 3.2.3. 与 [definition 3.1.1](#) 对照：拓扑空间 X 是 Hausdorff $\Leftrightarrow \Delta_X$ 是闭嵌入。将 $X \times X$ 替换为相对积 $X \times_Y X$ ，并把“闭嵌入”保留为 scheme 范畴中的闭浸入，就得到 X/Y 分离的定义。特别地， $Y = \mathrm{Spec} \mathbb{Z}$ 对应“绝对分离”：由于任意 scheme 都唯一地映射到 $\mathrm{Spec} \mathbb{Z}$ （其为 **Sch** 的终对象），这就是“分离概形”的默认含义。

The simplest non-trivial example of a non-separated scheme arises by a pathological gluing.

Example 3.2.4 (Affine line with doubled origin). Let k be a field. The *affine line with doubled origin* (双原点仿射直线) is

$$X := \mathbb{A}_k^1 \cup_{\mathbb{A}_k^1 \setminus \{0\}} \mathbb{A}_k^1,$$

obtained by gluing two copies U_1, U_2 of $\mathbb{A}_k^1$ along the common open subscheme $\mathbb{A}_k^1 \setminus \{0\}$ via the identity. Write $0_1 \in U_1, 0_2 \in U_2$ for the two origins. We claim $X \rightarrow \operatorname{Spec} k$ is *not* separated.

Since $\mathbb{A}_k^1 \times_k \mathbb{A}_k^1 = \operatorname{Spec}(k[x] \otimes_k k[y]) = \operatorname{Spec} k[x, y] \cong \mathbb{A}_k^2$ and $\{U_i\}$ covers each factor of X , we have an affine open cover

$$X \times_k X = \bigcup_{i,j \in \{1,2\}} U_i \times_k U_j, \quad U_i \times_k U_j \cong \mathbb{A}_k^2,$$

with four charts pairwise glued along the complements of the coordinate axes.

For the diagonal $\Delta : X \rightarrow X \times_k X$:

- if $t \in \mathbb{A}_k^1 \setminus \{0\}$ the two origins are identified off 0, so $\Delta(t) = (t, t)$ lies in every chart $U_i \times U_j$;
- if $t = 0_i$ is one of the two origins, $\Delta(0_i) = (0_i, 0_i)$ lies only in $U_i \times U_i$.

Hence $\text{Im}(\Delta)$ contains $(0_1, 0_1)$ and $(0_2, 0_2)$, but *not* $(0_1, 0_2)$ or $(0_2, 0_1)$.

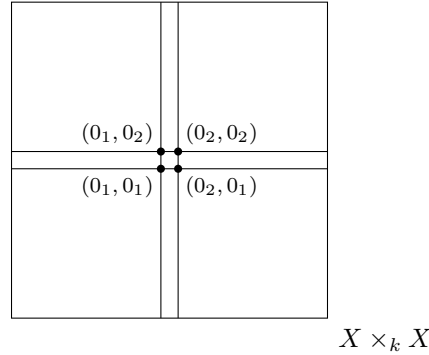
On the other hand, in any chart $U_i \times U_j \cong \text{Spec } k[x, y]$,

$$\Delta(\mathbb{A}_k^1 \setminus \{0\}) \cap (U_i \times U_j) = \{(t, t) : t \neq 0\} = V(x - y) \setminus V(x),$$

whose closure is the full diagonal $V(x - y)$, passing through $(0, 0)$. Identifying $(0, 0) \in U_i \times U_j$ with $(0_i, 0_j)$ gives

$$(0_i, 0_j) \in \overline{\text{Im}(\Delta)} \quad \text{for every } i, j \in \{1, 2\}.$$

Thus $\overline{\text{Im}(\Delta)}$ contains all four origins while $\text{Im}(\Delta)$ contains only two of them. So $\text{Im}(\Delta)$ is not closed in $X \times_k X$, Δ is not a closed immersion, and X is not separated over $\text{Spec } k$. $\square$



Remark 3.2.5. 这是 scheme 范畴中“非 Hausdorff”的标准范例，完全对应拓扑中双原点直线（两个 $\mathbb{R}$ 沿 $\mathbb{R} \setminus \{0\}$ 粘合，两个原点的任何邻域都相交）。值得强调一件事：仿射概形 $\mathbb{A}_k^1$ 自身是分离的；但把仿射概形沿开子集粘合后未必保持分离性。分离性并非自动成立，它是对粘合数据的一个实质约束。

3.2.1 The image of the diagonal

The verification of non-separatedness in [definition 3.2.4](#) only used that $\text{Im}(\Delta)$ fails to be closed; we did not need to check the sheaf-theoretic condition separately. The next proposition explains why: closedness of the image already forces the full closed-immersion condition.

Proposition 3.2.6. *Let $f : X \rightarrow Y$ be a morphism of schemes, and $\Delta = \Delta_{X/Y}$. Then f is separated if and only if $\text{Im}(\Delta) \subseteq X \times_Y X$ is closed as a subset.*

The direction $(\Rightarrow)$ is immediate from the definition. For the converse we first dispatch the affine case.

Lemma 3.2.7. *If X and Y are affine schemes, then every morphism $f : X \rightarrow Y$ is separated.*

Proof. Write $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, with f induced by a ring map $f^\# : B \rightarrow A$ making A a B -algebra. Then $X \times_Y X = \operatorname{Spec}(A \otimes_B A)$, and the diagonal Δ corresponds on rings to the *multiplication map* (乘法映射)

$$\Delta^\# : A \otimes_B A \longrightarrow A, \quad a \otimes b \longmapsto ab.$$

Every $a \in A$ is the image of $a \otimes 1$, so $\Delta^\#$ is surjective, inducing

$$(A \otimes_B A) / \ker \Delta^\# \xrightarrow{\sim} A.$$

Taking Spec yields

$$\operatorname{Spec} A \xrightarrow{\sim} V(\ker \Delta^\#) \subseteq \operatorname{Spec}(A \otimes_B A),$$

which exhibits Δ as a closed immersion. $\square$

Remark 3.2.8. 仿射情形下分离性自动成立——乘法映射 $A \otimes_B A \rightarrow A$ 总是满射。这也解释了为什么 [definition 3.2.4](#) 中双原点直线的非分离性必然来自粘合：非仿射是破坏分离性的前提。

Proof of [definition 3.2.6](#). $(\Rightarrow)$ By definition of closed immersion.

$(\Leftarrow)$ Assume $\operatorname{Im}(\Delta)$ is closed. A closed immersion is characterised by two conditions, which we verify separately.

(1) *Topological:* Δ is a homeomorphism onto a closed subset. The image is closed by hypothesis, and Δ is injective (with retraction p_1). For any open $U \subseteq X$,

$$\Delta(U) = p_1^{-1}(U) \cap \operatorname{Im}(\Delta),$$

which is open in $\operatorname{Im}(\Delta)$; thus Δ is open onto its image.

(2) *Sheaf-theoretic:* $\Delta^\# : \Delta^{-1}\mathcal{O}_{X \times_Y X} \rightarrow \mathcal{O}_X$ is surjective. It suffices to check surjectivity on stalks. Fix $P \in X$; we show

$$\Delta_P^\# : \mathcal{O}_{X \times_Y X, \Delta(P)} \longrightarrow \mathcal{O}_{X, P}$$

is surjective. Choose an affine open $V \subseteq Y$ containing $f(P)$, then an affine open $U \subseteq f^{-1}(V)$ containing P . Since V is open in Y , $U \times_V U = U \times_Y U$ is an open subscheme of $X \times_Y X$, and we have the commutative square

$$\begin{array}{ccc} U & \xrightarrow{\Delta_U} & U \times_V U \\ \text{open} \downarrow & & \downarrow \text{open} \\ X & \xrightarrow{\Delta_X} & X \times_Y X. \end{array}$$

By [definition 3.2.7](#), Δ_U is a closed immersion, so $\Delta_U^\#$ is surjective on stalks. The two vertical open immersions induce isomorphisms on the stalks at P and $\Delta(P)$, so $\Delta_X^\#$ is surjective on the stalk at P as well. $\square$

Remark 3.2.9. 证明第 (2) 步是“局部仿射化”的标准套路：把分离性这个整体条件落到一对仿射开集 $U \subseteq f^{-1}(V)$ 上验证，再用 $U \times_V U \hookrightarrow X \times_Y X$ 是开浸入将局部结果提升到整体。这一“开集-仿射-引理”的三段式以后会反复出现。

3.2.2 Affine morphisms are separated

Recall that $f : X \rightarrow Y$ is *affine* (仿射态射) if for every affine open $V \subseteq Y$, the preimage $f^{-1}(V) \subseteq X$ is affine. [Definition 3.2.7](#) upgrades immediately to this relative setting.

Corollary 3.2.10. *Every affine morphism of schemes is separated. In particular, any morphism between affine schemes is separated.*

Proof. Let $f : X \rightarrow Y$ be affine, and choose an affine open cover $Y = \bigcup_i V_i$. Set $U_i := f^{-1}(V_i)$; by hypothesis each U_i is affine. Then

$$U_i \times_{V_i} U_i = U_i \times_Y U_i \subseteq X \times_Y X$$

is open, and these charts cover $X \times_Y X$. For each i , the restriction $f|_{U_i} : U_i \rightarrow V_i$ is a morphism of affine schemes, so by [definition 3.2.7](#) the diagonal $\Delta_{U_i/V_i} : U_i \rightarrow U_i \times_{V_i} U_i$ is a closed immersion; hence

$$\Delta(X) \cap (U_i \times_{V_i} U_i) = \Delta(U_i)$$

is closed in $U_i \times_{V_i} U_i$. Closedness is local on the target, so $\Delta(X)$ is closed in $X \times_Y X$. By [definition 3.2.6](#), f is separated. $\square$

Remark 3.2.11. 证明的关键是 f 仿射： $U_i = f^{-1}(V_i)$ 本身是仿射。若 f 不仿射，只能退而求其次，取仿射细分

$$Y = \bigcup_i V_i \text{ affine}, \quad f^{-1}(V_i) = \bigcup_j U_{ij} \text{ affine}.$$

此时仍有 $\Delta(U_{ij}) \subseteq U_{ij} \times_{V_i} U_{ij}$ 闭，但 $\bigcup_{i,j} U_{ij} \times_{V_i} U_{ij} \subsetneq X \times_Y X$ 漏掉了“交叉项” $U_{ij_1} \times_{V_i} U_{ij_2}$ ($j_1 \neq j_2$)，所以论证断裂——这正是 [definition 3.2.10](#) 本质依赖“ f 仿射”的地方。

The following is a powerful test for detecting non-separatedness: finding two affine opens whose intersection is not affine.

Remark 3.2.12 (Hartshorne II, Exercise 4.3). Let X be separated over an affine base S . Then for any two affine opens $U, V \subseteq X$, the intersection $U \cap V$ is affine.

3.3 Non-affineness and the doubled plane

[Definition 3.2.12](#) provides a test: exhibiting affine opens whose intersection is *not* affine forces the scheme to be non-separated. To deploy it effectively we need a convenient way to certify non-affineness, and the most powerful such technique rests on an algebraic-geometric version of Hartogs' theorem: on a normal Noetherian scheme, regular functions extend across loci of codimension ≥ 2 . This entire section is a self-contained development of that tool, culminating in a second, cleaner proof of non-separatedness for a higher-dimensional variant of [definition 3.2.4](#).

3.3.1 A criterion for non-affineness

Lemma 3.3.1 (Non-affineness criterion). *Let $X = \operatorname{Spec} A$ be affine and $U \subsetneq X$ a proper open subscheme. If the restriction map*

$$\mathcal{O}_X(X) \longrightarrow \mathcal{O}_X(U)$$

is an isomorphism, then U is not affine.

Proof. Suppose for contradiction $U = \operatorname{Spec} B$ is affine. The open immersion $U \hookrightarrow X$ corresponds to the ring map $A = \mathcal{O}_X(X) \rightarrow \mathcal{O}_X(U) = B$, which is an isomorphism by assumption. Hence $U \hookrightarrow X$ is an isomorphism of schemes, contradicting $U \subsetneq X$. $\square$

Remark 3.3.2. 判别的实用性在于：只要我们能计算出从 X 去掉一个闭子集后的全局正则函数环没有变大，就立刻能断言这个开集不仿射。

3.3.2 Hartogs-type extension for normal schemes

To apply [definition 3.3.1](#) in practice we need conditions under which removing a closed subset does not enlarge the ring of global sections. This relies on a classical fact from commutative algebra.

Theorem 3.3.3 (Krull–Serre). *Let A be a Noetherian integrally closed domain. Then*

$$A = \bigcap_{\substack{P \in \operatorname{Spec} A \\ \operatorname{ht}(P)=1}} A_P \subseteq \operatorname{Frac}(A).$$

Remark 3.3.4. Noetherian 整环整闭当且仅当它等于自身在所有高度 1 素理想处局部化之交；参见 Matsumura, *Commutative Ring Theory*, Thm 11.5。这是 Serre 正规准则 $(R_1 + S_2)$ 的一半。

Proposition 3.3.5. *Let A be a Noetherian integrally closed domain, finitely generated as an algebra over a field (so A is catenary), and set $X = \operatorname{Spec} A$. For any closed subset $Z \subsetneq X$ with $\dim Z \leq \dim X - 2$,*

$$\mathcal{O}_X(X \setminus Z) = \mathcal{O}_X(X) = A.$$

Remark 3.3.6. 此命题是代数几何版本的 Hartogs 延拓定理：正规 Noetherian 仿射概形上，余维至少为 2 的开集上定义的正则函数自动延拓到整个空间——低维闭集不能“阻挡”正则函数。

The proof factors through two lemmas on integral schemes, both of independent interest. Throughout, X is an integral scheme with generic point η_X ; every non-empty open of X contains η_X .

Lemma 3.3.7. *For every non-empty open $U \subseteq X$, the restriction $\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X, \eta_X}$ is injective. If $X = \operatorname{Spec} A$, then $\mathcal{O}_{X, \eta_X} = \operatorname{Frac}(A)$.*

Proof. Fix an affine open cover $U = \bigcup_i U_i$ with $U_i = \operatorname{Spec} B_i$. Integrality of X means each B_i is a domain with a common fraction field $K := \operatorname{Frac}(B_i) = \mathcal{O}_{X, \eta_X}$. The sheaf axiom furnishes the top exact row of

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathcal{O}_X(U) & \rightarrow & \prod_i B_i & \rightarrow & \prod_{i \neq j} \mathcal{O}_X(U_i \cap U_j) \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & K & \xrightarrow{\text{diag}} & \prod_i K & \xrightarrow{\psi} & \prod_{i \neq j} K, \end{array}$$

where $\psi : (x_i) \mapsto (x_i - x_j)_{i \neq j}$ has kernel the diagonal copy of K . The middle vertical is injective (product of inclusions $B_i \hookrightarrow K$). If $s \in \mathcal{O}_X(U)$ has zero image in K , then $(s|_{U_i})_i \in \prod_i B_i$ has zero image in $\prod_i K$, forcing $s|_{U_i} = 0$ for all i , hence $s = 0$.

When $X = \operatorname{Spec} A$, the generic point corresponds to $(0) \in \operatorname{Spec} A$, so $\mathcal{O}_{X, \eta_X} = A_{(0)} = \operatorname{Frac}(A)$. $\square$

Lemma 3.3.8. *For every non-empty open U of an integral scheme X ,*

$$\mathcal{O}_X(U) \subseteq \bigcap_{P \in U} \mathcal{O}_{X, P} \subseteq \mathcal{O}_{X, \eta_X},$$

all regarded as subrings of $\mathcal{O}_{X, \eta_X}$.

Proof. Every local ring sits inside the generic stalk: choosing an affine open $V = \operatorname{Spec} B \ni P$,

$$\mathcal{O}_{X, P} = B_{\mathfrak{p}} \subseteq \operatorname{Frac}(B) = \mathcal{O}_{X, \eta_X}.$$

For each $P \in U$, pick an affine $U_i = \operatorname{Spec} B_i \subseteq U$ with $P \in U_i$. The restriction factors as

$$\mathcal{O}_X(U) \longrightarrow B_i \longrightarrow (B_i)_{\mathfrak{p}} = \mathcal{O}_{X,P} \hookrightarrow \mathcal{O}_{X,\eta_X},$$

compatible with the injection of [definition 3.3.7](#). Hence every section $s \in \mathcal{O}_X(U)$, viewed inside $\mathcal{O}_{X,\eta_X}$, lies in $\mathcal{O}_{X,P}$ for every $P \in U$. $\square$

Proof of [definition 3.3.5](#). Set $K = \operatorname{Frac}(A)$. By [definition 3.3.8](#),

$$\mathcal{O}_X(X \setminus Z) \subseteq \bigcap_{P \in X \setminus Z} A_P \subseteq K.$$

Claim: every height-one prime of A avoids Z . Suppose, for contradiction, $P \in Z$ with $\operatorname{ht}(P) = 1$. Then $V(P) \subseteq Z$, so

$$\dim A/P = \dim V(P) \leq \dim Z \leq \dim X - 2.$$

Since A is a catenary domain finitely generated over a field, $\operatorname{ht}(P) + \dim A/P = \dim A$, forcing $\dim A/P = \dim X - 1$ —contradiction.

Thus $\{P : \operatorname{ht}(P) = 1\} \subseteq X \setminus Z$, and

$$A \subseteq \mathcal{O}_X(X \setminus Z) \subseteq \bigcap_{P \in X \setminus Z} A_P \subseteq \bigcap_{\operatorname{ht}(P)=1} A_P \stackrel{3.3.3}{=} A.$$

Equality holds throughout. $\square$

Example 3.3.9 ($\mathbb{A}_k^2 \setminus \{(0,0)\}$ is not affine). Let $A = k[x, y]$, $X = \mathbb{A}_k^2 = \operatorname{Spec} A$, and $Z = \{(0,0)\} = V(x, y)$, a closed point with $\dim Z = 0 = \dim X - 2$. The ring A is a UFD, hence integrally closed, and $\dim A = 2$. [Definition 3.3.5](#) yields

$$\mathcal{O}_X(\mathbb{A}_k^2 \setminus \{(0,0)\}) = k[x, y].$$

The restriction $\mathcal{O}_X(\mathbb{A}_k^2) \rightarrow \mathcal{O}_X(\mathbb{A}_k^2 \setminus \{(0,0)\})$ is thus an isomorphism, and [definition 3.3.1](#) gives that $\mathbb{A}_k^2 \setminus \{(0,0)\}$ is *not* affine.

Remark 3.3.10. 对比 $\mathbb{A}_k^1 \setminus \{0\} = \operatorname{Spec} k[x, x^{-1}]$: 去掉一个点仍仿射, 因为点在 $\mathbb{A}^1$ 中余维为 1 ($= \dim X - 1$), 不满足 [definition 3.3.5](#) 的余维至少为 2 条件, Hartogs 型延拓失效——开集上多出了函数 x^{-1} 。到了 $\mathbb{A}^2$, 原点余维变成 2, 正则函数“穿透”低维缺口, $\mathcal{O}(X \setminus \{0\}) = k[x, y]$ 保持不变, 从而破坏了仿射性。

3.3.3 The affine plane with doubled origin

We are now in a position to give an alternative proof of non-separatedness for a doubled-origin construction—this time in dimension two, where the argument via [definition 3.2.12](#) is cleaner than the diagonal-closure computation of [definition 3.2.4](#).

Example 3.3.11 (Affine plane with doubled origin). Let X be the *affine plane with doubled origin* (双原点仿射平面) : two copies $U_1, U_2 \cong \mathbb{A}_k^2$ glued along $\mathbb{A}_k^2 \setminus \{(0, 0)\}$ via the identity. Then $X \rightarrow \operatorname{Spec} k$ is not separated.

The two charts $U_1, U_2 \subseteq X$ are affine, with intersection

$$U_1 \cap U_2 = \mathbb{A}_k^2 \setminus \{(0, 0)\},$$

which fails to be affine by [definition 3.3.9](#). Were X separated over $\operatorname{Spec} k$, [definition 3.2.12](#) would force $U_1 \cap U_2$ to be affine. Contradiction.

Remark 3.3.12 (Direct computation of $\mathcal{O}(\mathbb{A}_k^2 \setminus \{0\})$). One can also compute $\mathcal{O}_X(\mathbb{A}_k^2 \setminus \{(0, 0)\}) = k[x, y]$ directly without Krull–Serre. Cover by the basic opens

$$D(x) = \operatorname{Spec} k[x, y, x^{-1}], \quad D(y) = \operatorname{Spec} k[x, y, y^{-1}],$$

with intersection $D(xy) = \operatorname{Spec} k[x, y, x^{-1}, y^{-1}]$. The sheaf axiom gives the left-exact sequence

$$0 \rightarrow \mathcal{O}_X(\mathbb{A}_k^2 \setminus \{0\}) \rightarrow k[x, y, x^{-1}] \oplus k[x, y, y^{-1}] \xrightarrow{(f, g) \mapsto f - g} k[x, y, x^{-1}, y^{-1}],$$

hence

$$\mathcal{O}_X(\mathbb{A}_k^2 \setminus \{0\}) = k[x, y, x^{-1}] \cap k[x, y, y^{-1}] = k[x, y]$$

inside $k[x, y, x^{-1}, y^{-1}]$. The right-hand equality holds because an element of $k[x, y, x^{-1}]$ has no negative powers of y , while one of $k[x, y, y^{-1}]$ has no negative powers of x .

3.4 Valuation criterion of separatedness

Both the one- and two-dimensional doubled-origin constructions owe their non-separatedness to the same phenomenon: a single “generic” arc in X admits multiple “closed” completions. The valuation criterion promotes this observation to an equivalence.

3.4.1 Motivating example

Remark 3.4.1 (Motivation). If $X \rightarrow \operatorname{Spec} k$ is a separated curve, then X cannot have “double points”: two distinct points of X indistinguishable away from themselves. Any such pair would produce two distinct lifts of a common map from $\operatorname{Spec} K$ to $\operatorname{Spec} R$, where R is the DVR locally modelling the curve through the point. This observation suggests that separatedness should be equivalent to a uniqueness condition for extending maps from $\operatorname{Spec} K$ to $\operatorname{Spec} R$.

Example 3.4.2 (Two lifts from $\text{Spec } R$ for the doubled line). Let $X = \mathbb{A}_k^1 \cup \mathbb{A}_k^1$ be the affine line with doubled origin ([definition 3.2.4](#)), with origins x_1, x_2 and open immersions $i_1, i_2 : \mathbb{A}_k^1 \hookrightarrow X$. Set

$$R := \mathcal{O}_{\mathbb{A}_k^1, 0} = k[t]_{(t)}, \quad K := \text{Frac}(R) = k(t);$$

so R is a DVR. The canonical morphism $\text{Spec } R \rightarrow \mathbb{A}_k^1$ sends the closed point to the origin $0 \in \mathbb{A}_k^1$ and the generic point to the generic point of $\mathbb{A}_k^1$. Define

$$f_j := i_j \circ (\text{Spec } R \rightarrow \mathbb{A}_k^1) : \text{Spec } R \rightarrow X, \quad j = 1, 2.$$

$$\begin{array}{ccccc} & & f_j & & \\ & \text{Spec } R & \xrightarrow{\quad} & \mathbb{A}_k^1 & \xrightarrow{i_j} X \end{array}$$

The two morphisms f_1, f_2 are distinct, yet they agree on $\text{Spec } K$:

- *Closed point:* f_j sends the closed point of $\text{Spec } R$ to x_j , and $x_1 \neq x_2$;
- *Generic point:* on $\mathbb{A}_k^1 \setminus \{0\}$ the two charts are identified, so $f_1|_{\text{Spec } K} = f_2|_{\text{Spec } K}$.

Remark 3.4.3. 这个例子精确地刻画了“双原点”的非分离本质：从 DVR $R = k[t]_{(t)}$ 出发的一条“代数弧”，其 generic fibre ($\text{Spec } K$) 被 X 的粘合数据唯一确定，但 special fibre (闭点) 在 X 的两个原点之间存在选择余地——这种“尾部分叉”正是分离性所禁止的。valuation criterion 将此观察升格为一个等价条件： f 分离 $\Leftrightarrow$ 对任意 DVR R 及任意相容方块， $\text{Spec } R \rightarrow X$ 的延拓至多唯一。

3.4.2 Valuation rings

Definition 3.4.4 (Valuation). Let K be a field and G a totally ordered abelian group. A *valuation* (赋值) on K with values in G is a map

$$v : K \setminus \{0\} \rightarrow G$$

satisfying, for all $x, y \in K$ with $xy \neq 0$,

- (i) $v(xy) = v(x) + v(y)$;
- (ii) $v(x + y) \geq \min(v(x), v(y))$ whenever $x + y \neq 0$.

The customary value groups are $G = \mathbb{Z}, \mathbb{Q}, \mathbb{R}$. The associated *valuation ring* (赋值环) is

$$R_v := \{a \in K : a = 0 \text{ or } v(a) \geq 0\},$$

a local subring of K with maximal ideal $\{a : v(a) > 0\}$.

Fact 3.4.5 (Characteristic property). *A subring $R \subseteq K$ is a valuation ring of K if and only if for every $x \in K^\times$, at least one of x, x^{-1} lies in R .*

Remark 3.4.6 (Hartshorne II, Exercise 4.11). In the valuation criterion below, it suffices to test the condition against *discrete* valuation rings (DVRs, $G = \mathbb{Z}$). This reduction simplifies most applications: the DVRs relevant to algebraic geometry arise as local rings of codimension-one points on normal schemes.

3.4.3 Statement of the criterion

Theorem 3.4.7 (Valuation criterion for separatedness). *Let $f : X \rightarrow Y$ be a morphism of schemes with X Noetherian. Then f is separated if and only if the following condition holds:*

for every field K , every valuation ring R with $\text{Frac}(R) = K$, and every commutative square

$$\begin{array}{ccc} U = \text{Spec } K & \longrightarrow & X \\ \downarrow i & \nearrow \text{at most one} & \downarrow f \\ T = \text{Spec } R & \longrightarrow & Y \end{array}$$

*with i induced by the inclusion $R \xrightarrow{i^\#} K$, there is **at most one** morphism $T \rightarrow X$ making the full diagram commute.*

Remark 3.4.8. $\text{Spec } R$ 的点集只有两个：闭点（极大理想）与 generic 点（零理想）。给定 generic 点上的映射 $U \rightarrow X$ 以及 $T \rightarrow Y$ 的相容性，延拓到 $T \rightarrow X$ 相当于“把闭点放到 X 的某处”——分离性保证这个选择至多唯一。[definition 3.4.2](#) 中双原点直线给出了恰好两个不同的 lift，从而印证了那里的非分离性。值得注意判据只要求“至多”，而非“存在且唯一”——后者对应更强的性质 properness。

3.4.4 Applications over a base field

Example 3.4.9 (Base $Y = \text{Spec } k$). Take $Y = \text{Spec } k$ for a field k . Since every k -scheme has a unique structural morphism to $\text{Spec } k$, the outer square is automatic: any morphism $\text{Spec } K \rightarrow X$ already fits uniquely into the square. The criterion thus collapses to:

$$X(R) \longrightarrow X(K) \text{ is injective,}$$

where

$$X(R) := \text{Hom}_{\text{Spec } k}(\text{Spec } R, X), \quad X(K) := \text{Hom}_{\text{Spec } k}(\text{Spec } K, X).$$

Example 3.4.10 (Affine X over k). Suppose $X = \operatorname{Spec} k[x_1, \dots, x_n]/(f_1, \dots, f_m)$ is affine over k . For any k -algebra A , $\operatorname{Hom}_{\operatorname{Spec} k}(\operatorname{Spec} A, X)$ is in bijection with k -algebra maps $k[x_1, \dots, x_n]/(f_\bullet) \rightarrow A$, i.e. with

$$X(A) = \{ \underline{a} = (a_1, \dots, a_n) \in A^n : f_1(\underline{a}) = \dots = f_m(\underline{a}) = 0 \}.$$

The restriction map $X(R) \rightarrow X(K)$ is therefore the coordinate-wise inclusion

$$\{ \underline{a} \in R^n : f_\bullet(\underline{a}) = 0 \} \hookrightarrow \{ \underline{a} \in K^n : f_\bullet(\underline{a}) = 0 \},$$

induced by $R \hookrightarrow K$, which is manifestly injective. By [definition 3.4.9](#), affine X over $\operatorname{Spec} k$ is separated—consistent with [definition 3.2.10](#).

Remark 3.4.11 (Gluing can break injectivity). For a general k -scheme $X = \bigcup_i X_i$ with affine charts X_i , each $X_i(R) \hookrightarrow X_i(K)$ is injective by [definition 3.4.10](#), yet the assembled map

$$X(R) \longrightarrow X(K)$$

need *not* be injective: a single K -point may have multiple R -lifts after gluing. This failure is the non-separatedness.

The prototype is [definition 3.4.2](#): for X the affine line with doubled origin over k and $R = k[t]_{(t)}$, $K = k(t)$, the K -point arising from the composition $\operatorname{Spec} K \hookrightarrow \mathbb{A}_k^1 \hookrightarrow X$ (via either chart—they agree away from 0) admits two distinct lifts $f_1, f_2 \in X(R)$, distinguished by whether the closed point of $\operatorname{Spec} R$ maps to 0_1 or 0_2 .

Remark 3.4.12. 用 moduli 的语言：分离的 X/k 意味着 R -点由其 generic fibre (K -点) 唯一确定；非分离允许同一个 K -点有多种“特化”(specialization)。这也是判别 moduli 空间分离性的标准工具——检查“退化的极限是否唯一”。

3.4.5 Morphisms from T and U

The proof of [definition 3.4.7](#) rests on an explicit description of morphisms from T and U into an arbitrary scheme X . For the remainder of this section, fix a valuation ring R with fraction field $K = \operatorname{Frac}(R)$, and set

$$T := \operatorname{Spec} R, \quad U := \operatorname{Spec} K, \quad i : U \hookrightarrow T$$

the canonical open immersion (dual to $R \hookrightarrow K$).

Lemma 3.4.13. *Let X be a scheme.*

(1) *There is a natural bijection*

$$\operatorname{Hom}_{\operatorname{Sch}}(U, X) \xrightarrow{1:1} \{ (x_1, \iota) : x_1 \in X, \iota : k(x_1) \hookrightarrow K \text{ a field embedding} \},$$

sending $f : U \rightarrow X$ to $(f(\operatorname{pt}), f_{\operatorname{pt}}^\# : k(f(\operatorname{pt})) \hookrightarrow K)$.

(2) *There is a natural bijection*

$$\mathrm{Hom}_{\mathrm{Sch}}(T, X) \xrightarrow{1:1} \left\{ (x_0, x_1, \iota) \left| \begin{array}{l} x_0, x_1 \in X, \ x_0 \in \overline{\{x_1\}}, \\ \iota : k(x_1) \hookrightarrow K \text{ embedding,} \\ \iota^{-1}(R) = \mathcal{O}_{Z_1, x_0} \text{ inside } k(x_1) \end{array} \right. \right\},$$

where $Z_1 := \overline{\{x_1\}} \subseteq X$ is equipped with the reduced induced closed subscheme structure, so that $k(x_1) = \mathrm{Frac}(\mathcal{O}_{Z_1, x_0})$.

Remark 3.4.14. 直观: $U = \mathrm{Spec} K$ 出发的态射等价于“选定 X 中一点 x_1 及其剩余域到 K 的嵌入”。 $T = \mathrm{Spec} R$ 出发则多了“闭点 t_0 往哪儿去”的信息——那就是 x_1 的一个 specialization x_0 。兼容条件 $\iota^{-1}(R) = \mathcal{O}_{Z_1, x_0}$ 说: R 通过 $k(x_1)$ 嵌入 K 这一路径里截出来的“非负部分”恰好是 Z_1 在 x_0 处的局部环——也就是 R 主导 (dominates) 该局部环。

Proof of definition 3.4.13(1). Since $U = \mathrm{Spec} K$ consists of a single point, giving a morphism $(f, f^\#) : U \rightarrow X$ amounts to:

- a point $x_1 := f(\mathrm{pt}) \in X$, and
- a local ring homomorphism on stalks at pt:

$$f_{\mathrm{pt}}^\# : \mathcal{O}_{X, x_1} \longrightarrow \mathcal{O}_{U, \mathrm{pt}} = K.$$

Locality sends $\mathfrak{m}_{x_1}$ into $\mathfrak{m}_K = (0)$, so $f_{\mathrm{pt}}^\#$ factors through $k(x_1) = \mathcal{O}_{X, x_1} / \mathfrak{m}_{x_1}$. Since K is a field and the induced map $k(x_1) \rightarrow K$ is non-zero, it is injective. Thus

$$\mathrm{Hom}_{\mathrm{loc-Ring}}(\mathcal{O}_{X, x_1}, K) = \mathrm{Hom}_{\mathrm{Ring}}(k(x_1), K) = \{\text{field embeddings } k(x_1) \hookrightarrow K\}.$$

□

Proof of definition 3.4.13(2), direction ($\Rightarrow$). The ring R has two prime ideals: the maximal $\mathfrak{m}_R$ (closed point $t_0 \in T$) and (0) (generic point $t_1 \in T$). Given $f : T \rightarrow X$, set

$$x_0 := f(t_0), \quad x_1 := f(t_1).$$

The specialization $t_0 \in \overline{\{t_1\}} = T$ forces $x_0 \in \overline{\{x_1\}}$ (continuous maps preserve specializations); in particular $f(T) \subseteq \overline{\{x_1\}}$ as sets. Applying part (1) to $f \circ i : U \rightarrow X$ gives an embedding $\iota : k(x_1) \hookrightarrow K$.

Factorization through Z_1 . By [?, Ex. II.3.11], since T is reduced and $f(T) \subseteq Z_1 := \overline{\{x_1\}}$ set-theoretically, f factors uniquely:

$$T \xrightarrow{\exists!} Z_1 \hookrightarrow X.$$

$$\begin{array}{ccc}
 T & \xrightarrow{f} & X \\
 \searrow \exists! & & \uparrow \text{closed immersion} \\
 & & Z_1
 \end{array}$$

At the level of local rings this yields $\mathcal{O}_{Z_1, x_0} \rightarrow \mathcal{O}_{T, t_0} = R$, a local homomorphism compatible with $k(x_1) = \text{Frac}(\mathcal{O}_{Z_1, x_0}) \xrightarrow{\iota} K = \text{Frac}(R)$.

Compatibility. Locality plus compatibility yields the square

$$\begin{array}{ccc}
 \mathcal{O}_{Z_1, x_0} & \hookrightarrow & R \\
 \downarrow & & \downarrow \\
 k(x_1) & \xrightarrow{\iota} & K
 \end{array}$$

from which $\iota^{-1}(R) \supseteq \mathcal{O}_{Z_1, x_0}$. The reverse inclusion uses that R is a valuation ring: any $\alpha \in k(x_1) \setminus \mathcal{O}_{Z_1, x_0}$ satisfies $\alpha^{-1} \in \mathfrak{m}_{\mathcal{O}_{Z_1, x_0}}$, hence $\iota(\alpha^{-1}) \in \mathfrak{m}_R$, forcing $\iota(\alpha) \notin R$. Thus $\iota^{-1}(R) = \mathcal{O}_{Z_1, x_0}$, producing the required datum.

Conversely, given such a datum, the composition $\text{Spec } R \rightarrow \text{Spec } \mathcal{O}_{Z_1, x_0} \hookrightarrow Z_1 \hookrightarrow X$ reconstructs f , establishing the bijection. $\square$

Remark 3.4.15. [?, Ex. II.3.11] 说的是：对任意概形 T 和闭子集 $W \subseteq X$ ，若 T reduced 且拓扑上 $f(T) \subseteq W$ ，则 f 唯一地因子穿过 W 的 reduced induced 子概形结构。没有 reduced 条件不行，因为 T 上的 nilpotents 可以把非 reduced 信息传到 f 上；此处 $T = \text{Spec } R$ 是整环故 reduced。

3.4.6 Closedness from specialization

The other ingredient of the valuation criterion is a topological tool: *for quasi-compact morphisms, closedness of the image is a specialization-closure condition*. Before stating the result, one remark on the broader context.

Remark 3.4.16 (Constructibility of $f(X)$). For finite-type morphisms, Chevalley's theorem strengthens "closed image" to "constructible image": $f(X)$ is a finite boolean combination of locally closed subsets. Constructible + closed under specialization + quasi-compact together imply closed. The lemma below is the purely topological skeleton.

Lemma 3.4.17. *Let $f : X \rightarrow Y$ be a quasi-compact morphism of schemes. Then $f(X) \subseteq Y$ is closed if and only if $f(X)$ is stable under specialization, i.e.*

$$\forall x_1, x_0 \in Y : \quad x_0 \in \overline{\{x_1\}} \text{ and } x_1 \in f(X) \implies x_0 \in f(X).$$

Remark 3.4.18. 方向“ $\Rightarrow$ ”显然：闭集自然在特殊化下稳定。反方向需要 quasi-compact 把“局部特殊化稳定”拼成“全局闭”。在代数几何中这条引理极其有用：验证一个映射的像是闭集常常就归结为检查纤维的“极限点”是否落在像里——这也是 valuation criterion 做的事。

Before proving ($\Leftarrow$) we note the following example showing quasi-compactness cannot be dropped.

Example 3.4.19 (Quasi-compactness is essential). Let k be an infinite field, and set $S \subseteq \mathbb{A}_k^1$ to be the set of *all closed points*. Then:

1. $\eta_{\mathbb{A}_k^1} \notin S$ (the generic point is not closed);
2. $\overline{S} = \mathbb{A}_k^1$ (closed points are dense).

So S is stable under specialization (each closed point specializes only to itself) but not closed. Realize S as the image of

$$f : \coprod_{x \in S} \operatorname{Spec} \kappa(x) \longrightarrow \mathbb{A}_k^1,$$

a disjoint union indexed by the infinite set S . The source is not quasi-compact, so f is not quasi-compact. Thus $\operatorname{Im} f = S$ is stable under specialization but not closed.

Proof of definition 3.4.17, ($\Leftarrow$). Assume f quasi-compact with $f(X)$ stable under specialization; we show $f(X)$ is closed.

Reduction 1: reduce to X, Y reduced and $Y = \overline{f(X)}$. Replacing X with X_{red} does not change $|X|$ nor the image, so assume X reduced. By [?, Ex. II.3.11], f then factors through Y_{red} ; since $|Y_{\text{red}}| = |Y|$, it suffices to show $f(X)$ closed in Y_{red} . So assume Y reduced. Finally, $\overline{f(X)} \subseteq Y$ carries a reduced induced closed subscheme structure through which f factors (same exercise). Replacing Y by $\overline{f(X)}$, we assume

$$X, Y \text{ reduced}, \quad \overline{f(X)} = Y, \quad \text{and we must show } f(X) = Y.$$

Reduction 2: reduce to Y affine. Closedness is local on Y : cover $Y = \bigcup V_j$ by affines, and observe $f(X) \cap V_j$ is dense in V_j . Replacing Y by some V_j we may take Y affine; quasi-compactness of f then forces X quasi-compact, so

$$X = \bigcup_{i=1}^n X_i, \quad X_i = \operatorname{Spec} A_i \text{ affine, } n < \infty.$$

Reduction 3: reduce to $\operatorname{Spec} A \rightarrow \operatorname{Spec} B$ dominant with $B \hookrightarrow A$ reduced. From $Y = \overline{f(X)} = \bigcup_{i=1}^n \overline{f(X_i)}$, each $y \in Y$ lies in some $\overline{f(X_{i_0})}$. Set

$$X' := X_{i_0} = \operatorname{Spec} A, \quad Y' := \overline{f(X_{i_0})} \subseteq Y, \quad Y' = \operatorname{Spec} B.$$

$X' \rightarrow Y$ factors through Y' , giving a morphism of reduced affine schemes $\operatorname{Spec} A \rightarrow \operatorname{Spec} B$ with dense image, corresponding to a ring map $f^\# : B \rightarrow A$.

- (i) $f^\#$ is injective. If $\ker f^\# \neq 0$, then $\text{Im}(f|_{X'}) \subseteq V(\ker f^\#) \subsetneq \text{Spec } B$, contradicting density.
- (ii) Thus $B \hookrightarrow A$, with A, B reduced.

Heart of the argument: finding a generic lift. Fix $y \in Y$ with corresponding prime $P \subseteq B$; we show $y \in f(X)$. Pick a minimal prime $P' \subseteq P$ of B , corresponding to a generic point $y' \in Y'$ of an irreducible component. Then $y \in \overline{\{y'\}}$: y is a specialization of y' .

Claim 3.4.20. $B_{P'}$ is a field.

Proof of claim. Primes of $B_{P'}$ correspond bijectively to primes of B contained in P' . Minimality of P' forces this set to be $\{P'\}$, so $B_{P'}$ has a unique prime $P'B_{P'}$ —necessarily equal to the nilradical. Since B reduced implies $B_{P'}$ reduced, $P'B_{P'} = 0$, and $B_{P'}$ is a field. $\square$

Claim 3.4.21. $y' \in f(X)$.

Proof of claim. Localization is exact, so $B \hookrightarrow A$ induces $B_{P'} \hookrightarrow A \otimes_B B_{P'}$. In particular $A \otimes_B B_{P'} \neq 0$. The base-change square

$$\begin{array}{ccc} \text{Spec}(A \otimes_B B_{P'}) & \longrightarrow & \text{Spec } B_{P'} = \{y'\} \\ \downarrow & & \downarrow \\ \text{Spec } A & \xrightarrow{f} & \text{Spec } B \end{array}$$

shows any point of the non-empty $\text{Spec}(A \otimes_B B_{P'})$ maps to y' under f . Hence $y' \in f(X)$. $\square$

Conclusion via specialization stability. We have $y' \in f(X)$ and $y \in \overline{\{y'\}}$. By hypothesis $f(X)$ is stable under specialization, so $y \in f(X)$. As $y \in Y$ was arbitrary, $f(X) = Y$, which is closed. $\square$

Remark 3.4.22. 核心步骤是“先找一般点 (generic) 的 lift, 再用特殊化拉回到 y'' ”。极小素理想 P' 对应不可约分量的 generic 点 y' ; $B_{P'}$ 在 reduced 条件下退化成域, 使得 $\text{Spec}(A \otimes_B B_{P'})$ 的非空性 (来自 $B \hookrightarrow A$ 的局部化平坦性) 直接给出 $y' \in f(X)$ 。 “局部化到一般点 + 特殊化闭” 是交换代数证明几何命题的标准范式。